Steeper Simplicity Priors in Solomonoff Function Induction:
Absorption, the Teacher's Mistakes and the Model–Randomness Split

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Abstract

Solomonoff function induction (SFI) predicts labels from inputs with a universal prior over random functions. We study the proposal to penalise program length more steeply than log-likelihood, by a factor $\lambda > 1$, for hypotheses with private randomness at each input, so that learning selects a simple model plus noise rather than a model of a teacher's systematic errors. A standard objection is that a universal mixture, taken as one hypothesis, absorbs all structure. We prove that it does in every class containing its own mixture, such as Solomonoff induction and SFI with shared randomness, while no private-randomness hypothesis is universal. For a random parity on uniform inputs whose labels are flipped, at a rate below 0.056, on an error set of product form that is incompressible relative to the parity, we prove that for $\lambda = cn$ (n samples) and c in an explicit window, every near-optimal hypothesis of the population objective knows the parity and, when the error rule is large, gives the erroneous labels far less probability than a memoriser does. The window needs λ proportional to n; density and description length of the error rule do not suffice. At $\lambda = 1$ a computable sampler and the programs that fill in its sampled bits get similar posterior weight; for fixed $\lambda > 1$, private randomness selects samplers over fill-ins, while with shared randomness a universal mixture beats every sufficiently long sampler. For a random parity on $\{0,1\}^d$ whose labels are flipped on a random set of inputs of fixed density below $1/4$, the sophistication defined with product models exceeds the set sophistication of algorithmic statistics by $d - O(\log d)$ bits, for a range of sufficiency slacks.

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1 Introduction

Solomonoff function induction (SFI) is Bayesian prediction of a label y from an input x with a universal prior over random partial functions (Hänni, 2026a). It is the supervised-learning counterpart of Solomonoff induction (SI), and like SI it learns every computable regularity of the labels, including a labeller's systematic mistakes. A human learner, by contrast, can grasp what a teacher means by a word without learning to predict what the teacher would say, mistakes included. We study the following proposal for this purpose, a stronger simplicity prior: when the mistakes are explainable deviations from a simple model, select the simple model with noise, by penalising the length of a hypothesis more than its likelihood term.[1] In the form studied here, a hypothesis is a program π of length $\ell(\pi)$ with likelihood P_π, and one minimises

$$J_\lambda(\pi; D) = \lambda\, \ell(\pi) - \log P_\pi(D), \qquad \lambda > 1,$$

or uses the power prior $2^{-\lambda \ell(\pi)}$ (Definition 3.5). We consider it for function induction with private randomness at each input: a hypothesis is a probabilistic program that uses fresh coins at every input.

A standard objection, which we call the *absorption objection*, is that under a steeper penalty a hypothesis that is itself a universal mixture is short, penalises complexity inside itself only in the usual way, and therefore absorbs all the structure of the data, errors included. In sequence prediction the objection holds (Section 4). In function induction one may hope that it fails if each input samples its own random bits, so that a hypothesis pays its likelihood term separately at

[1] The proposal first appeared in the author's note (Hänni, 2026b) and in an online discussion (on LessWrong), where the absorption objection described below was also raised. So did the suggestion that private randomness at each input blocks this objection, the three-hypothesis calculation below (in both versions of Remark 5.1), the question whether its good model is a vertex of the convex hull, and an absolute complexity bound as an alternative. SFI itself first appeared in the author's working notes (Hänni, 2026c).

every input. A three-hypothesis calculation illustrates the proposal. Take a 100-bit simple model and systematic errors on a 2^{-10} fraction of the inputs, whose set takes 10,000 bits to specify; of three hypotheses (the simple model with noise, a model that learns the errors, and a fair coin) the first is selected for a range of λ/n (Remark 5.1). The example is favourable to the proposal, and three hypotheses are not all hypotheses: does another hypothesis do better, and is some other pathology present? The split between model and noise arises in its simplest form for labels drawn by a simple sampler, whose coins are the noise; programs that fill in the sampled bits store the noise as model. At $\lambda = 1$ the two get similar posterior weight, in SFI and in SI (Theorem 7.3 and Proposition E.3); does a steeper prior prefer the sampler? Table 1 lists these and further questions and where they are answered.

Conventions. Logarithms are to base 2. n is the number of samples, $N = 2^d$, $c = \lambda/n$, and κ is a constant that depends only on the reference machine and fixed programs; numbers computed with $\kappa = 0$ are illustrations, not results about a reference machine. A shared hypothesis P (Definition 3.3) gives the label y at a new input x after data D the probability $P(D{+}(x, y))/P(D)$; in this *semimeasure convention* the cumulative log loss telescopes, in every order, to at most $-\log P(D)$ bits. In the *normalised convention* the denominator is $\sum_{y'} P(D + (x, y')) \le P(D)$, so losses are never larger.

The main result. Theorem 5.7 makes the example rigorous for error rules that are incompressible relative to the model. Informally:

Theorem (informal). *Let the inputs lie in $\{0,1\}^d$, let the simple model be a random parity f_a, and let a teacher flip its labels on the inputs whose first m bits lie in an error rule $S \subseteq \{0,1\}^m$ of density $p < 0.056$, so that the error set $S \times \{0,1\}^{d-m}$ has product form. Score each hypothesis π with private randomness by the population objective $c\,\ell(\pi) + R(\pi)$, with $R(\pi)$ its expected log loss on the teacher's labels. If S is incompressible relative to the parity, the inputs are uniform and $\lambda = cn$ with c in an explicit window $(2^{-m}, c_{\mathrm{hi}})$, every near-optimal product hypothesis of the population objective knows the parity, is short, holds little information about S and gives the erroneous labels average probability at most 0.199 ($\kappa = 0$) or 0.274 ($\kappa = 30$) at $c = 10^{-4}$ and the example's sizes, against 1 for a memoriser. Below about 2^{-m} the memoriser beats every hypothesis that knows the parity and fits no better than noise.*

The example's sizes are $d = 100$, $m = 20$, $p = 2^{-10}$ and 11,710 bits for S; the numbers assume $\delta_S = 0$ for the deficiency of S. There the window is $(9.54 \cdot 10^{-7}, 8.06 \cdot 10^{-3}]$ at $\kappa = 0$ and $(9.54 \cdot 10^{-7}, 3.66 \cdot 10^{-3}]$ at $\kappa = 30$ (Remark 5.8).

Contributions.

- *Absorption* (Section 4). In every class that contains its own Bayes mixture, such as SI and SFI with shared randomness, every λ-MAP is shorter than a bound that does not depend on the data, and for every fixed $\lambda > 1$ a computable deterministic teacher's labels, errors included, are eventually predicted (Theorem 4.2). No hypothesis with private randomness is universal (Corollary 4.5); this protection costs a factor $\min(\lambda, 2)$ on shared latent coins and ends with labelled history (Propositions 4.7 and B.2).
- *The teacher's mistakes* (Section 5). The theorem above holds over all product hypotheses (Theorem 5.7), with versions for full truth tables and for sampled data (Corollaries 5.10 and 5.11).

- *Limits* (Section 6). Plain Bayes, MDL and SI learn the errors, and so does every schedule with $(\lambda_n/n)(d + L_S) \to 0$ along a sequence of instances, so steeper selection needs $\lambda = \Theta(n)$ (Proposition 6.1 and Remark 6.2). For an error rule with the example's density and about its description length, no hypothesis that fits no better than the good model is near-optimal on almost a third ($\kappa = 0$; a quarter at $\kappa = 30$) of the window's logarithmic width (Proposition 6.3). Cheap, frequent error rules defeat every c once the simple model is long enough (Proposition 6.4 and Lemma 6.5).
- *The model–randomness split* (Section 7). At $\lambda = 1$ the posterior odds of the fill-ins of a sampler of length k against the sampler lie in $[2^{-2\log(t_n+1)-\kappa}, 2^{k+\delta^*}]$, with $2\log(t_n+1)$ doubly logarithmic in the input length and at most triply logarithmic in n and in the sampled randomness, and δ^* a Ville deficiency (Theorem 7.3). For fixed $\lambda > 1$ with private randomness no λ-MAP is a fill-in once the sampled randomness $-\log \nu(D_n)$ exceeds $\lambda(k+\delta^*)/(\lambda-1)$ (Theorem 7.4); with shared randomness, and in SI, no sampler longer than a data-independent bound ℓ^*_λ is selected (Theorem 7.6).
- *Algorithmic statistics and computable classes* (Sections 8 and 9). For a noisy random parity, product sophistication exceeds set sophistication by $d - O(\log d)$ bits for a range of sufficiency slacks (Proposition 8.3). The lower bounds and the private-randomness split hold in every c.e. prefix-free product class (for the split, one that contains the sampler), with the class's own constants (Proposition 9.1).

Scope and reader's guide. The positive result is proved only for parities, error sets of product form, uniform inputs, flip rates below 0.056 and incompressible error rules (Remark 5.9); for simple, realistic error patterns we have counterexamples but no positive result. For sampled data, near-population constants need about 10^{11} samples; coarser guarantees hold from about 10^9 (Corollary 5.11). With $\lambda = cn$ the learner is a family of batch objectives, not one Bayesian predictor (Remark 3.7). After related work (Section 2) and the setting (Section 3), Sections 4 to 9 contain the results, each section with an appendix of proofs, and Section 10 collects the open problems. Notation follows the companion paper.

2 Related work

This section describes the closest work informally; the objects are defined in Section 3, and the basic theory of SFI is in the companion paper (Hänni, 2026a).

Steeper penalties and generalised posteriors. The proposal studied here weights the description length of a hypothesis by a factor $\lambda > 1$ against its log-likelihood, either in a two-part code or through the prior $2^{-\lambda\ell}$ (Definition 3.5). The tempered posterior $\Pi^{1/\lambda}$, which raises the likelihood to the power $1/\lambda$ instead, is a generalised (Gibbs, fractional or safe) posterior with learning rate $1/\lambda$ (Zhang, 2006; Bissiri et al., 2016; Bhattacharya et al., 2019; Grünwald, 2012; Grünwald and van Ommen, 2017), and the λ-MAP, its maximiser, is a two-part code with complexity weight λ. The power-prior posterior Π_λ has the same maximiser but is Bayes' rule with the prior $2^{-\lambda\ell}$; it is proportional to $(\Pi^{1/\lambda})^\lambda$. Our posterior results concern Π_λ unless stated. Unlike in these works, the target here deliberately differs from the data distribution (Remark 6.2).

λ-weighted two-part MDL in noisy classification. The closest prior work studies two-part codes in noisy binary classification, where a hypothesis is a classifier together with a code for the

Question	Answer	Where
Does absorption occur in the sequence case, and not if each input samples its own random bits?	Yes, in every class that contains its own mixture; and no hypothesis with private randomness is universal. In part: absorption returns with labelled history, and shared latent causes cannot be expressed	Thm. 4.2, Cor. 4.5, and Props. 4.7 and B.2
Can the good model of the example be selected for a range of c, as a vertex of the convex hull of (complexity, expected log loss)?	Yes for error rules incompressible relative to the model. Not for every rule with the same density and description length: for one such rule, no hypothesis that fits no better than the good model is near-optimal on about a third ($\kappa = 0$) to a quarter ($\kappa = 30$) of the window's logarithmic width; elsewhere only the weaker bounds of Thm. 5.7 with that rule's deficiency apply	Thm. 5.7 and Prop. 6.3
Does another hypothesis do better in the example?	No for incompressible rules: the theorem covers all product hypotheses. For structured rules, error envelopes do	Thm. 5.7 and Prop. 6.3
Is the three-hypothesis calculation realised by instances of the main theorem?	Yes. Its per-sample version is realised with an 11,710-bit rule; its total-bits version (Rem. 5.1) in population form, with $\lambda \in (9.77, 58.0]$ at $n = 10^4$, $\kappa = 0$ (three hypotheses: $(10, 100)$)	Rems. 5.1 and 5.8 and Ex. C.3
Must the length coefficient grow with the data?	Yes: $\lambda = \Theta(n)$ is needed	Prop. 6.1 and Rem. 6.2
Does an absolute bound on model complexity select the same hypotheses as the penalty?	In part: at vertices of the hull only	Lem. A.1
Is some other pathology present?	Yes: rate blindness, error envelopes, guard erosion, non-identifiability of λ, input-law dependence, labelled history, possible confident wrong extrapolation off the training support	§ 6, Prop. B.2
Do a sampler and its fill-ins get similar posterior weight at $\lambda = 1$, and does a steeper prior prefer the sampler?	Yes, with explicit factors, in SFI and in SI. Yes for fixed $\lambda > 1$ with private randomness (Rem. 7.5); no with shared randomness or in SI	Thms. 7.3, 7.4 and 7.6 and Prop. E.3
Against hypothesis length, does the entropy left given the hypothesis change slope at the model?	Yes for product models, at every flip rate below 1/4; not for set models	Prop. 8.3
Has a steeper penalty been studied for the model–noise split?	In part: λ-weighted MDL; restricted model families	§ 2

Table 1: Questions studied in this paper and where they are answered. Numbers with $\kappa = 0$ set the machine constant to 0 and are illustrations.

label noise, essentially a product hypothesis. Grünwald and Langford (2007) show that Bayes and MDL classifiers can behave suboptimally under misspecification. Zhu and Srebro (2025) quantify overfitting along the regularisation path of two-part-code MDL in supervised classification, and Zhu et al. (2026) study overfitting and generalisation of (PAC-)Bayesian prediction in noisy binary classification. We do not claim as new the qualitative finding that a bounded λ fits the noise and that λ must grow with n. Their target is the data distribution; ours deliberately differs from it, because the data contain the teacher's errors (Remark 6.2).

Algorithmic statistics. Algorithmic statistics splits the description of a string into a model and a data-to-model code through the structure function and sophistication (Koppel, 1987; Gács et al., 2001; Vereshchagin and Vitányi, 2004; Vereshchagin and Shen, 2017); restricting the model family, for example to Hamming balls (Vereshchagin and Vitányi, 2010), changes the structure function. Product models, a parameter with private noise at each input, are such a restriction,

and Section 8 compares the two splits for a noisy parity.

Cold posteriors. Wenzel et al. (2020) report that cold posteriors $\propto \exp(-U(\theta)/T)$ with $T < 1$, where $U(\theta) = -\sum_i \log p(y_i \mid x_i, \theta) - \log p(\theta)$, improve the predictions of Bayesian neural networks. Prior and likelihood get the same exponent, so their ratio stays that of $\lambda = 1$. The $\lambda > 1$ direction is that of fractional posteriors, with likelihood exponent $1/\lambda < 1$ (Bhattacharya et al., 2019).

An earlier report. Claude (Anthropic) (2026, Section 8) formalised the proposal for a general imitation channel, mostly by comparing a few hypotheses by rate. We re-derive the results we use from this earlier report: per-input and per-block pricing, the sequence-case threshold, the lemmas behind its kink theorem (here for semimeasures), the hull lemma, rate thresholds, rate blindness, non-identifiability, guard erosion and error envelopes (Appendix G.1). Two of its statements change here. Its full-table window at $d = 20$ has upper end about $(1 - H(p))/d \approx 0.049$ when logarithmic terms are ignored; with the $2 \log d$ terms, as large as d there, the value is $c_{\mathrm{hi}} = 0.01139$ with the machine constant set to 0 (Corollary 5.10). And its lemma that the randomness of the error set removes the slack $2p$, stated for computable hypotheses, holds for semimeasures (Lemma C.1).

Novelty. Known are λ-weighted MDL in noisy classification, the structure function, sophistication and restricted model families, generalised and tempered posteriors, and our tools (dominance bounds, Kakutani's dichotomy, Ville's inequality, hashing). To our knowledge the following are new: the contrast under a steeper prior between classes with a universal member and the private-randomness class, with its price (Section 4); the analysis over all hypotheses of Kolmogorov-complexity classes through list decoding (Theorem 5.7, Propositions 6.3 and 6.4); the model–randomness theorems (Section 7); and product sophistication with its gap to set sophistication (Proposition 8.3).

3 Setting: hypothesis classes and steeper objectives

This section fixes the data, the two kinds of hypotheses, hypothesis classes indexed by programs, and the steeper objectives. Proofs and standard facts are in Appendix A.

Objects from the companion paper. We use the following objects of Hänni (2026a).

T, ν_T	a *function machine* reads, on input x, a prefix of one random tape $\omega \sim \mathbb{U}$ that all inputs share, and outputs a label; its random partial function is f_ω^T, and its *machine law* is $\nu_T(D) = \mathbb{U}\{\omega : f_\omega^T \supseteq D\}$
M_F	Solomonoff function induction (SFI): the law of the universal function machine; $M_F(D) = \sum_i \mathbf{m}(i)\, \nu_{T_i}(D)$ for $D \neq \emptyset$
M_{prod}	the mixture of the product laws of lower-semicomputable (lsc) conditional semimeasures; product laws are machine laws, and they include the computably samplable predictors that use fresh random bits at each input
M^w	a universal lsc weak function semimeasure
M_H	the law of $\mathrm{SFI}^{\mathrm{hist}}$, SFI whose hypotheses see the earlier inputs; a universal lsc chronological semimeasure

M_{prod}, M_F and M^w agree up to $K(\mathrm{dom}\, D) + O(1)$ bits, but M_F/M_{prod} is unbounded (Hänni, 2026a, Theorem 4.4 and Proposition 4.7). If labels are redrawn at every query (fresh noise) and

inputs repeat, random-function SFI has infinite expected loss (Hänni, 2026a, Proposition 5.6); all noisy data in this paper draw each label once per distinct input (*frozen* noise).

3.1 Data and the two kinds of hypotheses

Definition 3.1 (Data)**.** The input set is $X = \{0,1\}^*$; statements about a fixed input length use $\{0,1\}^d \subseteq X$. The label set Y is *finite*, $|Y| \geq 2$. A *data set* is a finite partial function $D : X \rightharpoonup Y$ with $\operatorname{dom} D = \{x_1, \ldots, x_n\}$ (n distinct inputs) and $y_i = D(x_i)$. Noise is frozen: each distinct input has one label.

Definition 3.2 (Product hypothesis; private randomness)**.** A *product hypothesis* is an lsc conditional semimeasure $\nu(y \mid x)$ ($\sum_y \nu(y \mid x) \leq 1$). It is the law of a random partial function whose values $F(x) \sim \nu(\cdot \mid x)$ are independent across distinct x: a probabilistic machine that uses fresh coins of its own at each input. Its likelihood is the *product law* $\nu(D) := \prod_{x \in \operatorname{dom} D} \nu(D(x) \mid x)$, and its prediction at a new input x is $\nu(\cdot \mid x)$, whatever D is. It is *deterministic* if it is a partial computable function f; then $\nu(D) = \mathbf{1}[f \supseteq D]$. The randomness is private to each distinct input, not to each query; on data sets the two readings coincide (Hänni, 2026a, Proposition 5.7(a)).

Definition 3.3 (Shared hypothesis)**.** A *shared hypothesis* is an lsc weak function semimeasure P: $P(\emptyset) \leq 1$ and $\sum_y P(D + (x,y)) \leq P(D)$ for $x \notin \operatorname{dom} D$. Its prediction is $P(y \mid x, D) := P(D + (x,y))/P(D)$. Machine laws, Bayes mixtures $\sum_i w_i P_i$ with $\sum_i w_i \leq 1$, and product laws are shared hypotheses. By telescoping, the cumulative sequential log loss of P on D_n is $-\log P(D_n) + \log P(\emptyset) \leq -\log P(D_n)$ in every reveal order. Hypotheses of Solomonoff induction (SI), semimeasures on an interleaved string $x_1 y_1 x_2 y_2 \cdots$, are not weak function semimeasures: they depend on the order and assign probabilities to the inputs.

3.2 Hypothesis classes

A steeper penalty acts on single hypotheses, so its effect depends on what counts as one hypothesis and on its length. We therefore make the programs explicit.

Definition 3.4 (Hypothesis class)**.** Fix a countable set $\mathcal{O}$ of *observations*: data sets for SFI, binary strings for SI, input–label sequences for history-aware prediction. A *hypothesis class* is a c.e. prefix-free set $\mathcal{H} \subseteq \{0,1\}^*$ of *programs* π, of length $\ell(\pi) := |\pi|$, with a family $(P_\pi)_{\pi \in \mathcal{H}}$ of maps $\mathcal{O} \to [0,1]$ that are lsc uniformly in π. Two programs for the same law are different hypotheses. By Kraft's inequality, $\sum_\pi 2^{-\ell(\pi)} \leq 1$.

In a *canonical class*, $\mathcal{H} := \operatorname{dom} V$ for the fixed universal prefix machine V, and P_π is the member with index $V(\pi)$ of a fixed effective enumeration of a family, trimmed so that exactly the members of the family occur. The canonical classes and their mixtures $\xi_1 := \sum_\pi 2^{-\ell(\pi)} P_\pi$ are: $\mathcal{H}_\times$, lsc conditional semimeasures with their product laws ν_π, whose mixture is $\xi_\times = \sum_i \mathbf{m}(i)\nu_i = M_{\text{prod}}$ exactly; $\mathcal{H}_F$, function machines with their machine laws, whose mixture is M_F on nonempty data sets; $\mathcal{H}_w$, lsc weak function semimeasures, with mixture M^w; $\mathcal{H}_{\text{SI}}$, lsc semimeasures on binary strings, with the Solomonoff–Levin mixture M; $\mathcal{H}_{\text{chr}}$, history-oracle function machines with their chronological semimeasures $\rho(y_{1:n} \| x_{1:n})$, with mixture M_H, itself an lsc chronological semimeasure that dominates every one (Hänni, 2026a, Corollary 6.17(a)). We use only this universality of M_H, not its identification with Hutter's universal chronological semimeasure. Canonical classes are *closed under prefix-free simulation*: for every prefix machine V' whose outputs are indices of the family there is a string σ with $V(\sigma w) = V'(w)$ for all $w \in \operatorname{dom} V'$, so distinct w give distinct programs σw.

Throughout, κ is a constant that depends only on the reference machine, the class and the fixed programs built in the proofs, never on the data or the hypotheses; different occurrences may differ. We use five standard facts (F1)–(F5) of algorithmic information theory, stated in Appendix A.1: $K(\pi) \le \ell(\pi) + \kappa$ in every class (F1), the conditional coding theorem (F2), the chain rule (F3), symmetry of information (F4), and $K(j) \le 2\log(j+1) + \kappa$ (F5).

3.3 Steeper objectives

Definition 3.5 (Steeper objectives)**.** Fix a class, an observation o and a *steepness* $\lambda \ge 1$. The *two-part objective* is $J_\lambda(\pi; o) := \lambda\,\ell(\pi) - \log P_\pi(o) \in [0, \infty]$; a minimiser is a *$\lambda$-MAP*. The *power-prior posterior* is $\Pi_\lambda(\pi \mid o) := 2^{-\lambda\ell(\pi)} P_\pi(o)/\xi_\lambda(o)$, with the *steeper mixture* $\xi_\lambda(o) := \sum_\pi 2^{-\lambda\ell(\pi)} P_\pi(o)$: Bayes' rule with the prior $2^{-\lambda\ell}$, whose Kraft sum is at most 1. We write $\xi_\lambda^{\times}$, ξ_λ^{F} for $\mathcal{H}_\times$, $\mathcal{H}_F$, and $\xi_{\mathcal{C}} := \sum_{\pi\in\mathcal{C}} 2^{-\ell(\pi)} P_\pi$ for a class $\mathcal{C}$. The *tempered posterior* with exponent $\eta \in (0, 1]$ is $\Pi^\eta(\pi \mid o) \propto 2^{-\ell(\pi)} P_\pi(o)^\eta$.

The two-part objective weights the length of a program λ times as heavily as its negative log-likelihood; the power-prior posterior is its Bayesian form. For data-set classes, ξ_λ is a shared hypothesis. The tempered posterior $\Pi^{1/\lambda}$ is a generalised (Gibbs, fractional or safe) posterior with learning rate $1/\lambda$ (Zhang, 2006; Bissiri et al., 2016; Bhattacharya et al., 2019; Grünwald, 2012; Grünwald and van Ommen, 2017), and the λ-MAP, its maximiser (Lemma 3.6), is a two-part code with complexity weight λ. The power-prior posterior Π_λ has the same maximiser but is Bayes' rule with the prior $2^{-\lambda\ell}$; it is proportional to $(\Pi^{1/\lambda})^\lambda$. Our posterior results concern Π_λ unless stated. What we study is how $\lambda > 1$ interacts with universal hypotheses and with private versus shared randomness.

Lemma 3.6 (Basic properties)**.** *(i) If $J_\lambda(\pi; o) < \infty$ for some π, the set of λ-MAPs is finite and nonempty. (ii)* $\arg\min_\pi J_\lambda(\pi; o) = \arg\max_\pi \Pi_\lambda(\pi \mid o) = \arg\max_\pi \Pi^{1/\lambda}(\pi \mid o)$. *(iii) On deterministic hypotheses $P_\pi(D)^\eta = P_\pi(D) \in \{0, 1\}$, so on any set of deterministic hypotheses the tempered posterior has the relative weights of the plain ($\lambda = 1$) posterior.*

Proof. (i) $J_\lambda \ge \lambda\ell$, and by Kraft only finitely many programs have $\ell(\pi) \le J_\lambda(\pi_0; o)/\lambda$. (ii) $2^{-J_\lambda} = (2^{-\ell} P_\pi^{1/\lambda})^\lambda$. (iii) is immediate. □

Remark 3.7 (Per-sample form; $\lambda = cn$ gives batch objectives). For a product class and n data points put $c := \lambda/n$; then $J_\lambda/n = \hat{J}_c(\pi) := c\,\ell(\pi) + \frac{1}{n}\sum_{i\le n} -\log \nu_\pi(y_i \mid x_i)$. For an input law μ and a fixed teacher $y^* : X \to Y$ the population versions are the risk $R(\pi) := \mathbb{E}_{x\sim\mu}[-\log \nu_\pi(y^*(x) \mid x)]$, $J_c(\pi) := c\,\ell(\pi) + R(\pi)$ and $\Phi(c) := \inf_\pi J_c(\pi)$. The coefficient of the length term must grow with the size of the data set (Remark 6.2); $\lambda = cn$ with c fixed is this choice. For fixed λ, ξ_λ is one shared hypothesis and its sequential loss telescopes; with $\lambda = cn$ the prior exponent changes with n and this is not available. All statements with $\lambda = cn$ are therefore *batch* statements at a given n, not statements about one Bayesian predictor.

Selection by $c\,\ell + R$ has a simple geometry. The minimisers lie on the lower-left boundary of the convex hull of the attainable pairs $(\ell(\pi), R(\pi))$ of complexity and expected log loss, and exactly its vertices are unique minimisers on an open interval of c, so for the example of Section 5 the question is whether the good model is a vertex; a larger c selects shorter hypotheses with larger risk; and an absolute bound on model complexity, a natural alternative to the penalty, selects the same hypotheses at the vertices, but can also select points above the boundary, which the penalty never selects (Lemma A.1 and Remark A.2 in Appendix A.2).

The last lemma is the sufficiency line for data-set classes: one bit of description pays for at most one bit of likelihood.

Lemma 3.8 (Coding bound)**.** *In every data-set class (product or shared hypotheses), for every program π and data set D,*

$$\ell(\pi) - \log P_\pi(D) \;\geq\; K(y_D \mid \operatorname{dom} D) - \kappa,$$

where y_D lists the labels of D in the canonical order of $\operatorname{dom} D$*.*

Proof idea. Given π and $\operatorname{dom} D$, the labelling $y \mapsto P_\pi(\operatorname{dom} D \mapsto y)$ is an lsc semimeasure on $Y^{\operatorname{dom} D}$; apply (F2), (F3) and (F1) (Appendix A.3). What separates the two kinds of class is whether one member attains the class's best two-part code length $\min_\pi[\ell(\pi) - \log P_\pi(D)]$ up to a constant on every data set. Under (Dom) the member U does: $\ell_U - \log U(D) \leq \ell(\pi) - \log P_\pi(D) + \ell_U + \kappa_0$ for every π (Section 4). In the product class no member does (Corollary 4.5).

4 The absorption objection, and why private randomness blocks it

Under a steeper penalty, a hypothesis that is itself a universal mixture, and so penalises complexity only in the usual way inside itself, can absorb all the structure of the data. This is a standard point, familiar from algorithmic statistics, where the model classes are restricted, for example to finite sets (Gács et al., 2001; Vereshchagin and Shen, 2017). In sequence prediction it means that it is better to drop bits of the hypothesis and pay for them in likelihood. In function induction there is a natural way out: if the randomness is not shared between inputs, each input sampling its own random bits, a hypothesis pays its likelihood term separately at every input, and the pathology may be absent. The point is the absorption objection of Section 1. Theorem 4.2 makes it exact in every class that contains its own mixture; Proposition 4.4 and Corollary 4.5 show that the product class $\mathcal{H}_\times$ contains no such hypothesis; Propositions 4.7 and B.2 give the price and the boundary of this protection; Remark 4.8 assembles the answer. Proofs are in Appendix B.

4.1 Shared randomness: a universal hypothesis absorbs everything

A class satisfies (Dom) if it contains a program π_U, with $U := P_{\pi_U}$ and $\ell_U := \ell(\pi_U)$, such that

$$P_\pi(o) \;\leq\; 2^{\ell(\pi)+\kappa_0}\, U(o) \qquad \text{for all programs } \pi \text{ and observations } o. \tag{Dom}$$

Remark 4.1 (Classes with an absorbing member (proved in Appendix B.1))*.* (Dom) holds with $\kappa_0 = 0$, and $\xi_1 \leq U$, in every class that contains its own mixture as a member: in $\mathcal{H}_F$ with $U = M_F$, in $\mathcal{H}_w$ with $U = M^w$, in $\mathcal{H}_{\mathrm{SI}}$ with $U = M$, and in $\mathcal{H}_{\mathrm{chr}}$ with $U = M_H$ (only the universality of M_H is used).

Let $\ell^*_\lambda := (\lambda\ell_U + \kappa_0)/(\lambda - 1)$.

Theorem 4.2 (Absorption)**.** *Assume (Dom) and let $\lambda > 1$.*

(a) MAP. *For every observation o with $U(o) > 0$, every λ-MAP $\hat\pi$ has $\ell(\hat\pi) \leq \ell^*_\lambda$ and $-\log P_{\hat\pi}(o) \leq -\log U(o) + \lambda\ell_U - \lambda\ell(\hat\pi)$. So if $L > \ell_U$ and $\lambda > (L + \kappa_0)/(L - \ell_U)$, no hypothesis of length at least L is a λ-MAP for any such observation; this holds for fixed λ, and for $\lambda = cn$ once cn exceeds the bound.*

(b) Power posterior. $2^{-\lambda \ell_U} U(o) \le \xi_\lambda(o) \le \xi_1(o)$. *If* $\xi_1 \le 2^{\kappa_1} U$, *then* $-\kappa_1 \le -\log \xi_\lambda(o) + \log U(o) \le \lambda \ell_U$ *for every* o.

(c) Deterministic teacher, errors included (data-set classes). *Let* $\lambda > 1$ *be fixed. Let* $t : X \to Y$ *be total computable with a program* π_t *of length* ℓ_t *in the class, let* $x_1, x_2, \dots$ *be distinct and* $D_n := \{(x_i, t(x_i)) : i \le n\}$.

(c1) $\sum_{i \le n} -\log \xi_\lambda(t(x_i) \mid x_i, D_{i-1}) \le \min\{\lambda \ell_t,\ \lambda \ell_U + \kappa_0 + \ell_t\}$ *for every* n*; so at most that many inputs, over the whole sequence, get probability below* $1/2$ *for the teacher's label.*

(c2) Every λ*-MAP for* D_n *has* $-\log P_{\hat\pi}(D_n) \le \lambda \ell_U + \kappa_0 + \ell_t$.

(c3) There are n_0 *and a finite set* $\mathcal{H}_{\mathrm{fin}}$ *such that every* λ*-MAP for* D_n, $n \ge n_0$, *lies in* $\mathcal{H}_{\mathrm{fin}}$, *and at only finitely many* n *does a* λ*-MAP give* $t(x_{n+1})$ *probability below* $1/2$.

With $\lambda = cn$, *(c1) and (c3) are not available; (a) and (c2) hold at each* n.

(d) SI with random inputs. *Let the inputs lie in* $\{0,1\}^d$, *encode* $(x_1, y_1, x_2, \dots)$ *as a binary string with fixed-length blocks, let the* x_i *be i.i.d. from a computable law and* $y_i = t(x_i)$, *and let* $\mathbb{P}$ *be the resulting computable measure, with a program* $\pi_{\mathbb{P}} \in \mathcal{H}_{\mathrm{SI}}$. *Then* $\mathbb{E}_{\mathbb{P}} \sum_{i \le n} -\log \xi_\lambda(y_i \mid x_1 y_1 \cdots x_i) \le \lambda \ell_U + \kappa_0 + \ell(\pi_{\mathbb{P}})$.

Proof idea. (a) A λ-MAP is at least as good as π_U, and (Dom) bounds its gain in likelihood by $\ell(\hat\pi) + \kappa_0$, so $(\lambda - 1)\ell(\hat\pi) \le \lambda \ell_U + \kappa_0$. (b) Keep one term of the mixture. (c) Telescope $\xi_\lambda(D_n) \ge 2^{-\lambda \ell_U} U(D_n)$; for (c3), each of the finitely many short programs either has bounded loss along the sequence or is eventually beaten by π_U. (d) is Solomonoff's bound (Solomonoff, 1978) restricted to the label positions. See Appendix B.2.

Remark 4.3 (Reading). Part (a) is the absorption objection made exact, and in $\mathcal{H}_{\mathrm{SI}}$ the precise form of the sequence-case pathology; $\ell^*_\lambda = \ell_U + (\ell_U + \kappa_0)/(\lambda - 1)$ is the threshold of the earlier report (Claude (Anthropic), 2026). It tends to ℓ_U, not 0, as $\lambda \to \infty$. Whether a given model is longer than ℓ_U depends on the reference machine, but a model of millions of bits is longer than ℓ_U for any reasonable reference machine. For adaptive or non-computable inputs, (d) is replaced by $\mathcal{H}_{\mathrm{chr}}$, where (a)–(b) hold. Part (c3) has no explicit rate; explicit online bounds for two-part MDL are exponentially worse than for Bayes mixtures in general (Poland and Hutter, 2005).

4.2 Private randomness: no hypothesis absorbs anything

In the product class a hypothesis pays its likelihood term separately at every input. For a finite set F of total functions $X \to Y$ and a distribution Q on F, let Q_x be the law of $f(x)$ for $f \sim Q$.

Proposition 4.4 (Per-input pricing)**.** *Let* $x_1, \dots, x_n$ *be distinct.*

(i) For every product hypothesis ν *and every* Q, $\max_{f \in F} \sum_{i \le n} -\log \nu(f(x_i) \mid x_i) \ge \sum_{i \le n} H(Q_{x_i})$. *The value* $\inf_\nu \max_f \sum_i -\log \nu(f(x_i) \mid x_i)$ *equals* $\max_Q \sum_i H(Q_{x_i})$, *which can be strictly below* $\sum_i \log |\{f(x_i) : f \in F\}|$.

(ii) If F *is the set of parities on* $\{0,1\}^d$ *and the* x_i *are nonzero, every product hypothesis pays at least* n *bits on some parity, while the Bayes mixture over parities, a shared hypothesis, pays at most* d *bits on every parity.*

(iii) A per-input mixture $\nu(y \mid x) = \sum_q w_q\, q(y \mid x)$ *is a product hypothesis and pays up to* $\log(1/w_q)$ *at every input: with the two constant functions and* $w = 1/2$ *it pays* n *bits on* n *ones, where the shared mixture pays* 1 *bit.*

The bound in (i) is Gibbs' inequality applied to $\mathbb{E}_{f\sim Q}$, and the value follows from Sion's minimax theorem (Sion, 1958) (Appendix B.3, with an example of the strict inequality).

Corollary 4.5 (No universal product hypothesis)**.** *For every product hypothesis ν_U and every $d \geq 1$ there are a data set D with $\operatorname{dom} D = \{0,1\}^d \setminus \{0^d\}$ and a deterministic $f \supseteq D$ with a program of length at most $d + 2\log d + \kappa$ such that $\nu_U(D) \leq 2^{-(2^d-1)}$. Hence for every C there are such D and f with $\nu_U(D) < 2^{-C}2^{-\ell(\pi_f)}f(D)$, and (Dom) fails in $\mathcal{H}_\times$, and in every product class that contains, for every d, programs of length at most $d + 2\log d + \kappa$ for the parities on $\{0,1\}^d$.*

Proof. Take the parity given by Proposition 4.4(ii) on which ν_U loses at least $2^d - 1$ bits; it is described by d and $a \in \{0,1\}^d$, and (F5) gives the length bound. Choose d with $2^d - 1 - d - 2\log d - \kappa > C$. □

The natural candidate for a universal product hypothesis, the universal lsc conditional semimeasure, pays a constant at every input and never learns (Remark B.1 in Appendix B.3).

4.3 Why private randomness, and what it costs

Remark 4.6 (Why private randomness is what matters)*.* At $\lambda = 1$ the decomposition into hypotheses does not change the predictor ($\xi_\times = M_{\text{prod}}$ is one shared hypothesis); at $\lambda > 1$ it does. Over the class $\mathcal{H}_F$ of all machine laws, M_F would be a short member and absorption would return, so steeper SFI is defined over $\mathcal{H}_\times$, whose $\lambda \to 1$ limit is M_{prod}, not M_F. At $\lambda = 1$ this costs at most $K(\operatorname{dom} D) + O(1)$ bits against M_F, but not a constant: M_F/M_{prod} is unbounded already against the law ν_{len} of $F(x) = \omega_{|x|+1}$, one shared fair coin per input length, although on inputs of length at most k the gap to ν_{len} is at most $K^{\text{pad}}(k+1) + O(1)$ bits, $K^{\text{pad}}(N) := \min_{L\geq N} K(L)$ (Hänni, 2026a, Theorem 4.4, Proposition 4.7 and Remark 4.8). At $\lambda > 1$ the cost is a constant factor on shared latent causes (Proposition 4.7).

For $k \geq 1$ and $\beta \in \{0,1\}^k$ let $D_\beta := \{(0^i, \beta_i), (1^i, \beta_i) : i \leq k\}$: one fair coin per pair of inputs, shown on both.

Proposition 4.7 (The price of private randomness)**.** *Let $Y = \{0,1\}$. (i) For every $\lambda \geq 1$ and β: $-\log \xi^\times_\lambda(D_\beta) \geq \min(\lambda, 2)\,(K(\beta \mid k) - \kappa)$. (ii) Hence $J_\lambda(\pi; D_\beta) \geq \min(\lambda,2)(K(\beta \mid k) - \kappa)$ for every $\pi \in \mathcal{H}_\times$, and in every product class with a prefix-free code, with κ depending on the class. (iii) ν_{len} is a machine law of complexity $O(1)$ with $\nu_{\text{len}}(D_\beta) = 2^{-k}$, so $-\log M_F(D_\beta) \leq k + \kappa$ and $-\log \xi^F_\lambda(D_\beta) \leq k + \lambda\kappa$. (iv) For uniform β and $j \geq 0$, with probability at least $1 - 2^{-j}$ the steeper product predictor pays at least (in the semimeasure convention) $(\min(\lambda,2) - 1)k - 2j - (\lambda+2)\kappa$ bits more than M_F and than ξ^F_λ.*

Proof idea. By the inequality of arithmetic and geometric means, $\nu(y \mid 0^i)\nu(y \mid 1^i)$ is at most the square of their average, so every product law on D_β is at most the square of a semimeasure on β; mixing and the coding theorem give (i) (Appendix B.4). A product class must treat a coin shared by several inputs either as *model*, at λ bits per coin, or as *noise*, paid again at every input that shows it (2 bits here); a shared class pays 1 bit. Persistent latent causes in real labelling, such as a labeller's mood or a per-batch coin, are of this kind.

The protection also depends on what an input is. If an input contains the earlier labelled pairs of its block, an in-context Bayes mixture over a countable computable family is a single product hypothesis on the extended inputs that pays the log-weight of the best member once per block, not once per input, so labelled history restores universality (Proposition B.2); if it contains only the earlier inputs, per-input pricing persists (Proposition B.3); both are in Appendix B.5. With

shared randomness, as in $\mathrm{SFI}^{\mathrm{hist}}$, $\mathcal{H}_{\mathrm{chr}}$ has the universal member M_H and Theorem 4.2 applies: shared randomness, which makes $\mathrm{SFI}^{\mathrm{hist}}$ as strong as a universal chronological predictor, is what removes the protection against absorption.

4.4 Answer to the absorption objection

Remark 4.8 (Answer to the absorption objection). (i) *The objection is correct wherever a universal hypothesis is one member of the class*: in SI, in SFI with shared randomness, in the weak class and in history-aware SFI. There every λ-MAP has length at most ℓ^*_λ and fits almost as well as U; for fixed λ it reproduces a deterministic teacher's errors (Theorem 4.2(c)), and samplers longer than $\ell^*_\lambda = \ell_U + (\ell_U + \kappa_0)/(\lambda - 1)$ are never selected (Theorem 7.6); with $\lambda = cn$ this threshold tends to ℓ_U.

(ii) *The construction is unavailable in the product class $\mathcal{H}_\times$.* No product hypothesis is universal (Corollary 4.5); the universal conditional semimeasure never learns (Remark B.1); seeing earlier inputs does not help while the randomness stays private (Proposition B.3).

(iii) *The restriction has a price and a boundary.* The product class cannot represent latent randomness shared across inputs, which costs up to $K(\operatorname{dom} D) + O(1)$ bits against M_F at $\lambda = 1$, unboundedly in the worst case (Remark 4.6), and a factor $\min(\lambda, 2)$ on shared coins at $\lambda > 1$ (Proposition 4.7); and the protection ends where hypotheses see labelled history (Proposition B.2).

(iv) *The algorithmic-statistics reading.* For set models, which share their randomness, the d-bit parameter of a noisy random parity is noise; for product models it is model (Proposition 8.3 and Remark 8.4).

The restriction is a modelling choice. Two reasons support it: the product hypotheses are exactly the computably samplable predictors that use different random bits on different inputs, the natural class to compare SFI with, and SFI dominates them (Hänni, 2026a, Theorem 5.1(a)); and in supervised learning the model is naturally the part of the description that is reused across inputs.

Question 4.9 (shared hypotheses)**.** Is every lower-semicomputable law of a random partial function dominated by M_F up to a constant factor? This open question of the companion paper (Hänni, 2026a, Question 3.14) decides which laws with shared randomness the class $\mathcal{H}_F$ covers; Theorem 4.2 holds for $\mathcal{H}_w$ and for $\mathcal{H}_F$ either way.

5 The teacher's-mistakes theorem

A learner that grasps what a teacher means, rather than predicting what the teacher would say, should not reproduce the teacher's mistakes. When these are explainable deviations from a simple model, the steeper objective of Definition 3.5 over product hypotheses is meant to select the simple model with noise. This section makes this rigorous for the example of Remark 5.1. Theorem 5.7 quantifies over *all* product hypotheses and shows that, for error rules incompressible relative to the model, every near-optimal hypothesis is, up to explicit slack, the simple model with noise, on a window of $c = \lambda/n$ of more than three orders of magnitude at the example's sizes for $\kappa \le 30$; Section 6 shows where this stops. Proofs are in Appendix C.

5.1 The three-hypothesis calculation

Remark 5.1 (The three-hypothesis calculation). We give the calculation in two versions: in total bits, and per sample as in Section 1.

(1) *Total-bits version*: a simple model of 100 bits that predicts 100 bits of labels per bit of description, and on top of it a systematic error model of 100 bits that predicts 10 bits per bit. Coin, simple model, and simple model plus errors lie on a hull with slopes 100 and 10, so the simple model alone is selected exactly for $\lambda \in (10, 100)$.

(2) *Per-sample version*: a 100-bit simple model with systematic human error on a 2^{-10} fraction of the inputs, whose set takes 10,000 bits to specify. Of the good model (the simple model with flip probability $p = 2^{-10}$, 10 more bits), the model that learns the errors (10,100 bits, loss 0) and the fair coin, the good model is selected if and only if c lies between the rate $H(p)/9990 = 1.1185 \cdot 10^{-6}$ of the errors and the rate $(1 - H(p))/110 = 8.989 \cdot 10^{-3}$ of the model, a ratio of about 8000; plain MDL ($\lambda = 1$) switches to the memoriser at $n \approx 8.94 \cdot 10^5$.

A direct computation (code in the accompanying repository) gives these numbers, with the exact binary entropy $H(2^{-10}) = 0.0112$. The calculation compares three hypotheses only; it does not exclude partial models, partial memorisers, mixtures or some universal hypothesis, so it leaves open whether another hypothesis does better.

5.2 The instance

Notation for this section. The symbols in the statement of Theorem 5.7 are defined in Definition 5.2, in the display before the theorem and in the theorem itself:

$d,\ m$	input length, rule length	$p = 2^{-r},\ s_0$	flip rate, rule size
$a,\ f_a$	parity parameter, parity	$S,\ E,\ y^*$	error rule, error set, teacher
$L_S,\ B_S,\ \delta_S$	log-count, complexity, deficiency	$I(\pi)$	information of π about S
$B,\ R,\ R_B$	truncation, risk, truncated risk	$J_c,\ J_c^B,\ \Phi,\ \Phi^B$	$c\ell + R$, $c\ell + R_B$, infima
u, lev	agreement with f_a, level	$s(t),\ G_j$	list-decoding cost, risk floor
$\pi_{\text{coin}},\ \pi_g,\ \pi_b$	coin, good model, memoriser	$\kappa_g,\ \kappa_b,\ \Lambda_1$	length overheads, slack
$\Phi_1,\ \Phi_2,\ J_g,\ J_b$	bounds of Section 5.4	$\Delta,\ c_{\text{hi}}$	excess, end of the window

Definition 5.2 (The teacher's-mistakes instance)**.** Throughout this section and the next, $Y = \{0,1\}$. Fix integers $1 \le m \le d$ and $2 \le r \le m$, and put $p := 2^{-r}$ and $s_0 := p2^m$.

(a) *Model, errors, teacher.* The *simple model* is the parity $f_a(x) = \langle a, x\rangle \bmod 2$ on $\{0,1\}^d$, for some $a \in \{0,1\}^d$ with $K(a) \ge d$. The *error rule* is a set $S \subseteq \{0,1\}^m$ with $|S| = s_0$, and the *error set* is $E := S \times \{0,1\}^{d-m}$, the inputs whose first m bits lie in S. The *teacher* is $y^* := f_a \oplus \mathbf{1}_E$. For $m < d$ the errors are a systematic rule (a random junta) that generalises to new inputs; for $m = d$ they are an idiosyncratic random set.

(b) *Description lengths.* $L_S := \log \binom{2^m}{s_0}$, $B_S := K(S \mid a, K(a))$, and the *deficiency* $\delta_S := L_S - B_S$. The *information* a program π holds about the rule beyond the parity is $I(\pi) := B_S - K(S \mid a, \pi)$, so $L_S - K(S \mid a, \pi) = \delta_S + I(\pi)$. One has $2^m H(p) - \log(2^m + 1) \le L_S \le 2^m H(p)$ (Cover and Thomas, 2006); $\delta_S \ge -2\log m - 2\log s_0 - \kappa$, since S can be given by its index among the s_0-subsets; and $\Pr_S[\delta_S > j] < 2^{-j}$ for a uniformly random rule, since fewer than $2^{L_S - j}$ programs are shorter than $L_S - j$.

(c) *Population quantities.* Inputs are uniform on $\{0,1\}^d$, so $\Pr[x \in E] = p$. For a product program π and a truncation level $B \in [\log(1/p), \infty]$: $R(\pi) := \mathbb{E}_x[-\log \nu_\pi(y^*(x) \mid x)]$,

$R_B(\pi) := \mathbb{E}_x \min\{B, -\log \nu_\pi(y^*(x) \mid x)\} \le R(\pi)$, $J_c := c\,\ell + R$, $J_c^B := c\,\ell + R_B$, $\Phi(c) := \inf_\pi J_c(\pi)$ and $\Phi^B(c) := \inf_\pi J_c^B(\pi) \le \Phi(c)$, the infima over the class. We write $2^{-\infty} := 0$.

(d) *Agreement and level.* $u(\pi) := \mathbb{E}_x \nu_\pi(f_a(x) \mid x)$. With $\theta_t := \frac{1}{2} + 2^{-t-1}$, the *level* is $\mathrm{lev}(\pi) := \min\{t \ge 1 : u(\pi) > \theta_t\}$, and $\mathrm{lev}(\pi) := \infty$ if $u(\pi) \le \frac{1}{2}$; so $\mathrm{lev}(\pi) = 1$ if and only if $u(\pi) > \frac{3}{4}$. Put $s(t) := 2t + 2\log(t+1) \le 4t$ and $G_j := 1 - \log(1 + 2^{-j} + 2p + 2^{1-B})$ for $j \in [0, \infty]$; G_j is increasing and concave in j.

(e) *Reference programs* in $\mathcal{H}_\times$: the *coin* π_{coin}, with $\ell \le \kappa$ and $R = 1$; the *good model* π_g, which gives $f_a(x)$ probability $1 - p$, with $\ell(\pi_g) \le d + \kappa_g$, $\kappa_g := 2\log d + 2\log r + \kappa$, and $R(\pi_g) = R_B(\pi_g) = H(p)$; the *memoriser* π_b, the deterministic y^*, with $\ell(\pi_b) \le d + \lceil L_S \rceil + \kappa_b$, $\kappa_b := 2\log d + 2\log m + 2\log s_0 + \kappa$, and $R(\pi_b) = 0$.

(f) *Slack constant.* $\Lambda_1 := 2\log d + 2\log m + 2\log B \cdot \mathbf{1}[B < \infty] + 2^m \log(1 + 2^{1-B}) + \kappa$.

The *example's sizes* are $d = 100$, $m = 20$, $p = 2^{-10}$; then $s_0 = 1024$ and $L_S = 11{,}710.27$ bits, about the 10,000 bits of the example.

The objective is the population objective J_c; finite samples are treated in Corollary 5.11. The deficiency δ_S enters every slack term below, and every number we give states the δ_S it assumes; numbers with $\delta_S = 0$ concern error rules that are incompressible given the model, and say nothing about compressible ones (Proposition 6.3).

5.3 Four lemmas

The first lemma is the reason for choosing parities: a hypothesis that agrees with f_a noticeably more often than a coin determines a up to a few bits.

Lemma 5.3 (List decoding for semimeasures)**.** *If* $\mathrm{lev}(\pi) = t < \infty$*, then* $K(a \mid \pi) \le s(t) + 2\log d + \kappa$.

Proof idea. The agreement of ν_π with a parity $f_{a'}$ is $\frac{1}{2}$ plus half a Fourier coefficient of $\nu_\pi(0 \mid x) - \nu_\pi(1 \mid x)$, and by Parseval fewer than 2^{2t} parities have agreement above θ_t (Goldreich and Levin, 1989; Kushilevitz and Mansour, 1993). The agreement is lower semicomputable, so this list can be enumerated from π (Appendix C.1).

Lemma 5.4 (Risk floor)**.** $R_B(\pi) \ge -\log\left(u(\pi) + p + 2^{-B}\right)$*. Hence* $\mathrm{lev}(\pi) = t \ge 2$ *implies* $R_B(\pi) \ge G_{t-1}$*, and* $\mathrm{lev}(\pi) = \infty$ *implies* $R_B(\pi) \ge G_\infty$.

Lemma 5.5 (Coding the error rule with a hypothesis)**.** *For integer or infinite* B*,* $K(S \mid a, \pi) \le 2^m R_B(\pi) + 2^m \log(1 + 2^{1-B}) + 2\log m + 2\log B \cdot \mathbf{1}[B < \infty] + \kappa$.

Lemma 5.6 (Two-part lower bounds)**.** *If* $\mathrm{lev}(\pi) = t < \infty$*, then*

$$\textit{(D1)} \quad \ell(\pi) \ge d - s(t) - \Lambda_1, \qquad\qquad \textit{(D2)} \quad \ell(\pi) + 2^m R_B(\pi) \ge d + B_S - s(t) - \Lambda_1.$$

Proof ideas. Lemma 5.4 is Jensen's inequality; a hypothesis can be right on E for free only on a p-fraction of the inputs. Lemma 5.5 turns the per-z average likelihoods into a semimeasure on subsets of $\{0,1\}^m$; the product form of E makes the cost $2^m R_B$, not $2^d R_B$. (D1) is $d \le K(a) \le \ell(\pi) + K(a \mid \pi) + \kappa$; (D2) adds Lemma 5.5 and symmetry of information. See Appendix C.2, which also proves a sharper floor (Lemma C.1).

5.4 The theorem

Define

$$\psi_c(R) := c\max\{0, B_S - 2^m R\} + R, \qquad \Phi_1(c) := c(d-4-\Lambda_1) + B_S/2^m, \qquad t^* := (d-\Lambda_1)/4,$$

$$\Phi_2(c) := \begin{cases} \min\{c(d-8-\Lambda_1) + G_1,\ G_{t^*-1}\} & t^* \geq 2, \\ G_1 & t^* < 2, \end{cases}$$

$$J_g(c) := c(d+\kappa_g) + H(p) \geq J_c(\pi_g), \qquad J_b(c) := c(d + \lceil L_S \rceil + \kappa_b) \geq J_c(\pi_b),$$

$$\Delta(c) := \max\{0,\ c(4+\Lambda_1+\kappa_g) + H(p) - B_S/2^m\}.$$

J_g minus each piece of Φ_2 is affine and increasing in c, so $\{c > 0 : J_g(c) \leq \Phi_2(c)\} = (0, c_{\mathrm{hi}}]$ (empty if $c_{\mathrm{hi}} \leq 0$) with

$$c_{\mathrm{hi}} := \min\Big\{\frac{G_1 - H(p)}{8+\Lambda_1+\kappa_g},\ \frac{G_{t^*-1} - H(p)}{d+\kappa_g}\Big\} \ \ (t^* \geq 2), \qquad c_{\mathrm{hi}} := \frac{G_1 - H(p)}{d+\kappa_g} \ \ (t^* < 2).$$

$c_{\mathrm{hi}} > 0$ requires $G_1 > H(p)$; with $B = \infty$ this holds if and only if $p < 0.0560$.

Theorem 5.7 (The teacher's-mistakes theorem)**.** *In the setting of Definition 5.2, with $B \in [\log(1/p), \infty]$, the following hold for every program π of $\mathcal{H}_\times$, and of every product class covered by Proposition 9.1.*

() Level bounds. *If* $\mathrm{lev}(\pi) = 1$, *then* $J_c^B(\pi) \geq c(d-4-\Lambda_1) + \psi_c(R_B(\pi))$. *If* $\mathrm{lev}(\pi) \geq 2$, *including* ∞, *then* $J_c^B(\pi) \geq \Phi_2(c)$.

(i) Window. *If* $c\,2^m > 1$ *and* $c \leq c_{\mathrm{hi}}$, *then* $\Phi(c) \geq \Phi^B(c) \geq \min\{\Phi_1(c), \Phi_2(c)\}$ *and* $J_c(\pi_g) \leq \Phi(c) + \Delta(c)$, *where* $\Delta(c) \leq c(4+\Lambda_1+\kappa_g) + (\log(2^m+1) + \delta_S)/2^m$.

(ii) Identification. *Let* $c\,2^m > 1$, $\epsilon \geq 0$ *and* $J_g(c) + \epsilon < \Phi_2(c)$, *and let* $J_c^B(\pi) \leq J_g(c) + \epsilon$; *this includes every* π *with* $J_c(\pi) \leq \Phi(c) + \epsilon$. *Then* $\mathrm{lev}(\pi) = 1$. *Put* $\Lambda_2 := 2^m H(p) - B_S + 4 + \Lambda_1 + \kappa_g \leq \log(2^m+1) + \delta_S + 4 + \Lambda_1 + \kappa_g$, $\Lambda_3 := 4 + 2\log d + \kappa$ *and*

$$\Delta_\ell := \kappa_g + \frac{\epsilon}{c} + \frac{\Lambda_2 + \epsilon/c}{c2^m - 1}.$$

(a) It knows the simple model: $u(\pi) \geq \max\{\frac{3}{4},\ 2^{-(J_g(c)+\epsilon)} - p - 2^{-B}\}$.

(b) It fits no better than noise: $R_B(\pi) \geq H(p) - \dfrac{c\Lambda_2 + \epsilon}{c2^m - 1}$.

(c) It is short: $\ell(\pi) \leq d + \Delta_\ell$.

(d) It holds little information about the error rule: $I(\pi) \leq \Delta_\ell + \Lambda_3$, *so* $L_S - K(S \mid a, \pi) \leq \delta_S + \Delta_\ell + \Lambda_3$.

(e) It does not predict the errors: *for every integer* $g \geq 1$,

$$e(\pi) := \mathbb{E}\big[\nu_\pi(y^*(x) \mid x) \,\big|\, x \in E\big] \ \leq\ 1 - u(\pi) + \gamma_g + 2^{-g},$$

where $\gamma_g := \sqrt{j_g \ln 2/(2s_0)}$ *and* j_g *is the least integer* $j \geq 0$ *with* $j - 2\log(j+1) > \delta_S + \Delta_\ell + \Lambda_3 + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$.

(iii) Memoriser regime. *Let* $c\,2^m < 1$, *and let* $J_c^B(\pi) \leq J_b(c) + \epsilon$ *with* $J_b(c) + \epsilon < \Phi_2(c)$. *Then* $\mathrm{lev}(\pi) = 1$ *and* $(1 - c2^m)\min\{R_B(\pi), B_S/2^m\} \leq \varrho := c(\lceil L_S \rceil - L_S + \delta_S + 4 + \Lambda_1 + \kappa_b) + \epsilon$. *If* $\varrho < (1 - c2^m)B_S/2^m$, *then* $R_B(\pi) \leq \varrho/(1 - c2^m)$ *and* $e(\pi) \geq 2^{-R_B(\pi)/p} - 2^{-B}$.

κ	δ_S	$u \geq$	$\ell \leq$	$I \leq$	$L_S - K(S \mid a, \pi) \leq$	$e \leq$
0	0	0.983	$d + 20.4$	37.7	37.7	0.199
0	20	0.983	$d + 20.6$	37.9	57.9	0.218
30	0	0.981	$d + 51.0$	98.3	98.3	0.274
30	20	0.981	$d + 51.2$	98.5	118.5	0.287

Table 2: Identification at $c = 10^{-4}$ (Theorem 5.7(ii), g optimised) for the running example: $d = 100$, $m = 20$, $p = 2^{-10}$ ($L_S = 11{,}710$ bits), $B = \infty$, $\epsilon = 0$. Rows with $\kappa = 0$ set the machine constant to 0 and are illustrations. Information bounds are in bits, out of L_S. Numbers are rounded to the nearest unit in the last digit.

(iv) The lower end is sharp. *If* $c < H(p)/(\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1)$, *then* $J_c(\pi_b) < J_c(\pi)$ *for every* π *with* $\mathrm{lev}(\pi) = 1$ *and* $R(\pi) \geq H(p)$. *Since* $L_S = 2^m H(p) - O(m)$, *this threshold is* $2^{-m}\big(1 + O((m + \log d + \kappa)/L_S)\big)$.

Proof idea. For a program of level 1, (D1) and (D2) give $J_c^B \geq c(d - 4 - \Lambda_1) + \psi_c(R_B)$, and when $c2^m > 1$ the function ψ_c is minimised at $R = B_S/2^m$: each bit of rule information costs c per sample in length and saves only 2^{-m} in risk. A program of level $t \geq 2$ pays in length through (D1) or in risk through Lemma 5.4; concavity of G reduces the t-dependence to two endpoints. Part (ii) reads these inequalities backwards. For (e), if a near-optimal program gave the erroneous labels much probability, S would lie in a small set enumerable from the program (Hoeffding's inequality for sampling without replacement (Hoeffding, 1963)), contradicting (d). Parts (iii) and (iv) follow from (D2) and (D1). See Appendix C.3.

So in the window $2^{-m} < c < c_{\mathrm{hi}}$, for error rules that are incompressible relative to the model, no product hypothesis of any kind beats the good model by more than $\Delta(c)$, and every near-optimal hypothesis knows the parity, is at most Δ_ℓ bits longer than d, holds little information about the rule and gives the erroneous labels average probability at most $1 - u + \gamma_g + 2^{-g}$ (at most 0.2 to 0.29 at the example's sizes, against 1 for the memoriser and 2^{-10} for the good model), when the error rule is large compared with the information budget. Below 2^{-m}, by a relative margin, near-optimal hypotheses reproduce the errors. For such rules this settles the question left open by the three-hypothesis calculation of Remark 5.1.

Remark 5.8 (Numbers for the running example). At $d = 100$, $m = 20$, $p = 2^{-10}$, $B = \infty$ and $\epsilon = 0$, a direct computation (code in the accompanying repository) gives Table 2, rounded to the nearest unit in the last digit. Rows with the machine constant set to 0 ($\kappa = 0$) are illustrations, not results about any reference machine; rows with $\kappa = 30$ show how the bounds degrade, and every bound is explicit in κ and δ_S. The window is $(9.54 \cdot 10^{-7},\, 8.06 \cdot 10^{-3}]$ at $\kappa = 0$ (a ratio of 8453) and $(9.54 \cdot 10^{-7},\, 3.66 \cdot 10^{-3}]$ at $\kappa = 30$ (ratio 3837), against the three-hypothesis window $(1.12 \cdot 10^{-6}, 8.99 \cdot 10^{-3})$ of Remark 5.1. The memoriser has $e = 1$ and the good model $e = 2^{-10}$; for $m = 24$ ($L_S = 187{,}457$) the bound at $c = 10^{-4}$ is $e \leq 0.065$ ($\kappa = 0$, $\delta_S = 0$). Near 2^{-m} the memoriser regime is only partly covered (lower bounds on e in Appendix C.3): with $\delta_S = 0$, (iii) applies only for $c < 0.99418 \cdot 2^{-m}$ at $\kappa = 0$ ($c < 0.98914 \cdot 2^{-m}$ at $\kappa = 30$), and $e \geq \frac{1}{2}$ needs $c \leq 0.937 \cdot 2^{-m}$ ($0.888 \cdot 2^{-m}$ at $\kappa = 30$); in a band of relative width about 6% below 2^{-m} nothing useful is proved.

Remark 5.9 (What the theorem relies on). Without one of the first four of the following hypotheses a step of the proof fails as written; the fifth enters every bound. (1) *A list-decodable model class*: parities admit Lemma 5.3, while simple models with short approximations, such as majority-like

functions, do not. (2) *The product form* $E = S \times \{0,1\}^{d-m}$: Lemma 5.5 averages each $z \in \{0,1\}^m$ over 2^{d-m} inputs; for an unstructured $E \subseteq \{0,1\}^d$ it would give $2^d R_B$ bits, useless at $d = 100$. (3) *Uniform inputs*: rates, and with them the window, scale with the input law (Proposition D.3). (4) $p < 0.056$, so that $G_1 > H(p)$ (Remark D.4). (5) *A small deficiency* δ_S: every slack term contains it, and Proposition 6.3 shows that this is not an artefact of the proof.

5.5 Idiosyncratic errors and finite samples

With idiosyncratic errors and the whole truth table, the window becomes a window in λ itself. Write L_E, B_E, δ_E for L_S, B_S, δ_S when $m = d$.

Corollary 5.10 (Idiosyncratic errors, full truth table)**.** *Take $m = d$, $N = 2^d$, $B = \infty$, and let the data be the whole truth table of y^*, so $n = N$. A λ-MAP minimises $\lambda\ell + NR = N\, J_{\lambda/N}$.*

(i) For every $\lambda > 1$ with $J_g(\lambda/N) < \Phi_2(\lambda/N)$, in particular every $\lambda \in (1, Nc_{\mathrm{hi}})$, every λ-MAP π satisfies Theorem 5.7(ii) with $c = \lambda/N$, $\epsilon = 0$: $u(\pi) \geq \max\{\frac{3}{4}, 2^{-J_g(\lambda/N)} - p\}$, $NR(\pi) \geq NH(p) - \lambda\Lambda_2/(\lambda - 1)$, $\ell(\pi) \leq d + \kappa_g + \Lambda_2/(\lambda - 1)$, $I(\pi) \leq \kappa_g + \Lambda_2/(\lambda - 1) + \Lambda_3$, and (e).

(ii) If $G_1 > H(p)$ and $N(G_1 - H(p)) > d + \kappa_g + \log(N + 1)$, then at $\lambda = 1$ the optimum of $\ell + NR$, and the values $\ell(\pi_g) + NR(\pi_g)$ and $\ell(\pi_b) + NR(\pi_b)$, lie within $O(\log N) + \delta_E$ of each other, so $\lambda = 1$ does not decide whether the errors are memorised; lower-order terms and the reference machine do.

For instance, at $d = 20$ and $p = 2^{-10}$ ($L_E = 11{,}710$ bits) a direct computation (code in the accompanying repository) gives the window $\lambda \in (1, 1.19 \cdot 10^4)$ with the machine constant set to 0 (an illustration) and $\lambda \in (1, 6456)$ at $\kappa = 30$; the bounds at $\lambda = 2$ and $\lambda = 105$ are in Appendix C.4.

The theorem is a population statement. For sampled data, let $n \leq 2^d$ inputs be drawn uniformly without replacement, so that the data form a partial function, and labelled by y^*; let $B < \infty$, $\delta \in (0,1)$, $\epsilon_1 := \log(1/p)\sqrt{\ln(2/\delta)/(2n)}$, $\hat{L} := (J_c(\pi_g) + \epsilon_1)/c$ and $\epsilon_n := \epsilon_1 + B\sqrt{(\hat{L}\ln 2 + \ln(2/\delta))/(2n)}$.

Corollary 5.11 (Finite samples)**.** *With probability at least $1 - \delta$, every λ-MAP with $\lambda = cn$ satisfies $J_c^B \leq J_g(c) + \epsilon_n$, and hence the conclusions of Theorem 5.7(ii) with $\epsilon = \epsilon_n$, whenever $c2^m > 1$ and $J_g(c) + \epsilon_n < \Phi_2(c)$.*

The proof is uniform convergence over all programs with Kraft weights (Lemma C.2), which is pessimistic. At the example's sizes, with $B = 30$, $\delta = 0.01$, $\delta_S = 0$, $c = 10^{-4}$ and the machine constant set to 0 (an illustration), it gives near-population constants only at about 10^{11} samples ($\epsilon_n/c = 9.2$ bits at $n = 10^{11}$, 0.9 bits at $n = 10^{13}$); coarser guarantees hold earlier (at $n = 10^9$: $\ell \leq d + 113.9$, $e \leq 0.284$; Appendix C.5). In Appendix C, the power posterior with $\lambda = cn$, not only its maximiser, is shown to concentrate on near-optimal programs (Proposition C.4), and the total-bits version of the example is realised in population form (Example C.3).

Question 5.12 (other models, flip rates and sample sizes)**.** (a) For which classes of simple models other than parities is there a window as in Theorem 5.7? We expect that majority-like functions, which have short approximations, give none. (b) Is there a window for flip rates $0.056 \leq p < 1/4$, where the kink of Proposition 8.3 exists (for $p < 0.123$ see Question 6.8)? (c) Do the conclusions of Theorem 5.7(ii), with constants close to the population ones, hold for the λ-MAPs of sampled data at sample sizes far below 10^{11}?

6 Limits of steeper selection

Theorem 5.7 needs λ to grow linearly in n and the error rule to be incompressible relative to the model. Neither requirement can be dropped, and steeper selection by description length has further failure modes. The setting is Definition 5.2 unless stated otherwise; proofs are in Appendix D.

6.1 Why λ must be proportional to n

Proposition 6.1 (Plain Bayes, MDL and SI learn the errors)**.** *(a) Let P be a shared hypothesis with $P \geq 2^{-\ell(\pi)-\kappa}\nu_\pi$ for every $\pi \in \mathcal{H}_\times$, such as M_{prod} or M_F. For every sequence of distinct inputs and every n, $\sum_{i\leq n} -\log P(y^*(x_i) \mid x_i, D_{i-1}) \leq \ell(\pi_b)+\kappa \leq d+\lceil L_S\rceil+\kappa_b+\kappa$. So at most $d + L_S + O(\log d + \log m)$ inputs in total (the memoriser can name s_0 through r) get probability below $\frac{1}{2}$ for the teacher's label (semimeasure convention, hence also normalised), and every other revealed error input gets the erroneous label with probability at least $\frac{1}{2}$.*

(b) Let $B < \infty$, let $\lambda = \lambda_n \geq 1$ be fixed or depend on n, and draw $n \leq 2^d$ inputs uniformly without replacement. With probability at least $1-\delta$, every λ-MAP $\hat{\pi}$ has $R_B(\hat{\pi}) \leq \lambda\hat{L}/n + B\sqrt{(\hat{L}\ln 2 + \ln(2/\delta))/(2n)}$ with $\hat{L} := d + \lceil L_S\rceil + \kappa_b$, hence $e(\hat{\pi}) \geq 2^{-R_B(\hat{\pi})/p} - 2^{-B}$. In particular $e(\hat{\pi}) \geq 1 - 2^{-B} - o(1)$ whenever $(\lambda_n/n)\hat{L} \to 0$ and $\hat{L}/n \to 0$, along any sequence of instances (d_n) with fixed m, r and error rule and $n \leq 2^{d_n}$. On a fixed instance the bound is informative only once $(\lambda_n/n)(d+L_S)$ is small compared with p; for slowly decaying schedules such as $\lambda_n = n^{0.9}$ at the example's sizes this never happens for $n \leq 2^d$.

(c) On the interleaved string with i.i.d. uniform inputs, Solomonoff's M has expected total label loss at most $d + L_S + O(\log d + \log m)$, and for fixed λ the steeper SI predictor pays at most $\lambda\ell_U$ more (Theorem 4.2(d)).

For instance, at the example's sizes with all inputs revealed and the machine constant set to 0 (an illustration), a direct computation (code in the accompanying repository) shows that the exceptions in (a) are a fraction at most $9.6 \cdot 10^{-24}$ of the error inputs.

Proof idea. (a) Telescope $P(D_n) \geq 2^{-\ell(\pi_b)-\kappa}$; for M_F this is Hänni (2026a, Corollary 5.4). (b) is Lemma C.2 with the memoriser as reference. With the machine constant set to 0, $B = 30$ and $\delta = 0.01$, a direct computation gives $e(\hat{\pi}) \geq 0.649$ at $n = 10^{13}$ for $\lambda = 1$, and $e(\hat{\pi}) \geq 0.734$ at $n = 10^{15}$ for $\lambda_n = \sqrt{n}$, whatever δ_S (Appendix D.1).

Remark 6.2 (Fixed λ versus $\lambda \propto n$). So steeper selection needs $\lambda = \Theta(n)$: λ/n must stay above 2^{-m} (Theorem 5.7(iii),(iv)), and every schedule with $(\lambda_n/n)(d + L_S) \to 0$ along a sequence of instances eventually reproduces the errors (Proposition 6.1(b)). A fixed $\lambda > 1$ removes noise (Theorem 7.4), not finite-complexity errors. With $\lambda = cn$ the learner discards every component whose rate is below c, valid or not (Proposition 6.6): it is deliberately inconsistent for the teacher's label distribution. Its price is a batch excess: $-\log \xi_\lambda(D_n) \leq n\,(c\,\ell(\pi) + \hat{R}(\pi))$ for every π, with $\hat{R}$ the empirical risk, so the code length exceeds π's own loss by at most $c\,\ell(\pi)$ bits per sample (the guaranteed excess), against $\ell(\pi)/n$ at $\lambda = 1$. This is not a sequential regret, since $\lambda = cn$ is not one Bayesian predictor (Remark 3.7). We do not claim as new the qualitative finding that a bounded λ fits the noise and that λ must grow with n; λ-weighted two-part MDL and PAC-Bayes prediction in noisy binary classification are studied by Zhu and Srebro (2025); Zhu et al. (2026) (Section 2). Their target is the data distribution, which here contains the teacher's errors, and by

Proposition 6.1(b) every schedule with $(\lambda_n/n)(d + L_S) \to 0$ along a sequence of instances learns them.

6.2 Which error rules defeat steeper selection

The example of Section 5 fixes only a density and a description length, and realistic systematic errors are often low-complexity patterns. Split the rule as $S = S_1 \sqcup S_2$, $|S| = s_0 = p2^m$, with $s_i := |S_i|$, $p_1 := s_1/2^m$ and $p_2 := s_2/(2^m - s_1)$, where S_1 is *cheap*, $K(S_1 \mid m) \le k_1$; every input of $E_1 := S_1 \times \{0,1\}^{d-m}$ is an error input. The *envelope* π_e predicts $f_a \oplus 1$ with probability 1 on E_1 and f_a with flip probability p_2 elsewhere; it has risk $R_e = (1 - p_1)H(p_2) < H(p)$ and length at most $d + k_1 + \kappa_e$, $\kappa_e := 2\log d + 2\log m + 2\log(s_2 + 1) + \kappa$. Put $J_e(c) := c(d + k_1 + \kappa_e) + R_e$, $\zeta := c(k_1 + \kappa_e + 4 + \Lambda_1) + \epsilon$ and $c_{\mathrm{env}} := (H(p) - R_e)/(k_1 + \kappa_e + 4 + \Lambda_1)$.

Proposition 6.3 (Error envelopes)**.** *Let $B = \infty$, and let $J_c(\pi) \le J_e(c) + \epsilon$ with $J_e(c) + \epsilon < \Phi_2(c)$.*

(i) $\mathrm{lev}(\pi) = 1$ *and $R(\pi) \le R_e + \zeta$. So for $c < c_{\mathrm{env}}$ and ϵ small enough that $\zeta < H(p) - R_e$, no program with $R(\pi) \ge H(p)$, the good model included, is ϵ-near-optimal.*

(ii) With $B_2 := K(S_2 \mid a, K(a))$, the average probability of reproducing the cheap errors is

$$e_1(\pi) := \mathbb{E}\big[\nu_\pi(y^*(x) \mid x) \,\big|\, x \in E_1\big] \;\ge\; 2^{-\bar{R}_1},$$
$$\bar{R}_1 := \tfrac{1}{p_1}\big(\zeta + (1 - p_1)H(p_2) - \min(c, 2^{-m})\,(B_2 - K(S_1))^+\big).$$

Proof idea. (i) is Theorem 5.7(o) with (D1). For (ii), code S_2 given S_1 with the hypothesis, as in Lemma 5.5; this bounds its risk off E_1 from below, so its risk on E_1 is small (Appendix D.4).

At the example's sizes take $S_1 = \{z : z_1 = \cdots = z_j = 0\}$ and S_2 uniformly random. For $j = 13$ (128 cheap and 896 random points; $\log \binom{2^m - s_1}{s_2} = 10{,}418$ bits), a direct computation (code in the accompanying repository) gives $c_{\mathrm{env}} = 1.64 \cdot 10^{-5}$, which is 31% of the logarithmic width of the window $(9.54 \cdot 10^{-7}, 8.06 \cdot 10^{-3}]$ of Theorem 5.7 with the machine constant set to 0 (25% at $\kappa = 30$), and $e_1 \ge 0.545$ at $c = 1.01 \cdot 2^{-m}$ ($\kappa = 0$, B_2 typical). For $j = 11$ ($512 + 512$ points, 6,364 bits): 47% of the log-width and $e_1 \ge 0.862$ (table in Appendix D.4).

So density and description length alone do not suffice: the rule with $j = 13$ has density 2^{-10} and about 10,400 bits, close to the 10,000 bits of the example, yet on almost a third of the window's log-width no hypothesis that fits no better than the good model is near-optimal, and near the lower end near-optimal hypotheses reproduce the cheap errors. (Its deficiency is about 1,300 bits; Theorem 5.7 applies to it with this value of δ_S in its bounds, which are correspondingly weaker, and Proposition 6.3 says nothing about it above c_{env}.) The example is favourable to the proposal; its conclusion holds for error rules incompressible relative to the model (Theorem 5.7). For structured rules the dangerous hypotheses are envelopes, not universal ones: a cheap, dense error region has a high rate and is learned like any valid component. The extreme case is a cheap, frequent rule.

Proposition 6.4 (Rate blindness)**.** *Let $B = \infty$ and $\ell(\pi_{\mathrm{coin}}) \le \kappa$. (i) If $H(p)/(\ell(\pi_b) - \ell(\pi_g)) \ge (1 - H(p))/(\ell(\pi_g) - \ell(\pi_{\mathrm{coin}}))$, then at every c the good model is beaten by the memoriser or beaten or tied by the coin. (ii) If*

$$\frac{H(p)}{\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1} \;\ge\; \frac{1 - H(p)}{d - 4 - \Lambda_1 - \kappa} \quad \textit{and} \quad d - 4 - \Lambda_1 - \kappa > 0, \tag{$\star$}$$

then at every c every program with $u > \frac{3}{4}$ and $R \ge H(p)$ is beaten by the memoriser or beaten or tied by the coin. (iii) For $c < 2^{-m}$, near-optimal programs reproduce the errors, under the condition of Theorem 5.7(iii).

Lemma 6.5 (The criterion is robust in the machine constant)**.** *Fix m and r, let κ bound the machine constants in Λ_1, κ_b and $\ell(\pi_{\mathrm{coin}})$, and put $H := H(2^{-r})$, $q := (1-H)/H$, $\alpha_0 := 2+4q$, $\beta_0 := 4+2\log m+2\kappa+q(\lceil L_S\rceil+4\log m+2\log s_0+2\kappa+4)$ and $F(d) := d-\alpha_0\log d-\beta_0$. (i) For $d\geq 1$, $(\star)$ holds if and only if $F(d)\geq 0$. (ii) If $\beta_0\geq\alpha_0/\ln 2$, then $\{d\geq 1 : F(d)\geq 0\} = [d_0,\infty)$ for a unique real $d_0 = d_0(\kappa)$. (iii) $d_0(\kappa)$ grows linearly in κ, with slope $(2+2q)(1+o(1))$.*

For a single 3-bit prefix as error rule ($m = 3$, $p = 1/8$), a direct computation gives $d_0 = 49$ with the machine constant set to 0 and 169 at $\kappa = 30$; for a single 6-bit prefix, 450 and 1005 (Appendix D.2). So for each fixed cheap rule and each κ, once the simple model is long enough, no program that knows the parity and fits no better than noise is ever the unique optimum: at every c it is beaten by the memoriser or beaten or tied by the coin. The incompressible 11,710-bit rule of the example, by contrast, leaves a window of a factor of about 8500 ($\kappa = 0$) or 3800 ($\kappa = 30$).

6.3 Other pathologies

Proposition 6.6 (Separable components)**.** *Let the inputs split into regions $X_1,\dots,X_k$ and the hypotheses be products of per-region choices with additive code lengths. Each region selects a vertex of its own hull for slope c (Lemma A.1). If region j offers a default or a component j of extra length k_j and population risk reduction r_j, component j is selected if and only if $r_j/k_j > c$ (ties arbitrary). So a valid component whose rate is at most an error component's rate cannot be kept without the error, except at a common tie value. Two worlds with the same teacher channel and input law, one where a component is an error and one where it is a rare valid rule, get the same selection at every c.*

The steeper penalty sorts components by *rate*, risk reduction per bit, not by validity. Three consequences follow (proofs in Appendix D.3). A guard that restricts a rule to its proper contexts is dropped when its rate is below the rule's: on an interval of c the unguarded rule is the unique selection and applies the rule where the teacher never did (Proposition D.1; in a numerical instance the interval spans a factor of 67 in c). The selection depends on the input law (Proposition D.3); Theorem 5.7's window depends on it by the same mechanism, which is proved here only in the separable case. And λ cannot be chosen from imitation data: every strictly proper scoring rule prefers the teacher's channel, errors included (Gneiting and Raftery, 2007); log-loss validation along the selection path picks the smallest, error-memorising, value of c; and in the twin worlds of Proposition 6.6 every rule that chooses c from imitation data alone is wrong in one of them (Proposition D.2).

6.4 Unseen properties

For M_F, a simple property absent from the training data gets a constant probability of bad behaviour, $2^{-K(Q,b)-O(1)}$, where $K(Q,b)$ is the length of a shortest self-delimiting code for a decider of a set Q and a program for a behaviour b (Hänni, 2026a, Proposition 8.1). The steeper counterpart is as follows (proof in Appendix D.5).

Proposition 6.7 (Steeper simplicity and unseen properties)**.** *In $\mathcal{H}_\times$, let $Q\subseteq X$ be decidable, b partial computable, D a data set with $\operatorname{dom} D\cap Q = \emptyset$, and $Z\subseteq Q\cap\operatorname{dom} b$ finite. For every $\lambda\geq 1$: (i) a family of programs that answer $b(x)$ on Q and run an arbitrary program of the class elsewhere (one for each such program) carries posterior mass exactly $2^{-\lambda(K(Q,b)+\kappa)}$; hence all programs with this behaviour together carry at least this mass, and $\xi_\lambda(D\cup b|_Z)\geq 2^{-\lambda(K(Q,b)+\kappa)}\xi_\lambda(D)$; (ii) if $\nu_{\pi_1}(D) = \nu_{\pi_2}(D) > 0$, then $\Pi_\lambda(\pi_1\mid D)/\Pi_\lambda(\pi_2\mid D) = 2^{-\lambda(\ell(\pi_1)-\ell(\pi_2))}$.*

At $\lambda = 1$, (i) is the statement of Hänni (2026a, Proposition 8.1(a)) with M_{prod} in place of W and M_F. With $\lambda = cn$ the lower bound of (i) degrades to $2^{-cn(K(Q,b)+\kappa)}$, so a constant lower bound no longer follows; whether the probability of b on Q vanishes depends, by (ii), on what the shortest hypotheses that fit the training data do on Q. If the shortest one extrapolates well, steeper SFI is more reliable out of distribution than SFI; if a shorter one fits the training data and is wrong on Q, it is more confidently wrong. No guarantee about unseen properties follows.

Question 6.8 (structured errors and the choice of c)**.** (a) For envelopes whose cheap region has error density $\theta < 1$, only part (i) of Proposition 6.3 carries over (Appendix D.4). Do near-optimal hypotheses then memorise the random part S_2, and what is selected above c_{env}? (b) Is there a window as in Theorem 5.7 for $p < 0.123$ and c well above 2^{-m} (Remark D.4)? (c) Is there a rule for choosing c, using information other than the imitation data, that keeps rare valid rules and drops errors? By Propositions 6.6 and D.2 imitation data alone cannot tell them apart.

7 The model–randomness split

Labels drawn from a computable sampler are explained both by the sampler, with private coins, and by programs that fill in the sampled bits. At $\lambda = 1$, Solomonoff induction has no preference between the two, although only the sampler separates model from noise; a steeper prior might prefer it. This section makes both halves precise. At $\lambda = 1$ the sampler and the programs that fill in its sampled bits get similar posterior, up to explicit factors: $2^{k+\delta^*}$ in one direction and $2^{2\log(t_n+1)+\kappa}$ in the other, where $2\log(t_n+1)$ grows at most triply logarithmically in n and in the sampled randomness (not at all if the sampler's label probabilities are bounded below) (Theorem 7.3 and Proposition E.3). At fixed $\lambda > 1$ with private randomness samplers beat fill-ins (not necessarily ν itself; Remark 7.5), in every countable product class (Theorem 7.4). With shared randomness, or in SI, the universal mixture wins instead and no split is made (Theorem 7.6). Proofs are in Appendix E.

7.1 Setting

Definition 7.1 (Samplers and fill-ins)**.** A *sampler* is a computable conditional measure ν ($\nu(y \mid x)$ computable uniformly in (x, y), $\sum_y \nu(y \mid x) = 1$) with a program π_ν of length k in the class considered. Distinct inputs $x_1, x_2, \ldots$ are chosen by any rule, possibly adaptive, randomised or non-computable, and then $y_i \sim \nu(\cdot \mid x_i)$; $\mathbb{P}$ is the joint law and D_n the first n pairs. The *sampled randomness* is $-\log \nu(D_n)$. *Fill-ins* are deterministic programs consistent with D_n; in a product class their total weight is $W_{\mathrm{det}}(D) := \sum\{2^{-\ell(\pi)} : \pi \text{ deterministic},\ \nu_\pi \supseteq D\}$.

The first lemma bounds how much any program, or the whole mixture, can gain over the sampler on its own data.

Lemma 7.2 (Ville deficiency)**.** *Let ξ be a shared hypothesis with $\xi(\emptyset) \le 1$, for example the mixture $\xi_{\mathcal{C}}$ of a countable product class $\mathcal{C}$ with $\sum_{\pi\in\mathcal{C}} 2^{-\ell(\pi)} \le 1$ (for $\mathcal{H}_\times$ this is M_{prod}). Let $\delta^* := \sup_n \log\left(\xi(D_n)/\nu(D_n)\right)$. Then $\mathbb{P}[\delta^* \ge j] \le 2^{-j}$ for all $j \ge 0$. Consequently, for $\xi = \xi_{\mathcal{C}}$, every $\pi \in \mathcal{C}$ and every n,*

$$\nu_\pi(D_n) \le 2^{\ell(\pi)+\delta^*}\nu(D_n).$$

Adaptive inputs are allowed.

Proof idea. $\xi(D_n)/\nu(D_n)$ is a nonnegative supermartingale with initial value at most 1, by the one-point inequality; apply Ville's maximal inequality (Ville, 1939) (Appendix E.1).

Fill-ins, on the other hand, are cheap: a deterministic program can store a finite piece of each input's private random stream. Adapting the tape-length bound of Hänni (2026a, Proposition 4.6(a)) to product classes and to every computable sampler gives, for every data set D with n inputs of length at most d and $\beta_D := \min_{(x,y)\in D} \nu(y \mid x) > 0$, the bound $W_{\det}(D) \geq 2^{-k-2\log(t_D+1)-\kappa}\nu(D)$, where $2\log(t_D+1)$ is doubly logarithmic in d and triply logarithmic in n and in the sampled randomness $-\log \nu(D)$ (Lemma E.1 in Appendix E.2). This needs a computable *measure*; for dyadic samplers a hashing argument gives a weaker bound (Lemma E.2).

7.2 $\lambda = 1$: the sampler and its fill-ins get similar posterior

For a run of the data process put $d_n := \max_{i\leq n} |x_i|$, $\beta_n := \min_{i\leq n} \nu(y_i \mid x_i)$, $i_n := \lceil \log(6n/\beta_n) \rceil$ and $t_n := \lceil \log(2d_n + 2 + 2\log(i_n+1)) \rceil$. Let $\Pi_1(\det \mid D_n)$ be the posterior mass of all deterministic programs.

Theorem 7.3 (SFI at $\lambda = 1$)**.** *In the setting of Definition 7.1 with the class $\mathcal{H}_\times$, for every sampler ν with a program of length k, simultaneously for all n:*

(i) $\Pi_1(\pi_\nu \mid D_n) \geq 2^{-k-\delta^*}$;

(ii) $\Pi_1(\det \mid D_n) = W_{\det}(D_n)/\xi_\times(D_n) \geq 2^{-k-2\log(t_n+1)-\kappa-\delta^*}$;

(iii) the posterior odds of all fill-ins against the sampler program lie in $\left[2^{-2\log(t_n+1)-\kappa},\, 2^{k+\delta^*}\right]$, *with a lower end that depends on the sampled randomness only through t_n;*

(iv) for a sampler whose values are multiples of 2^{-r}, *Lemma E.2 gives the alternative lower ends* $2^{-k-2\log(n+1)-2\log(d_n+1)-2\log(r+1)-\kappa-\delta^*}$ *in (ii) and* $2^{-2\log(n+1)-2\log(d_n+1)-2\log(r+1)-\kappa}$ *in (iii).*

Proof. $\Pi_1(\pi_\nu \mid D_n) = 2^{-k}\nu(D_n)/\xi_\times(D_n)$ and $\Pi_1(\det \mid D_n) = W_{\det}(D_n)/\xi_\times(D_n)$. Bound $\xi_\times(D_n) \leq 2^{\delta^*}\nu(D_n)$ by Lemma 7.2, $W_{\det}(D_n)$ from below by Lemma E.1 (or Lemma E.2 for (iv)) and from above by $\xi_\times(D_n)$. □

The lower ends of (ii) and (iii) hold for every data set, not only for samples from ν, and cover non-dyadic samplers: one that gives every label probability at least 1/3, such as the Bern(1/3) threshold hypotheses in Sam Eisenstat's coin example (personal communication; see also Hänni (2026a, Section 7.3)), has $i_n \leq \lceil \log(18n) \rceil$. The hashing bound of (iv) is weaker at realistic sizes (32 to 143 bits; Appendix E.3).

In SI, the total weight of the deterministic hypotheses consistent with the data $z_{1:n}$ is at least $2^{-k-K(n)-\kappa}\nu(z_{1:n})$, and in SI's representation by deterministic programs every hypothesis is a fill-in (Proposition E.3 in Appendix E.4). So at $\lambda = 1$ the sampler and its fill-ins get similar posterior in SFI and in SI, up to factors 2^k, $2^{2\log(t_n+1)+\kappa}$ (SFI) or $2^{O(k+K(n))}$ (SI), and the Ville deficiency, with the dependence on the sampled randomness only through t_n.

7.3 $\lambda > 1$ with private randomness: samplers beat fill-ins

Call a program π *ν-equivalent on the run* if $\sum_n d_{\mathrm{H}}^2\big(\nu(\cdot \mid x_n), \nu_\pi(\cdot \mid x_n)\big) < \infty$, where $d_{\mathrm{H}}^2(P,Q) := 1 - \sum_y \sqrt{P(y)Q(y)} \in [0,1]$, also for a sub-probability Q.

Theorem 7.4 (Private randomness, $\lambda > 1$)**.** *Let $\mathcal{C}$ be a countable class of product programs with $\sum_{\pi\in\mathcal{C}} 2^{-\ell(\pi)} \leq 1$ that contains π_ν (for example $\mathcal{H}_\times$, or a time-bounded or circuit subclass). Fix $\lambda > 1$ and define δ^* from $\xi_\mathcal{C}$. Simultaneously for all n:*

(i) MAP. *Every λ-MAP $\hat{\pi}_n$ has $\ell(\hat{\pi}_n) \le (\lambda k+\delta^*)/(\lambda-1) = k+(k+\delta^*)/(\lambda-1)$ and $-\log \nu(D_n) - \ell(\hat{\pi}_n) - \delta^* \le -\log \nu_{\hat{\pi}_n}(D_n) \le -\log \nu(D_n) + \lambda k$.*

(ii) Length tail. $\Pi_\lambda(\{\pi : \ell(\pi) > L\} \mid D_n) \le 2^{\lambda k+\delta^*-(\lambda-1)L}$ *for every L.*

(iii) Fill-ins. *Every deterministic program consistent with D_n has $\ell \ge -\log \nu(D_n) - \delta^*$; $\Pi_\lambda(\det \mid D_n) \le 2^{\lambda k+\lambda\delta^*-(\lambda-1)(-\log \nu(D_n))}$; and for the tempered posterior with $\eta = 1/\lambda$, $\Pi^\eta(\det \mid D_n) \le 2^{k+\delta^*-(1-\eta)(-\log \nu(D_n))}$.*

(iv) Identification. *Suppose the input sequence is infinite. Then, almost surely, from some n on every λ-MAP is ν-equivalent on the run, and $\Pi_\lambda(\nu$-equivalent on the run $\mid D_n) \to 1$.*

Proof idea. (i)–(iii) compare with the sampler program, $J_\lambda(\pi_\nu) = \lambda k - \log \nu(D_n)$, and bound every program's likelihood by Lemma 7.2. For (iv), the likelihood ratio $\Lambda_n := \nu_\pi(D_n)/\nu(D_n)$ of a program that is not ν-equivalent on the run tends to 0 almost surely, by the martingale argument behind Kakutani's dichotomy (Kakutani, 1948), which holds for adaptive inputs (Appendix E.5).

Remark 7.5 (What the theorem says, and what it does not). This gives the split for every fixed $\lambda > 1$, in a precise and limited sense. The selected model is a sampler, not a fill-in, but not necessarily ν: it may carry up to $(k+\delta^*)/(\lambda-1)$ bits beyond the sampler, for example memorised noise. In particular no λ-MAP is a fill-in once $-\log \nu(D_n) > \lambda(k+\delta^*)/(\lambda-1)$. The fill-ins' posterior is exponentially small in the sampled randomness, but positive. "ν-equivalent on the run" is not equality of predictive laws and says nothing about unseen inputs. On a finite domain such as $\{0,1\}^d$, (iv) is vacuous.

7.4 $\lambda > 1$ with shared randomness: no split

Theorem 7.6 (SI, or SFI with shared randomness, $\lambda > 1$)**.** *Assume (Dom) and $\xi_1 \le 2^{\kappa_1} U$, let $\lambda > 1$, and let π_ν be a program of length k for a sampler in the class. For every observation o with $U(o) > 0$:*

(i) if $(\lambda-1)k > \lambda\ell_U + \kappa_0$, then $J_\lambda(\pi_U; o) < J_\lambda(\pi_\nu; o)$: the sampler program is never a λ-MAP, and every λ-MAP has length at most $\ell^*_\lambda < k$;

(ii) for every set A of programs, $\Pi_\lambda(A \mid o) \le 2^{-(\lambda-1)\min_{\pi\in A}\ell(\pi)+\lambda\ell_U+\kappa_1}$. In particular the power posterior puts at most $2^{-(\lambda-1)L+\lambda\ell_U+\kappa_1}$ on all programs of length at least L.

For SI with adaptive or non-computable inputs the sampler is not an SI hypothesis, since it does not model the inputs; use $\mathcal{H}_{\mathrm{chr}}$, where it is a chronological semimeasure and (Dom) holds with $U = M_H$. With i.i.d. computable inputs, $\mathcal{H}_{\mathrm{SI}}$ with $U = M$ works.

Proof idea. (i) compares π_U with π_ν using (Dom); (ii) bounds the numerator by $2^{-(\lambda-1)\min_A \ell}\xi_1(o)$ and the normaliser by $2^{-\lambda\ell_U}U(o)$ (Appendix E.6).

Compare Theorem 7.4. In a shared class the steeper posterior lives, up to mass 2^{-s}, on programs of length at most $(\lambda\ell_U + \kappa_1 + s)/(\lambda-1)$, *whatever the data*; in the product class it concentrates on samplers ν-equivalent on the run, whose length is unbounded over ν. For fixed λ, both steeper predictors dominate ν ($\xi^\times_\lambda \ge 2^{-\lambda k}\nu$; $\xi^F_\lambda \ge 2^{-\lambda\ell_U-\kappa_0-k}\nu$), so by the argument of Hänni (2026a, Theorem 5.3) their summed expected KL divergences to ν's predictions are at most λk and $\lambda\ell_U + \kappa_0 + k$. The difference is *what the selected hypothesis is*: a model plus private noise, or a short universal learner that contains no model of ν.

Appendix E.6 illustrates the contrast on a small noisy parity (Example E.4) and shows that in SI's representation by deterministic programs no form of steepness separates a sampler from its fill-ins (Remark E.5).

8 Relation to algorithmic statistics

Besides a steeper penalty, one can look for the split between model and noise by plotting the entropy left given the hypothesis against the hypothesis length and looking for a change of slope. With shared randomness, as in the sequence case, one expects none where a random parameter of the model ends, because each of its bits pays for about one bit of data (Remark 8.5).[2] For set models this graph is Kolmogorov's structure function. We define a product version and show, for a noisy random parity, that the two place the split about d bits apart. Proofs are in Appendix F.

8.1 Set models and steeper two-part codes

Definition 8.1 (Set models)**.** For a string z, a *set model* is a finite set $A \ni z$, with two-part length $K(A) + \log|A|$. The *structure function* is $h_z(\alpha) := \min\{\log|A| : A \ni z,\ K(A) \le \alpha\}$. The *sufficiency line* is the bound $K(A) + \log|A| \ge K(z) - \kappa$, valid for every $A \ni z$ since $K(z \mid A) \le \log|A| + \kappa$. A is *b-sufficient* if $K(A) + \log|A| \le K(z) + b$; a *minimal sufficient statistic* minimises $K(A)$ among b-sufficient models; and the *set sophistication* is $\mathrm{soph}_b^{\mathrm{set}}(z) := \min\{K(A) : A \ni z,\ K(A) + \log|A| \le K(z) + b\}$. Conditional versions given a string w (for example a domain) replace K by $K(\cdot \mid w)$.

These notions are standard (Gács et al., 2001; Vereshchagin and Vitányi, 2004; Vereshchagin and Shen, 2017); we use them as stated. Sophistication goes back to Koppel (1987); Koppel and Atlan (1991), with total programs as models; see also Vitányi (2006) and Antunes and Fortnow (2009). We do not use the equivalences between the versions.

Steeper two-part codes over sets select an approximately minimal sufficient statistic only in a limited sense: a minimiser A_λ of $\lambda K(A) + \log|A|$ has $K(A_\lambda) \le \mathrm{soph}_b^{\mathrm{set}}(z) + (b+\kappa)/(\lambda-1)$ and is $(b+(\lambda-1)\,\mathrm{soph}_b^{\mathrm{set}}(z))$-sufficient, and as $\lambda \to 1$ the length bound becomes vacuous (Proposition F.1 and Remark F.2 in Appendix F.1).

8.2 Product models

Definition 8.2 (Product structure function and sophistication)**.** For a data set D and the class $\mathcal{H}_\times$, the *product structure function* is $\rho_D^\times(\alpha) := \min\{-\log \nu_\pi(D) : \ell(\pi) \le \alpha\}$; π is *product-b-sufficient* if $\ell(\pi) - \log \nu_\pi(D) \le K(y_D \mid \mathrm{dom}\, D) + b$; and the *product sophistication* is $\mathrm{soph}_b^\times(D) := \min\{\ell(\pi) : \pi \text{ product-}b\text{-sufficient}\}$. Lemma 3.8 is the product sufficiency line. Product notions are conditional on the domain by construction, since product hypotheses do not model the inputs.

The argument of Proposition F.1, with Lemma 3.8 as the sufficiency line, bounds the length of every λ-MAP over $\mathcal{H}_\times$ by the product sophistication plus $(b+\kappa)/(\lambda-1)$ (Proposition F.3). Thresholding a computable product hypothesis with rational values gives a set model that is at most logarithmically longer, so set models are at least as coarse as product models, at least for computable hypotheses (Proposition F.4; whether this holds for lower-semicomputable hypotheses is Question F.5). Both are in Appendix F.1.

8.3 A noisy random parity

We use the instance of Definition 5.2 with $m = d$ (idiosyncratic errors) and $B = \infty$, so $E = S$, $N := 2^d$ and $|E| = pN$, and write L_E, δ_E for L_S, δ_S. Let $z = y^* \in \{0,1\}^N$ be the teacher's table

[2] The suggestion to look for this change of slope in function induction, and the reason why none is expected with shared randomness, first appeared in the author's note (Hänni, 2026b).

and D the full data set, so $\operatorname{dom} D = \{0,1\}^d$ and $K(y_D \mid \operatorname{dom} D) = K(z \mid d) \pm \kappa$; for a product program $R(\pi) = -\frac{1}{N}\log \nu_\pi(D)$ is the risk under uniform inputs. Here $G_1 = 1 - \log(\frac{3}{2} + 2p)$, which exceeds $H(p)$ exactly for $p < 0.0560$ (Section 5.4), $\Lambda_1 = 4\log d + \kappa$ and $\kappa_b = 4\log d + 2\log(pN) + \kappa$. We use list decoding (Lemma 5.3), the risk floor (Lemma 5.4) and the length bound (D1) of Lemma 5.6; part (iv) also uses the sharper floor Lemma C.1.

Proposition 8.3 (The two splits differ by about d bits)**.** *Let $p < 1/4$.*

(i) Sets: the parity is noise. *$A_0 := \{f_{a'} \oplus \mathbf{1}_{E'} : a' \in \{0,1\}^d,\ |E'| = pN\}$ has $|A_0| = 2^{d+L_E}$ and $K(A_0) \le 2\log d + 2\log r + \kappa$, and $K(z) \ge d + L_E - \delta_E - \kappa$. So A_0 is $(\delta_E + 2\log d + 2\log r + \kappa)$-sufficient, and $\operatorname{soph}_b^{\mathrm{set}}(z) \le 2\log d + 2\log r + \kappa$ for every such b. The structure function lies on the sufficiency line, within $\delta_E + O(\log d)$, for α between $O(\log d)$ and $d + O(\log d)$. Every λ-optimal set model A_λ with $\lambda > 1$ holds at most $O(\log d) + (\delta_E + O(\log d))/(\lambda - 1)$ bits of information about a.*

(ii) Products: the parity is model. *Assume also $G_1 > H(p)$. For $\delta_E + 2\log d + \kappa_g + \log(N+1) + \kappa \le b < N(G_1 - H(p)) - d - 2\log N - \kappa$,*

$$d - 4 - \Lambda_1 \ \le\ \operatorname{soph}_b^{\times}(D) \ \le\ d + \kappa_g,$$

and $\rho_D^{\times}(\alpha) \ge NG_1$ for every $\alpha < d - 4 - \Lambda_1$, while $\rho_D^{\times}(\alpha) \le NH(p)$ for $\alpha \ge d + \kappa_g$ (the good model): the product structure function has a kink at $\alpha \approx d$.

(iii) The gap $\operatorname{soph}^{\times} - \operatorname{soph}^{\mathrm{set}} \ge d - O(\log d)$ is unbounded in d.

(iv) Every flip rate below $1/4$. *Drop the condition $G_1 > H(p)$; let $p = |E|/N < 1/4$ have small complexity $K(p)$, with $\kappa_g := 2\log d + K(p) + \kappa$. For integers $t, g \ge 1$ let $\gamma_d := \sqrt{j_d \ln 2/(2pN)}$, where j_d is the least integer with $j_d - 2\log(j_d + 1) > \delta_E + d + 1 + 2\log d + 2\log(pN) + g + 2\log g + 2\kappa$, and*

$$F_t := 1 - \log\Big(1 + (1-2p)2^{-t} + 2p\big(2\gamma_d + 2^{1-g}\big)\Big).$$

Every product program with $\ell(\pi) < d - s(t) - \Lambda_1$ has $R(\pi) \ge F_t$, so $\rho_D^{\times}(\alpha) \ge NF_t$ for $\alpha < d - s(t) - \Lambda_1$, while $\rho_D^{\times}(\alpha) \le NH(p)$ for $\alpha \ge d + \kappa_g$. Moreover $d - s(t) - \Lambda_1 \le \operatorname{soph}_b^{\times}(D) \le d + \kappa_g$ for every b with

$$\delta_E + 2\log d + \kappa_g + \log(N+1) + \kappa \ \le\ b \ <\ NF_t - d - \lceil L_E \rceil - \kappa_b - 2\kappa.$$

For fixed $p < 1/4$ and $\delta_E = O(d)$, $\gamma_d \to 0$ as $d \to \infty$, and with g growing slowly $F_t \to 1 - \log(1 + (1-2p)2^{-t})$, which exceeds $H(p)$ for some fixed $t = t_p$. So the admissible range of b has length of order $N(F_{t_p} - H(p))$, product sophistication is $d \pm O(\log d)$, and the gap of (iii) holds for every $p < 1/4$.

Proof idea. (i) Minimum-distance decoding recovers (a, E) from z, since two parities differ on $N/2 > 2pN$ inputs, so $K(z) \ge d + L_E - \delta_E - \kappa$, while A_0 is described by d and r. (ii) The good model is product-sufficient; a program with $\operatorname{lev}(\pi) \ge 2$ pays NG_1, more than the memoriser's two-part length; so a product-sufficient program has $\operatorname{lev}(\pi) = 1$, and (D1) bounds its length. (iv) uses the sharper floor (Appendix F.2; numerical values in Table 3).

Remark 8.4 (Restricted model families)*.* Restricting the model family changes the structure function; Hamming-ball models (Vereshchagin and Vitányi, 2010) are the closest case. For the full-table noisy parity the Hamming ball of radius pN around f_a is a set model of complexity about d and log-size about $NH(p)$, that is, the parity plus independent flips as a set. See also Milovanov (2016).

Remark 8.5 (Reading: one bit for one bit). With shared randomness (a set model, or any shared model) a uniformly random d-bit parameter costs the same in the model as in the data-to-model code, one bit for one bit. So the minimal sufficient statistic treats the parity's parameter as noise. With private randomness, not knowing the parameter costs about one bit *per input*, so the product split keeps it in the model, and the slope of $\rho_D^{\times}$ changes at $\alpha \approx d$. For supervised learning the product split is the relevant one, because the model is the part of the description reused across inputs. This is also why steeper simplicity is useless over set models in the parity example (Proposition 8.3(i)) and decisive over product models (Theorem 5.7 and Corollary 5.10).

9 Computable hypothesis classes

The lower bounds of Section 5 use only coding properties of the class, so they survive in resource-bounded classes (proofs in Appendix G).

Proposition 9.1 (Computable product classes)**.** *Let $\mathcal{H}'$ be a hypothesis class of product hypotheses (Definition 3.4). (a) Lemmas 5.3 to 5.6 and the lower-bound parts of Theorem 5.7 ((o), the bound on Φ in (i), (ii), and the level and R_B statements of (iii)) hold for every $\pi \in \mathcal{H}'$, with κ replaced by a constant that depends on $\mathcal{H}'$. The parts that use reference programs ($\Delta(c)$, J_g, J_b, (iv)) hold if $\mathcal{H}'$ contains a good model π'_g and a memoriser π'_b, with $\kappa_g := \ell(\pi'_g) - d$ and $\kappa_b := \ell(\pi'_b) - d - \lceil L_S \rceil$. (b) Theorem 7.4 holds in every countable product class with Kraft weights that contains a program for the sampler. (c) Proposition 4.7(ii) holds in every product class with a prefix-free code.*

Remark 9.2 (Examples). Examples are probabilistic programs clocked at $t(|x|)$ steps, as in time-bounded SFI (Hänni, 2026a, Section 9.1), where generous time bounds change κ_g, κ_b by logarithmic terms, and circuits with private output noise, where a parity costs about $d \log d$ bits, so κ_g is of order $d \log d$ and c_{hi} shrinks by a factor of order $\log d$. Theorem 4.2 does not transfer, since a mixture of all time-t programs is not a time-t program; so the contrast between product and shared classes is established only for unbounded classes.

Proposition 9.3 (Computability of the steeper predictor)**.** *Let $\lambda \geq 1$ be computable, the class $\mathcal{H}_{\times}$ or $\mathcal{H}_F$, X infinite and $\{0,1\} \subseteq Y$. (a) $\xi_\lambda(D)$ is lower semicomputable uniformly in D. (b) Neither the raw conditional $\xi_\lambda(D + (x,y))/\xi_\lambda(D)$ nor its normalised version is computable. (c) Both are limit-computable.*

Part (b) is the diagonal argument for M_F (Hänni, 2026a, Proposition A.10(c)): a computable conditional yields a computable labelling on which the predictor loses a constant per input, against the weight $2^{-\lambda(K(f)+\kappa)}$ of a program for it.

Remark 9.4 (Neural-network Bayes). With labels independent given the parameters θ, a network's Bayes posterior is a mixture of product laws, an object of the type of M_{prod} (Hänni, 2026a, Proposition 10.1). Theorem 7.4 applies to countable discretisations with weights $p(\theta)$: under the power prior $p(\theta)^\lambda$, $\lambda > 1$, deterministic parameter settings that reproduce the noisy labels get posterior mass exponentially small in the sampled randomness, if the noise model is in the family. Whether a parameter prior behaves like a description-length prior, as Theorem 5.7 needs, is the bridge question of the companion paper (Hänni, 2026a, Question 10.9 and Remark H.2), which the slides of Vaintrob and Hänni (2026) motivate; nothing here settles it.

Question 9.5 (resource bounds and computability)**.** (a) Does some time-bounded shared class contain a short member that absorbs everything at $\lambda > 1$, as in Theorem 4.2? If not, steeper simplicity could also work in time-bounded Solomonoff induction. (b) For computable $\lambda \geq 1$ and the class $\mathcal{H}_{\times}$, is there a computable map from data sets to λ-MAPs?

10 Discussion

What counts as one hypothesis. Whether a steeper simplicity prior can separate a model from its noise depends on what counts as one hypothesis. When a universal mixture is a member of the class, as in SI and in SFI with shared randomness, it cannot for any model longer than ℓ^*_λ, a bound that does not depend on the data, and for fixed λ a computable teacher's labels, errors included, are eventually predicted (Theorem 4.2). With private randomness at each input no member is universal, and steeper selection can work; as in algorithmic statistics (Proposition 8.3), the product view counts as model what is reused across inputs and as noise what is private to each input. This is a modelling choice with a price on shared latent causes (Remark 4.6 and Proposition 4.7), and the protection ends when inputs carry labelled history (Proposition B.2).

The window and its price. For error rules incompressible relative to the model, near-optimal product hypotheses are, up to explicit slack, the simple model with noise for $2^{-m} < c < c_{\mathrm{hi}}$ (Theorem 5.7). The lower end is sharp because a bit of information about the error rule costs c per sample in length and saves about 2^{-m} per sample in risk. So λ must grow linearly with n, and the learner is then deliberately inconsistent for the teacher's labels: it discards every component whose rate is below c, valid or not, and its guaranteed code length exceeds a program's own loss by $c\,\ell(\pi)$ bits per sample (Remark 6.2). Imitation data cannot justify this price, because strictly proper scores prefer the teacher's channel, errors included (Proposition D.2).

The gap to realistic errors. Human labelling errors are often cheap, frequent patterns rather than incompressible sets. In the parity instance steeper selection provably fails for such errors in two ways: a fixed cheap rule defeats every c once the simple model is long enough (Proposition 6.4 and Lemma 6.5), and a rule made of a cheap region plus random points, with the original density and description length, makes the good model non-optimal on about a third of the window (Proposition 6.3). Whether any window survives for structured errors in general is open. Steeper simplicity sorts components by rate, not by validity (Proposition 6.6), so by itself it does not tell errors from rare valid rules. We do not expect steeper penalisation alone to solve the general problem of learning a norm from imperfect imitation; it works in one toy case and fails in others.

The model–randomness split. At $\lambda = 1$ a Bayes mixture is indifferent, up to factors that grow at most triply logarithmically in the amount of data, between a sampler and the programs that fill in its sampled bits (Theorem 7.3). For fixed $\lambda > 1$ a steeper prior over product hypotheses breaks the tie in favour of samplers, in the limited sense of Remark 7.5; in SI's representation by deterministic programs no form of steepness does (Remark E.5).

Relation to other work. The λ-MAP is a two-part code with complexity weight λ, as in λ-weighted MDL in noisy classification, and maximises the tempered posterior, a generalised posterior with learning rate $1/\lambda$; the power-prior posterior has the same maximiser but is Bayes' rule with the prior $2^{-\lambda\ell}$. Those works aim at the data distribution; our target differs from it by design (Section 2). A neural network whose labels are independent given its parameters has a Bayes posterior that mixes product laws, so Theorem 7.4 applies to countable discretisations; whether a parameter prior behaves like a description-length prior, as Theorem 5.7 would need, is the bridge question of the companion paper (Hänni, 2026a, Question 10.9), which the slides of Vaintrob and Hänni (2026) motivate (Remark 9.4).

10.1 Open problems

(1) *Structured error rules.* For envelopes whose cheap region has error density below one, only part (i) of Proposition 6.3 carries over; whether near-optimal hypotheses memorise the random part, and what is selected above c_{env}, are open (Question 6.8).

(2) *Other model classes and flip rates.* Which simple-model classes give a window? Parities do; we expect that majority-like functions do not give a window. Is there a window for flip rates up to 1/4, where the kink exists (Questions 5.12 and 6.8)?

(3) *Non-uniform inputs and finite samples.* A version of Theorem 5.7 for non-uniform input laws (only the separable case is proved, Proposition D.3); near-population constants at realistic sample sizes: they need about 10^{11} samples, and coarser guarantees hold from about 10^9 (Corollary 5.11 and Question 5.12).

(4) *Choosing c.* A principled choice of c that does not use the imitation data, and the extra information that would tell errors from rare valid rules (Question 6.8).

(5) *Shared hypotheses.* Is every lower-semicomputable law of a random partial function dominated by M_F up to a constant (Question 4.9)?

(6) *Resource bounds and computability.* Does a time-bounded shared class contain a short member that absorbs everything at $\lambda > 1$? Is there a computable map from data sets to λ-MAPs (Question 9.5)?

(7) *Lower-semicomputable product models.* Does Proposition F.4 hold for them (Question F.5)?

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Code and data

Every numerical statement comes from a deterministic or seeded Python script (Python 3.11 with NumPy and SciPy) whose output is saved next to it. The scripts, their outputs and independent re-implementations of the main computations are in the folder `solomonoff-function-induction/` of the repository `https://github.com/kaarelh/AI-works`; its README lists the script behind each number.

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A Proofs for Section 3 (setting)

A.1 Standard facts

The following facts are standard (Li and Vitányi, 2008); κ is as in Section 3.2.

(F1) $K(\pi) \le \ell(\pi) + \kappa$ for every program π of a class $\mathcal{H}$. The identity map on the c.e. prefix-free set $\mathcal{H}$ is a prefix machine, which the universal machine simulates with a fixed prefix; for canonical classes, $\mathcal{H} = \operatorname{dom} V$ and the fact is immediate.

(F2) *Conditional coding theorem.* If a fixed program lower-semicomputes from z a semimeasure Q_z on a countable set, then $K(w \mid z) \le -\log Q_z(w) + \kappa$.

(F3) $K(x) \le K(y) + K(x \mid y) + \kappa$, and $K(x, y \mid z) \le K(x \mid z) + K(y \mid x, z) + \kappa$.

(F4) *Symmetry of information.* $K(x, y) = K(x) + K(y \mid x, K(x)) + O(1)$.

(F5) $K(j) \le 2\log(j+1) + \kappa$ for every integer $j \ge 0$, by a doubling code.

A.2 The hull lemma

Lemma A.1 (Hull lemma)**.** *Let $\mathcal{A} := \{(\ell(\pi), R(\pi))\}$ with $R \in [0, \infty]$ and finitely many points below each ℓ-level, let $C := \operatorname{conv}(\mathcal{A} \cap \{R < \infty\}) + \mathbb{R}^2_{\ge 0}$, and call the extreme points of C its* vertices*.*

(i) For $c > 0$, every minimiser of $c\,\ell + R$ lies on the boundary of C, on a supporting line with normal $(c, 1)$.

(ii) If $c < c'$ and h, h' minimise $J_c, J_{c'}$, then $\ell(h) \ge \ell(h')$ and $R(h) \le R(h')$.

(iii) A point of $\mathcal{A}$ is the unique minimising point of $c\,\ell + R$ for every c in some nonempty open interval if and only if it is a vertex of C; the least-ℓ vertex and the least-R vertex, if it exists, count. A hypothesis is then the unique minimiser if and only if no other hypothesis has the same point.

(iv) (Absolute length bound.) Every minimiser h_c of $c\,\ell + R$ ($c > 0$) solves $\min\{R(h) : \ell(h) \le \ell(h_c)\}$. *Conversely, a solution of* $\min\{R(h) : \ell(h) \le L\}$ *minimises $c\,\ell + R$ for some $c > 0$ if and only if its point lies on the boundary of C and not on the open horizontal ray to the right of the least-R vertex.*

Remark A.2 (The converse in (iv)). The converse in (iv) cannot be weakened to "on the boundary of C": in $\mathcal{A} = \{(0,1),(10,0),(20,0)\}$ the point $(20,0)$ lies on the boundary and solves the constrained problem for $L \ge 20$, but $(10,0)$ is strictly better for every $c > 0$.

Examples. In $\{(0,1),(10,0.5),(20,0)\}$ the middle point lies on the segment between the other two, so it is never the unique minimiser; the end points are the unique minimisers on $(0.05,\infty)$ and on $(0,0.05)$. In $\{(0,1),(5,0.9),(10,0)\}$ the point $(5,0.9)$ lies above the boundary of C; it solves the constrained problem for $L \in [5,10)$ and minimises no penalised objective.

We prove Lemma A.1. Write $\mathcal{A}' := \mathcal{A} \cap \{R < \infty\}$ and $\langle (c,1),(\ell,R)\rangle = c\,\ell + R$.

(i). Since $(c,1) \ge 0$, the infimum of $c\,\ell + R$ over $C = \operatorname{conv}(\mathcal{A}') + \mathbb{R}^2_{\ge 0}$ equals its infimum over $\mathcal{A}'$. A minimiser a over $\mathcal{A}$ therefore minimises a nonzero linear functional over C. So the line $\{c\,\ell + R = c\,\ell(a) + R(a)\}$ supports C at a, and a lies on the boundary of C.

Structure of the boundary. Fix $a \in \mathcal{A}'$ and $s > 0$. A point $b \in \mathcal{A}'$ with $\ell(b) > \ell(a)$ and $R(b) \le R(a) - s(\ell(b) - \ell(a))$ has $\ell(b) \le \ell(a) + R(a)/s$, because $R(b) \ge 0$; only finitely many points satisfy this. So from each point the steepest descending direction to the right is attained, and the lower-left boundary of C is a polygonal line. It consists of a vertical ray above the least-ℓ vertex, segments between consecutive vertices with slopes $-s$, $s \in (0,\infty)$, strictly increasing from left to right, and, if a point of least R exists, a horizontal ray to the right of the least-R vertex. Its corners are extreme points of C, and every extreme point of C lies in $\mathcal{A}'$, because the extreme points of a convex hull belong to the generating set and adding the orthant creates none.

(iii). Let a be a vertex. Its two incident boundary pieces have slopes $-s_- < -s_+ \le 0$, with $s_- = \infty$ at the least-ℓ vertex and $s_+ = 0$ at the least-R vertex. For $c \in (s_+, s_-)$ the line with normal $(c,1)$ through a meets C only at a, so a is the unique minimising point. Conversely, if a is the unique minimising point for every c in an open interval, then a is an exposed point of C, hence an extreme point, hence a vertex. A boundary point that is not a vertex lies in the relative interior of a segment between two vertices, or on one of the two rays. In the first case, whenever it minimises $c\,\ell + R$, so do both endpoints of its segment. A point on the vertical ray is beaten by the vertex below it (smaller R), and a point on the horizontal ray by the least-R vertex (smaller ℓ, same R). So no such point is ever the unique minimising point. The statement about hypotheses is immediate: a hypothesis is the unique minimiser exactly when its point is the unique minimising point and no other hypothesis has that point.

(ii). Add $c\,\ell(h) + R(h) \le c\,\ell(h') + R(h')$ and $c'\ell(h') + R(h') \le c'\ell(h) + R(h)$ to get $(c'-c)(\ell(h') - \ell(h)) \le 0$, so $\ell(h') \le \ell(h)$. Then the first inequality gives $R(h) \le R(h') + c(\ell(h') - \ell(h)) \le R(h')$.

(iv). If $\ell(h) \le \ell(h_c)$, then $R(h) \ge R(h_c) + c(\ell(h_c) - \ell(h)) \ge R(h_c)$, so h_c solves the constrained problem with $L = \ell(h_c)$. For the converse, by the description of the boundary: a vertex minimises $c\,\ell + R$ for every c in its interval (s_+, s_-), which contains positive values; a point on a segment between two vertices, of slope $-s$ with $s > 0$, lies on the supporting line with normal $(s,1)$ and so minimises $s\,\ell + R$. A point on the open horizontal ray is beaten, for every $c > 0$, by the least-R vertex. A point on the vertical ray above the least-ℓ vertex is never a constrained solution, since that vertex is feasible whenever the point is and has smaller R. A point not on the boundary minimises no linear functional with normal $(c,1)$ over $\mathcal{A}$, by (i).

Remark A.2. In $\mathcal{A} = \{(0,1),(10,0),(20,0)\}$ the point $(20,0)$ lies on the open horizontal ray to the right of the least-R vertex $(10,0)$. It solves $\min\{R : \ell \le L\}$ for $L \ge 20$, since it ties with $(10,0)$ there, but $20c > 10c$ for every $c > 0$.

A direct computation (code in the accompanying repository) confirms the equivalence in (iii) on 401 random point sets with duplicate and collinear points, and the minimisers on 300 further point sets.

A.3 Proof of the coding bound

We prove Lemma 3.8. Given π and the finite set $\operatorname{dom} D$, the map $y \mapsto P_\pi(\operatorname{dom} D \mapsto y)$ on $Y^{\operatorname{dom} D}$ is lower semicomputable, uniformly in $(\pi, \operatorname{dom} D)$. It is a semimeasure. For a product hypothesis its total mass is $\prod_{x \in \operatorname{dom} D} \sum_y \nu_\pi(y \mid x) \le 1$. For a shared hypothesis, repeated use of the one-point inequality gives $\sum_y P_\pi(\operatorname{dom} D \mapsto y) \le P_\pi(\emptyset) \le 1$. By (F2), $K(y_D \mid \pi, \operatorname{dom} D) \le -\log P_\pi(D) + \kappa$. By (F3) and (F1),

$$K(y_D \mid \operatorname{dom} D) \le K(\pi) + K(y_D \mid \pi, \operatorname{dom} D) + \kappa \le \ell(\pi) - \log P_\pi(D) + \kappa.$$

□

B Proofs for Section 4 (absorption)

B.1 Classes with an absorbing member

We prove Remark 4.1. Suppose first that the mixture $\xi_1 = \sum_\pi 2^{-\ell(\pi)} P_\pi$ is itself a member, $\xi_1 = P_{\pi_U}$. Then $P_\pi \le 2^{\ell(\pi)} \xi_1 = 2^{\ell(\pi)} U$ for every π, which is (Dom) with $\kappa_0 = 0$, and $\xi_1 = U$. This covers $\mathcal{H}_w$ ($M^w = \sum_i \mathbf{m}(i)\nu_i$ is an lsc weak function semimeasure (Hänni, 2026a, Theorem 3.11), hence occurs in the trimmed enumeration), $\mathcal{H}_{\mathrm{SI}}$ (M is an lsc semimeasure) and $\mathcal{H}_{\mathrm{chr}}$ (M_H is an lsc chronological semimeasure (Hänni, 2026a, Corollary 6.17) and, by Hänni (2026a, Theorem 6.16), the law of a history-oracle function machine at every nonempty history; at the empty history both sides of (Dom) are at most 1 and $P_{\pi_U} = 1$). For $\mathcal{H}_F$: M_F is a machine law (Hänni, 2026a, Theorem 3.8), so it has an index; let π_U be a shortest program for it. For $D \neq \emptyset$, $P_\pi(D) = \nu_{T_{V(\pi)}}(D) \le M_F(D)/\mathbf{m}(V(\pi)) \le 2^{\ell(\pi)} M_F(D)$, and at $\emptyset$ both sides equal 1. Here $\xi_1 = M_F$ on nonempty data sets and $\xi_1(\emptyset) \le 1 = M_F(\emptyset)$, so $\xi_1 \le U$. □

B.2 Proof of the absorption theorem

We prove Theorem 4.2. Throughout, (Dom) holds and $\lambda > 1$.

(a). Let $\hat\pi$ be a λ-MAP for o. Then $J_\lambda(\hat\pi; o) \le J_\lambda(\pi_U; o) = \lambda \ell_U - \log U(o)$. By (Dom), $-\log U(o) \le \ell(\hat\pi) + \kappa_0 - \log P_{\hat\pi}(o)$. Substituting,

$$\lambda \ell(\hat\pi) - \log P_{\hat\pi}(o) \le \lambda \ell_U + \ell(\hat\pi) + \kappa_0 - \log P_{\hat\pi}(o),$$

so $(\lambda - 1)\ell(\hat\pi) \le \lambda \ell_U + \kappa_0$, which is the length bound. (Since $U(o) > 0$, $J_\lambda(\pi_U; o) < \infty$, so $P_{\hat\pi}(o) > 0$ and the subtraction is legitimate.) The loss bound is $J_\lambda(\hat\pi; o) \le J_\lambda(\pi_U; o)$ rearranged. Finally, $\ell^*_\lambda < L$ if and only if $\lambda(L - \ell_U) > L + \kappa_0$.

(b). The first inequality keeps the single term of π_U in $\xi_\lambda(o) = \sum_\pi 2^{-\lambda \ell(\pi)} P_\pi(o)$. The second holds because $2^{-\lambda \ell} \le 2^{-\ell}$. If $\xi_1 \le 2^{\kappa_1} U$, then $-\log \xi_\lambda(o) \ge -\log \xi_1(o) \ge -\log U(o) - \kappa_1$, and the first inequality gives $-\log \xi_\lambda(o) \le \lambda \ell_U - \log U(o)$. For a data-set class, ξ_λ is a weak function semimeasure with $\xi_\lambda(\emptyset) \le \sum_\pi 2^{-\lambda \ell(\pi)} \le 1$, because each P_π is one and the weak inequalities are linear. Its sequential log loss on D_n is $-\log \xi_\lambda(D_n) + \log \xi_\lambda(\emptyset)$ in every order (Definition 3.3), and the same holds for U.

(c1). Since π_t is deterministic and consistent with D_n, $P_{\pi_t}(D_n) = 1$, so $\xi_\lambda(D_n) \geq 2^{-\lambda \ell_t}$. Also, by (Dom), $U(D_n) \geq 2^{-\ell_t - \kappa_0} P_{\pi_t}(D_n) = 2^{-\ell_t - \kappa_0}$, so $\xi_\lambda(D_n) \geq 2^{-\lambda \ell_U} U(D_n) \geq 2^{-\lambda \ell_U - \kappa_0 - \ell_t}$. By telescoping, $\sum_{i \leq n} -\log \xi_\lambda(t(x_i) \mid x_i, D_{i-1}) = -\log \xi_\lambda(D_n) + \log \xi_\lambda(\emptyset) \leq -\log \xi_\lambda(D_n)$. Each conditional is at most 1 by the one-point inequality, so each term is nonnegative, and a label with probability below $1/2$ adds more than one bit.

(c2). Combine (a) with $-\log U(D_n) \leq \ell_t + \kappa_0$.

(c3). Let $\mathcal{H}_{\leq \ell^*_\lambda} := \{\pi : \ell(\pi) \leq \ell^*_\lambda\}$. It is finite by Kraft, and by (a) it contains every λ-MAP for every D_n. For $\pi \in \mathcal{H}_{\leq \ell^*_\lambda}$ let $L_\pi(n) := -\log P_\pi(D_n)$. It is nondecreasing in n, because weak function semimeasures are monotone, so it has a limit $L_\pi(\infty) \in [0, \infty]$. Let $\mathcal{H}_{\text{fin}} := \{\pi \in \mathcal{H}_{\leq \ell^*_\lambda} : L_\pi(\infty) < \infty\}$. It contains π_U, because $\ell_U \leq \ell^*_\lambda$ and $L_{\pi_U}(\infty) \leq \ell_t + \kappa_0$. For $\pi \in \mathcal{H}_{\leq \ell^*_\lambda} \setminus \mathcal{H}_{\text{fin}}$, $J_\lambda(\pi; D_n) \to \infty$, while $J_\lambda(\pi_U; D_n) \leq \lambda \ell_U + \ell_t + \kappa_0$ for every n. Since there are finitely many such π, there is n_0 beyond which none of them is a λ-MAP. For $\pi \in \mathcal{H}_{\text{fin}}$, $\sum_n [L_\pi(n+1) - L_\pi(n)] \leq L_\pi(\infty)$, and each increment equals $-\log\left(P_\pi(D_{n+1})/P_\pi(D_n)\right)$, the loss of π's prediction of the next label. So π gives the next label probability below $1/2$ fewer than $L_\pi(\infty) + 1$ times. Taking the union over the finite set $\mathcal{H}_{\text{fin}}$ bounds the number of $n \geq n_0$ at which some λ-MAP does so.

(d). By (Dom), $\xi_\lambda \geq 2^{-\lambda \ell_U} U \geq 2^{-\lambda \ell_U - \kappa_0 - \ell(\pi_\mathbb{P})} \mathbb{P}$ on every string. Hence, for every length L and $\omega_{1:L}$ drawn from $\mathbb{P}$,

$$\mathbb{E}_\mathbb{P} \log \frac{\mathbb{P}(\omega_{1:L})}{\xi_\lambda(\omega_{1:L})} \leq \lambda \ell_U + \kappa_0 + \ell(\pi_\mathbb{P}).$$

Write $\xi_\lambda(b \mid h) := \xi_\lambda(hb)/\xi_\lambda(h)$ and expand by the chain rule over bit positions: $\log(\mathbb{P}/\xi_\lambda)(\omega_{1:L}) = \sum_{j \leq L} \log\left(\mathbb{P}(\omega_j \mid \omega_{<j})/\xi_\lambda(\omega_j \mid \omega_{<j})\right) - \log \xi_\lambda(\varepsilon)$, and $-\log \xi_\lambda(\varepsilon) \geq 0$. At each position the conditional expectation of the summand is $\mathrm{KL}\left(\mathbb{P}(\cdot \mid h) \,\|\, \xi_\lambda(\cdot \mid h)\right)$, which is at least $-\log \sum_b \xi_\lambda(b \mid h) \geq 0$ because $\xi_\lambda(\cdot \mid h)$ is a sub-probability. With fixed-length blocks the label positions are fixed. At a label position $\mathbb{P}(\cdot \mid h)$ is a point mass, because $y_i = t(x_i)$, so the summand equals $-\log \xi_\lambda(\omega_j \mid \omega_{<j})$; the sum of these over the bits of one label block is $-\log \xi_\lambda(y_i \mid x_1 y_1 \cdots x_i)$. Dropping the nonnegative terms of the input positions gives the claim. This is Solomonoff's bound (Solomonoff, 1978) restricted to the label positions. □

B.3 Proof of per-input pricing

We prove Proposition 4.4.

(i), the bound. $\max_f \geq \mathbb{E}_{f \sim Q}$. For each i,

$$\mathbb{E}_{f \sim Q}\big[-\log \nu(f(x_i) \mid x_i)\big] = H(Q_{x_i}) + \sum_y Q_{x_i}(y) \log \frac{Q_{x_i}(y)}{\nu(y \mid x_i)},$$

and the last sum is at least $-\log \sum_y \nu(y \mid x_i) \geq 0$, by Gibbs' inequality with a sub-probability.

(i), the value. For $\varepsilon \in (0, 1/|Y|)$ let Δ_ε be the set of conditional sub-probabilities on $\{x_1, \ldots, x_n\}$ with $\nu(y \mid x_i) \geq \varepsilon$ for all y, i. It is compact and convex, and $L(\nu, Q) := \mathbb{E}_{f \sim Q} \sum_i -\log \nu(f(x_i) \mid x_i)$ is continuous on $\Delta_\varepsilon \times \Delta(F)$, convex in ν and linear in Q. By Sion's minimax theorem (Sion, 1958), $\min_{\Delta_\varepsilon} \max_Q L = \max_Q \min_{\Delta_\varepsilon} L$. Since L is linear in Q, $\max_Q L(\nu, Q) = \max_{f \in F} \sum_i -\log \nu(f(x_i) \mid x_i)$. For each Q, $\min_{\Delta_\varepsilon} L(\cdot, Q)$ lies between $\sum_i H(Q_{x_i})$, by the bound, and $\sum_i H(Q_{x_i}) - n \log(1 - |Y|\varepsilon)$, attained up to that error by $\nu = (1 - |Y|\varepsilon) Q_x + \varepsilon$, which lies in Δ_ε and sums to 1. The error term is uniform in Q. Rational conditional probabilities, which are computable, are dense in Δ_ε and $L(\cdot, Q)$ is continuous there, so the infimum over product hypotheses equals the infimum over Δ_ε. Hence

$$\max_Q \sum_i H(Q_{x_i}) \leq \inf_\nu \max_f \sum_i -\log \nu(f(x_i) \mid x_i) \leq \max_Q \sum_i H(Q_{x_i}) - n \log(1 - |Y|\varepsilon),$$

where the first inequality is the bound, applied to every ν. Let $\varepsilon \to 0$.

The example. For $F = \{(a,a),(a,b),(b,a),(c,a)\}$ a direct computation (code in the accompanying repository) gives $\max_Q \sum_i H(Q_{x_i}) = 2.5431$, and the same value for $\min_\nu \max_f$ by constrained optimisation over ν. The naive bound is $\log 3 + \log 2 = 2.5850$.

(ii). For $x \neq 0$ exactly half of the parities give each label at x, so the uniform Q has $H(Q_x) = 1$, and (i) gives a parity with loss at least n. The Bayes mixture with weight 2^{-d} on each parity gives every parity's data probability at least 2^{-d}.

(iii). Each term of the per-input mixture assigns $1/2$ to the observed label 1, at every input; the shared mixture gives the constant-1 function weight $1/2$. □

Remark B.1 (The universal conditional semimeasure never learns). The universal lsc conditional semimeasure satisfies $\mathbf{m}(y \mid x) \geq 2^{-K(y|x)-\kappa}$. For $Y = \{0,1\}$ this gives a constant $\kappa_{\mathbf{m}}$ with $\mathbf{m}(y \mid x) \geq 2^{-\kappa_{\mathbf{m}}}$, hence $\mathbf{m}(y \mid x) \leq 1 - 2^{-\kappa_{\mathbf{m}}}$, for both labels, so it pays at least $\log \frac{1}{1-2^{-\kappa_{\mathbf{m}}}} > 0$ bits for every label on every data set. It dominates each product hypothesis input by input, $\mathbf{m} \geq 2^{-K(\nu)-\kappa}\nu$, but pays the factor $2^{K(\nu)}$ again at every input; its prediction does not depend on the data.

B.4 Proof of the price of private randomness

We prove Proposition 4.7.

Write $a_i := 0^i$ and $b_i := 1^i$, so $D_\beta = \{(a_i, \beta_i), (b_i, \beta_i) : i \leq k\}$.

(i). Fix $\pi \in \mathcal{H}_\times$ with conditional semimeasure $\nu = \nu_\pi$. Then $\nu(D_\beta) = \prod_{i \leq k} s_i(\beta_i)$ with $s_i(y) := \nu(y \mid a_i)\,\nu(y \mid b_i)$. By the inequality of arithmetic and geometric means, $s_i(y) \leq q_i(y)^2$ with $q_i(y) := \frac{1}{2}(\nu(y \mid a_i) + \nu(y \mid b_i))$, and each q_i is a sub-probability on $\{0,1\}$. So $\nu(D_\beta) \leq Q_\pi(\beta)^2$ with $Q_\pi(\beta) := \prod_i q_i(\beta_i) \leq 1$. Since $\sum_\beta Q_\pi(\beta) = \prod_i \sum_y q_i(y) \leq 1$, Q_π is a semimeasure on $\{0,1\}^k$, lower semicomputable uniformly in (π, k). Let $\tilde{R}(\beta) := \sum_{\pi \in \mathcal{H}_\times} 2^{-\ell(\pi)} Q_\pi(\beta)$. It is an lsc semimeasure on $\{0,1\}^k$ given k, so $\tilde{R}(\beta) \leq 2^{-K(\beta|k)+\kappa}$ by (F2). Now let $\gamma := \min(\lambda, 2)$. If $1 \leq \lambda \leq 2$, then $2^{-\lambda\ell(\pi)} Q_\pi^2 = (2^{-\ell(\pi)} Q_\pi)^\lambda Q_\pi^{2-\lambda} \leq (2^{-\ell(\pi)} Q_\pi)^\lambda$. If $\lambda \geq 2$, then $2^{-\lambda\ell(\pi)} Q_\pi^2 \leq (2^{-\ell(\pi)} Q_\pi)^2$. For nonnegative t_π and $\gamma \geq 1$, $\sum_\pi t_\pi^\gamma \leq (\sum_\pi t_\pi)^\gamma$. Therefore

$$\xi_\lambda^\times(D_\beta) = \sum_\pi 2^{-\lambda\ell(\pi)} \nu_\pi(D_\beta) \leq \sum_\pi 2^{-\lambda\ell(\pi)} Q_\pi(\beta)^2 \leq \tilde{R}(\beta)^\gamma \leq 2^{-\gamma(K(\beta|k)-\kappa)}.$$

(ii). $2^{-J_\lambda(\pi; D_\beta)}$ is one term of $\xi_\lambda^\times(D_\beta)$. For a product class with a prefix-free code, define $\tilde{R}$ over that class; (F2) then holds with a constant for the class.

(iii). ν_{len} is the law of the function machine $T = \{(x, w, b) : |w| = |x| + 1,\ w_{|x|+1} = b\}$, which outputs the tape bit $\omega_{|x|+1}$. Inputs a_i, b_i have the same length i, so they receive the same bit ω_{i+1}, and distinct i receive independent bits; so $\nu_{\text{len}}(D_\beta) = 2^{-k}$. Universality gives $M_F \geq 2^{-K(\nu_{\text{len}})}\nu_{\text{len}}$ (Hänni, 2026a, Theorem 3.8), so $-\log M_F(D_\beta) \leq k + \kappa$. For ξ_λ^F keep the single term of a shortest program for ν_{len}.

(iv). Fewer than 2^{k-j} strings β have $K(\beta \mid k) < k - j$, so $\Pr[K(\beta \mid k) < k - j] < 2^{-j}$. On the complement, (i) gives $-\log \xi_\lambda^\times(D_\beta) \geq \gamma(k - j - \kappa)$, while (iii) gives $-\log M_F(D_\beta) \leq k + \kappa$ and $-\log \xi_\lambda^F(D_\beta) \leq k + \lambda\kappa$. Subtracting, and using $\gamma \leq 2$ and $\gamma \leq \lambda$, the excess is at least $(\gamma - 1)k - 2j - (\lambda + 2)\kappa$ in both cases. □

A direct computation (code in the accompanying repository) confirms the key inequality $\xi_\lambda^\times(D_\beta) \leq \tilde{R}(\beta)^{\min(\lambda,2)}$ on random finite product classes with Kraft weights, for λ from 1 to 4; equality occurs.

Reading. At $\lambda = 1$ the first option of the product class, writing the coins into the program, costs the same as the shared class up to lower-order terms, which is why M_{prod} and M_F are close; at $\lambda > 1$ the product class pays $\min(\lambda, 2)$ bits per coin, against 1 for the shared class.

B.5 Hypotheses that see the history

Proposition B.2 (Labelled history restores universality)**.** *Suppose a hypothesis's input at position i of a block contains the block's earlier labelled pairs. Let $(\nu_\theta)_{\theta\in\Theta}$, Θ countable, be uniformly computable history-dependent conditional measures with positive values, and w computable positive weights with a computable sum $\sum_\theta w_\theta \le 1$. With $Z_i := \sum_\theta w_\theta \prod_{j<i} \nu_\theta(y_j \mid \cdot)$, the in-context Bayes mixture $h_w(y_i \mid \cdot) := Z_{i+1}/Z_i$ is a single product hypothesis on the extended inputs, with a program of length at most $K(w, (\nu_\theta)) + \kappa$, and on every block*

$$\textstyle\sum_i -\log h_w(y_i \mid \cdot) \;\le\; \min_\theta \big[\log(1/w_\theta) + \sum_i -\log \nu_\theta(y_i \mid \cdot)\big].$$

So it pays $\log(1/w_\theta)$ once per block, not once per input.

Proof. Each Z_i is a positive real, computable uniformly in the block's history: its terms $w_\theta \prod_{j<i} \nu_\theta(y_j \mid \cdot)$ are computable and positive, and the tail $\sum_{\theta>T} w_\theta \prod_{j<i} \nu_\theta(\cdot)$ is at most $\sum_{\theta>T} w_\theta$, which is computable from T because w and $\sum_\theta w_\theta$ are computable. So $h_w = Z_{i+1}/Z_i$ is computable. It sums to 1 over y_i, because the ν_θ are measures. So h_w is a computable conditional measure on extended inputs; a program for it is obtained from a description of w and (ν_θ), which gives the length bound in the canonical class. The product of the predictive probabilities over a block of length n' telescopes to $Z_{n'+1}/Z_1$, and $Z_1 = \sum_\theta w_\theta \le 1$, so

$$-\log \frac{Z_{n'+1}}{Z_1} \le -\log \frac{w_\theta \prod_i \nu_\theta(y_i \mid \cdot)}{\sum_{\theta'} w_{\theta'}} \le \log \frac{1}{w_\theta} + \sum_i -\log \nu_\theta(y_i \mid \cdot)$$

for every θ. □

Without the computability assumptions, Z_{i+1}/Z_i is a ratio of lsc sums, which need not be lsc; this is why they are made. A finite family with rational values and rational weights also satisfies them. A direct computation (code in the accompanying repository) confirms the block identity on 2000 random instances with history-dependent inputs.

Proposition B.3 (A history of inputs only)**.** *A hypothesis that sees the earlier inputs but not the earlier labels, and draws fresh randomness for each extended input $(x_{<i}, x_i)$, is a product hypothesis on the extended inputs, which are pairwise distinct; so Proposition 4.4 and Corollary 4.5 apply and per-input pricing persists. With shared randomness instead, as in* $\mathrm{SFI}^{\mathrm{hist}}$*, $\mathcal{H}_{\mathrm{chr}}$ has the universal member M_H and Theorem 4.2 applies.*

Proof. A hypothesis of the first kind is an lsc conditional semimeasure on the set of extended inputs, with private randomness per extended input, so it is a product hypothesis there. Extended inputs $(x_{<i}, x_i)$ at different positions are distinct. In Proposition 4.4(ii) take F to be the functions $(x_{<i}, x_i) \mapsto \langle a, x_i\rangle \bmod 2$ with nonzero x_i; under uniform a each label is a fair coin, and the argument is unchanged, as is that of Corollary 4.5. The second claim is Remark 4.1 for $\mathcal{H}_{\mathrm{chr}}$. □

C Proofs for Section 5 (the teacher's mistakes)

Throughout, the setting is Definition 5.2, and (F1)–(F5) are the standard facts of Appendix A.1. For $z \in \{0,1\}^m$ and $w \in \{0,1\}^{d-m}$ we write (z,w) for the input with prefix z; w is uniform on $\{0,1\}^{d-m}$ in every expectation $\mathbb{E}_w$. Since $a \in \{0,1\}^d$, the length d can be computed from a. In the proofs we write $\tau_B := 2^{1-B}$ and $\beta_B := 2^m \log(1+\tau_B)$, both 0 if $B = \infty$, so that $G_j = 1 - \log(1 + 2^{-j} + 2p + \tau_B)$ and $\Lambda_1 = 2\log d + 2\log m + 2\log B \cdot \mathbf{1}[B < \infty] + \beta_B + \kappa$; and $\Gamma_g := \delta_S + \Delta_\ell + \Lambda_3 + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$, so that j_g in Theorem 5.7(e) is the least integer $j \geq 0$ with $j - 2\log(j+1) > \Gamma_g$.

C.1 List decoding

Proof of Lemma 5.3. Write $h_y(x) := \nu_\pi(y \mid x)$, $s_h := h_0 - h_1 \in [-1,1]$, $\chi_{a'}(x) := (-1)^{\langle a', x\rangle}$ and $\hat{s}(a') := \mathbb{E}_x[s_h \chi_{a'}]$. Since $\nu_\pi(f_{a'}(x) \mid x) = \frac{1}{2}(h_0 + h_1) + \frac{1}{2}\chi_{a'}(x) s_h(x)$, the agreement with $f_{a'}$ is

$$u(\pi; a') := \mathbb{E}_x \nu_\pi(f_{a'}(x) \mid x) = \tfrac{1}{2}\big(\mathbb{E}_x[h_0 + h_1] + \hat{s}(a')\big) \leq \tfrac{1}{2}\big(1 + \hat{s}(a')\big).$$

The characters are orthonormal under the uniform law on $\{0,1\}^d$, so Parseval's identity gives $\sum_{a'} \hat{s}(a')^2 = \mathbb{E}_x s_h^2 \leq 1$. Hence $A_t := \{a' : u(\pi; a') > \theta_t\} \subseteq \{a' : \hat{s}(a') > 2^{-t}\}$ has fewer than 2^{2t} elements. This is the list-size bound for the Hadamard code (Goldreich and Levin, 1989; Kushilevitz and Mansour, 1993). The value $u(\pi; a')$ is a finite average of lower semicomputable values, so it is lower semicomputable uniformly in (π, a', d), and A_t is computably enumerable given (π, d, t). If $\mathrm{lev}(\pi) = t$, then $a \in A_t$. So a is described, given π, by d, by t, and by its index in the enumeration of A_t, which takes $2t$ bits given t. By (F5), $K(a \mid \pi) \leq 2t + 2\log(t+1) + 2\log d + \kappa = s(t) + 2\log d + \kappa$. □

The argument uses the lower-semicomputable agreement u rather than the correlation $\hat{s}$, which is not computable for semimeasures; this is what extends the computable-hypothesis version of the lemma in the earlier report (Claude (Anthropic), 2026) to all product hypotheses. A direct computation (code in the accompanying repository) on 400 random semimeasure hypotheses at $d = 7, 8$ confirms the identity for u, the list bound and the risk floor.

C.2 The risk floors, the coding lemma and the two-part bounds

Proof of Lemma 5.4. Since $\min\{B, -\log v\} = -\log \max\{v, 2^{-B}\}$ and $-\log$ is convex, Jensen's inequality gives

$$R_B(\pi) \geq -\log \mathbb{E}_x \max\{\nu_\pi(y^*(x) \mid x), 2^{-B}\} \geq -\log\big(\mathbb{E}_x \nu_\pi(y^*(x) \mid x) + 2^{-B}\big).$$

Off E, $y^* = f_a$; on E, $\nu_\pi(y^* \mid x) \leq 1$, and $\Pr[E] = p$. So $\mathbb{E}_x \nu_\pi(y^*(x) \mid x) \leq u(\pi) + p$. If $\mathrm{lev}(\pi) = t \geq 2$, then $u(\pi) \leq \theta_{t-1} = \frac{1}{2} + 2^{-t}$, and $\frac{1}{2} + 2^{-t} + p + 2^{-B} = \frac{1}{2}(1 + 2^{-(t-1)} + 2p + \tau_B)$, which gives $R_B \geq G_{t-1}$. If $\mathrm{lev}(\pi) = \infty$, then $u \leq \frac{1}{2}$ and $R_B \geq G_\infty$. □

Lemma 5.4 is tight as a function of u alone: a hypothesis that gives the teacher's label probability 1 on E and is uniform elsewhere has $u = \frac{1-p}{2}$ and $\mathbb{E}_x \nu(y^* \mid x) = \frac{1}{2} + \frac{p}{2} = u + p$. Such a hypothesis knows S. Removing the additive p therefore needs the randomness of S relative to the hypothesis, which the next lemma uses. It is the semimeasure version of a lemma of the earlier report, which handled computable hypotheses through their exact correlation; for semimeasures only lower-semicomputable quantities are available. The lemma is used for the kink of Proposition 8.3 at every flip rate below 1/4, not for Theorem 5.7 (see Remark D.4).

Lemma C.1 (Randomness of the error rule sharpens the floor)**.** *For every program π, every integer $g \geq 1$ and every $B \in [\log(1/p), \infty]$,*

$$R_B(\pi) \;\geq\; -\log\Big(p + (1-2p)\,u(\pi) + p(2\gamma(\pi) + 2^{1-g}) + 2^{-B}\Big),$$

where $\gamma(\pi) := \sqrt{j \ln 2/(2s_0)}$ and j is the least integer with $j - 2\log(j+1) > \delta_S + I(\pi) + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$. Always $I(\pi) \leq \ell(\pi) + \kappa$, so the bound also holds with $\ell(\pi)$ in place of $I(\pi)$. The proof does not use $p = 2^{-r}$; it holds for every rule size s_0, with $p := s_0/2^m$.

Proof. For $z \in \{0,1\}^m$ let $A(z) := \mathbb{E}_w \nu_\pi(f_a(z,w) \mid (z,w))$ and $V(z) := \mathbb{E}_w \nu_\pi(f_a(z,w) \oplus 1 \mid (z,w))$. Then $A + V \leq 1$, $u = 2^{-m}\sum_z A(z)$, and $\bar{v} := 2^{-m}\sum_z V(z) \leq 1 - u$. Off E the teacher's label is f_a and on E it is $f_a \oplus 1$, so

$$\mathbb{E}_x \nu_\pi(y^*(x) \mid x) = 2^{-m}\Big[\sum_{z \notin S} A(z) + \sum_{z \in S} V(z)\Big] = u - p\,\mathrm{avg}_S A + p\,\mathrm{avg}_S V. \tag{1}$$

Information budget. By definition $K(S \mid a, \pi) = B_S - I(\pi) = L_S - \delta_S - I(\pi)$. By (F3), $B_S = K(S \mid a, K(a)) \leq K(\pi \mid a, K(a)) + K(S \mid a, \pi) + \kappa$, so $I(\pi) \leq K(\pi \mid a, K(a)) + \kappa \leq \ell(\pi) + \kappa$ by (F1).

Coding step. Suppose $\mathcal{B}$ is a set of s_0-subsets of $\{0,1\}^m$ with $|\mathcal{B}| \leq 2^{L_S - j}$ that contains S and is computably enumerable given (π, a, m, s_0) and a parameter θ with $K(\theta) \leq g + 2\log g + \kappa$. Then S is described by j, m, s_0, θ and its index among the first $2^{\lceil L_S \rceil - j}$ enumerated elements, so $K(S \mid a, \pi) \leq \lceil L_S \rceil - j + 2\log(j+1) + 2\log m + 2\log s_0 + g + 2\log g + \kappa$. With the budget this gives $j - 2\log(j+1) \leq \delta_S + I(\pi) + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$, which the choice of j excludes. So no such $\mathcal{B}$ exists.

The V side. If $1 - u + \gamma \geq 1$, then $\mathrm{avg}_S V \leq 1 \leq 1 - u + \gamma$. Otherwise let $\theta \leq 1$ be the least multiple of 2^{-g} with $\theta \geq 1 - u + \gamma$; then $K(\theta) \leq g + 2\log g + \kappa$ and $\theta \geq \bar{v} + \gamma$. If $\mathrm{avg}_S V > \theta$, the set $\mathcal{B} := \{S' : |S'| = s_0,\ \mathrm{avg}_{S'} V > \theta\}$ contains S and is computably enumerable, because V is lower semicomputable. For S' uniform among the s_0-subsets, Hoeffding's inequality for sampling without replacement (Hoeffding, 1963) gives $\Pr[\mathrm{avg}_{S'} V - \bar{v} \geq \gamma] \leq e^{-2s_0\gamma^2} = 2^{-j}$, so $|\mathcal{B}| \leq 2^{L_S - j}$, which the coding step excludes. Hence $\mathrm{avg}_S V \leq 1 - u + \gamma + 2^{-g}$.

The A side. A is only lower semicomputable, so a set of the form "$\mathrm{avg}_{S'} A$ is small" need not be enumerable; we use a finite stage. Let $\nu_s \uparrow \nu_\pi$ be the stage-s lower approximations, $A_s(z) := \mathbb{E}_w \nu_s(f_a(z,w) \mid (z,w))$ and $u_s := 2^{-m}\sum_z A_s(z) \uparrow u$. If $u \leq 2^{-g}$, then $\mathrm{avg}_S A \geq 0 > u - 2^{-g} - \gamma$. Otherwise let θ' be the largest multiple of 2^{-g} strictly below u, so $\theta' \geq u - 2^{-g}$ and $K(\theta') \leq g + 2\log g + \kappa$, and let s^* be the first stage with $u_{s^*} > \theta'$. Then A_{s^*} is computable from (π, a, m, θ'). If $\mathrm{avg}_S A_{s^*} < u_{s^*} - \gamma$, the set $\mathcal{B} := \{S' : |S'| = s_0,\ \mathrm{avg}_{S'} A_{s^*} < u_{s^*} - \gamma\}$ is decidable from these data and contains S; the lower-tail form of the same inequality gives $|\mathcal{B}| \leq 2^{L_S - j}$, which the coding step excludes. Hence $\mathrm{avg}_S A \geq \mathrm{avg}_S A_{s^*} \geq u_{s^*} - \gamma > u - 2^{-g} - \gamma$.

Combination. By (1), $\mathbb{E}_x \nu_\pi(y^* \mid x) \leq u - p(u - 2^{-g} - \gamma) + p(1 - u + \gamma + 2^{-g}) = p + (1-2p)u + p(2\gamma + 2^{1-g})$. Conclude with Jensen's inequality as in Lemma 5.4. □

A direct computation (code in the accompanying repository) on 400 random semimeasure hypotheses at $d = 8$, $m \in \{6, 8\}$, a third of them correlated with S, confirms (1) and the form $\mathbb{E}\nu(y^*) \leq p + (1-2p)u + p(\mathrm{dev}_A + \mathrm{dev}_V)$ with the actual deviations.

Proof of Lemma 5.5. For $z \in \{0,1\}^m$ and $b \in \{0,1\}$ let

$$q_z(b) := 2^{-\mathbb{E}_w \min\{B,\, -\log \nu_\pi(f_a(z,w) \oplus b \mid (z,w))\}}.$$

By Jensen's inequality for the convex function $t \mapsto 2^{-t}$, $q_z(b) \le \mathbb{E}_w \max\{\nu_\pi(f_a(z,w) \oplus b \mid (z,w)), 2^{-B}\}$, so $\sum_b q_z(b) \le 1 + 2 \cdot 2^{-B} = 1 + \tau_B$. Each $q_z(b)$ is lower semicomputable from (π, a, m, B), because the exponent is an average of upper semicomputable values. Define, on subsets $S' \subseteq \{0,1\}^m$,

$$Q(S') := \prod_{z \in \{0,1\}^m} \frac{q_z(\mathbf{1}_{S'}(z))}{1+\tau_B},$$

normalised factor by factor. Its total mass is $\prod_z \sum_b q_z(b)/(1+\tau_B) \le 1$, so Q is a lower-semicomputable semimeasure given (π, a, m, B). Since $y^*(z,w) = f_a(z,w) \oplus \mathbf{1}_S(z)$,

$$-\log Q(S) = \sum_z \mathbb{E}_w \min\{B, -\log \nu_\pi(y^*(z,w) \mid (z,w))\} + 2^m \log(1+\tau_B) = 2^m R_B(\pi) + \beta_B.$$

By (F2), $K(S \mid \pi, a, m, B) \le 2^m R_B(\pi) + \beta_B + \kappa$, and by (F3) and (F5) the descriptions of m and of the integer B cost at most $2\log m + 2\log B + \kappa$ more. For $B = \infty$ the terms in B vanish. □

The normalisation must be per factor: with a single division by $1+\tau_B$, Q is in general not a semimeasure (for a deterministic hypothesis at the smallest admissible B its total mass is 1.119).

Proof of Lemma 5.6. (D1). By (F3), (F1) and Lemma 5.3, $d \le K(a) \le K(\pi) + K(a \mid \pi) + \kappa \le \ell(\pi) + s(t) + 2\log d + \kappa$. All terms of Λ_1 other than $2\log d + \kappa$ are nonnegative.

(D2). By symmetry of information (F4), $K(a,S) = K(a) + K(S \mid a, K(a)) + O(1) \ge d + B_S - \kappa$. By (F3), $K(a,S) \le K(\pi) + K(a \mid \pi) + K(S \mid a, \pi) + \kappa$. Insert (F1), Lemma 5.3 and Lemma 5.5: $d + B_S \le \ell(\pi) + s(t) + 2\log d + 2^m R_B(\pi) + \beta_B + 2\log m + 2\log B \cdot \mathbf{1}[B < \infty] + \kappa$. □

C.3 Proof of the theorem

Proof of Theorem 5.7. (o). If $\mathrm{lev}(\pi) = 1$, then (D1) and (D2) with $s(1) = 4$ give $\ell(\pi) \ge d - 4 - \Lambda_1 + \max\{0, B_S - 2^m R_B(\pi)\}$; multiply by c and add $R_B(\pi)$. If $\mathrm{lev}(\pi) = t \in [2, \infty)$, then (D1), $\ell \ge 0$ and Lemma 5.4 give

$$J_c^B(\pi) \ge c\,(d - s(t) - \Lambda_1)^+ + G_{t-1} \ge \varphi(t) := c\,(d - 4t - \Lambda_1)^+ + G_{t-1},$$

using $s(t) \le 4t$. Extend φ to real $t \ge 2$. If $t^* \ge 2$: on $[2, t^*]$ the first term is linear and G_{t-1} is concave, so $\varphi \ge \min\{\varphi(2), \varphi(t^*)\} = \min\{c(d - 8 - \Lambda_1) + G_1,\, G_{t^*-1}\}$; for $t > t^*$, $\varphi(t) \ge G_{t-1} \ge G_{t^*-1}$ because G is increasing. If $t^* < 2$: $\varphi(t) \ge G_{t-1} \ge G_1$. If $\mathrm{lev}(\pi) = \infty$: $J_c^B \ge R_B \ge G_\infty \ge G_j$ for every j, so $J_c^B \ge \Phi_2(c)$ in both cases. (G_j is increasing and concave: with $v = 2^{-j}$, $\mathrm{d}G/\mathrm{d}j = v/(1 + v + 2p + \tau_B) > 0$ and $\mathrm{d}^2G/\mathrm{d}j^2 = -\ln 2 \cdot v(1 + 2p + \tau_B)/(1 + v + 2p + \tau_B)^2 < 0$.)

(i). When $c2^m > 1$, ψ_c has slope $1 - c2^m < 0$ on $[0, B_S/2^m]$ and slope 1 after it, so $\min_{R \ge 0} \psi_c(R) = \psi_c(B_S/2^m) = B_S/2^m$, and every program of level 1 has $J_c^B \ge \Phi_1(c)$. With (o), $\Phi^B(c) \ge \min\{\Phi_1(c), \Phi_2(c)\}$, and $\Phi \ge \Phi^B$ because $R \ge R_B$. Since $c \le c_{\mathrm{hi}}$, $J_g(c) \le \Phi_2(c)$, so

$$J_c(\pi_g) - \Phi(c) \le J_g(c) - \min\{\Phi_1, \Phi_2\} = \max\{J_g - \Phi_1, J_g - \Phi_2\} \le \max\{J_g - \Phi_1, 0\} = \Delta(c).$$

Finally $H(p) - B_S/2^m = (2^m H(p) - L_S + \delta_S)/2^m \le (\log(2^m + 1) + \delta_S)/2^m$, by the lower bound on L_S.

(ii). $J_c^B(\pi) \le J_g(c) + \epsilon < \Phi_2(c)$, so $\mathrm{lev}(\pi) = 1$ by (o). Every π with $J_c(\pi) \le \Phi(c) + \epsilon$ qualifies, since $J_c^B \le J_c$ and $\Phi(c) \le J_c(\pi_g) \le J_g(c)$.

(a) By Lemma 5.4, $-\log(u + p + 2^{-B}) \le R_B \le J_c^B \le J_g + \epsilon$; and $\mathrm{lev}(\pi) = 1$ gives $u > \frac{3}{4}$.

(b) By (o), $c(d-4-\Lambda_1)+\psi_c(R_B)\le J_g+\epsilon$, that is, $\psi_c(R_B)\le c(4+\Lambda_1+\kappa_g)+H(p)+\epsilon$. If $R_B<B_S/2^m$, then $\psi_c(R_B)=cB_S-(c2^m-1)R_B$. Using the identity $cB_S-H(p)=(c2^m-1)H(p)-c(2^mH(p)-B_S)$,

$$(c2^m-1)R_B\ge(c2^m-1)H(p)-c\big(2^mH(p)-B_S+4+\Lambda_1+\kappa_g\big)-\epsilon=(c2^m-1)H(p)-c\Lambda_2-\epsilon.$$

If $R_B\ge B_S/2^m$ and $2^mH(p)-B_S>0$, then $R_B\ge H(p)-(2^mH(p)-B_S)/2^m\ge H(p)-c\Lambda_2/(c2^m-1)$, because $\Lambda_2\ge 2^mH(p)-B_S$ and $c/(c2^m-1)>2^{-m}$; if $2^mH(p)-B_S\le 0$ the claim is immediate.

(c) $c\,\ell=J_c^B-R_B\le c(d+\kappa_g)+H(p)+\epsilon-R_B$; apply (b) and divide by c.

(d) By (F3), $B_S\le K(\pi\mid a,K(a))+K(S\mid a,\pi)+\kappa$. By (F4) and (F3), $K(\pi\mid a,K(a))=K(\pi,a)-K(a)+O(1)\le K(\pi)+K(a\mid\pi)-K(a)+O(1)$. By (F1), Lemma 5.3 with $t=1$, $K(a)\ge d$ and (c), this is at most $\ell(\pi)-d+4+2\log d+\kappa\le\Delta_\ell+\Lambda_3$.

(e) Let $v(z):=\mathbb{E}_w\nu_\pi(f_a(z,w)\oplus 1\mid(z,w))\in[0,1]$. Given $x\in E$, z is uniform on S and $y^*=f_a\oplus 1$, so $e(\pi)=\mathrm{avg}_S v$. Since ν_π is a semimeasure, $\bar v:=2^{-m}\sum_z v(z)\le 1-u(\pi)$. If $1-u+\gamma_g\ge 1$ there is nothing to prove. Otherwise let $\theta\le 1$ be the least multiple of 2^{-g} with $\theta\ge 1-u(\pi)+\gamma_g$; then $K(\theta)\le g+2\log g+\kappa$ and $\theta\ge\bar v+\gamma_g$. Suppose $e(\pi)>\theta$. The set $\mathcal{B}_\theta:=\{S':|S'|=s_0,\ \mathrm{avg}_{S'}v>\theta\}$ is computably enumerable given (π,a,m,s_0,θ), because v is lower semicomputable, and it contains S. By Hoeffding's inequality for sampling without replacement, $|\mathcal{B}_\theta|\le\binom{2^m}{s_0}e^{-2s_0\gamma_g^2}=2^{L_S-j_g}$. Describing S by j_g and its index among the first $2^{\lceil L_S\rceil-j_g}$ enumerated elements gives $K(S\mid a,\pi)\le\lceil L_S\rceil-j_g+2\log(j_g+1)+2\log m+2\log s_0+g+2\log g+\kappa$. By (d), $K(S\mid a,\pi)\ge L_S-\delta_S-\Delta_\ell-\Lambda_3$. Together, $j_g-2\log(j_g+1)\le\Gamma_g$, contradicting the choice of j_g. So $e(\pi)\le\theta\le 1-u(\pi)+\gamma_g+2^{-g}$.

(iii). $\mathrm{lev}(\pi)=1$ follows from (o) as in (ii). For $c2^m<1$ and $R\ge 0$, $\psi_c(R)\ge cB_S+(1-c2^m)\min\{R,B_S/2^m\}$: for $R\le B_S/2^m$ this is an equality, and for $R\ge B_S/2^m$, $\psi_c(R)=R\ge B_S/2^m=cB_S+(1-c2^m)B_S/2^m$. Compare with $J_c^B\le J_b(c)+\epsilon$ and (o), and use $L_S=B_S+\delta_S$:

$$(1-c2^m)\min\{R_B,B_S/2^m\}\le c\big(\lceil L_S\rceil+\kappa_b-B_S+4+\Lambda_1\big)+\epsilon=\varrho.$$

If $\varrho<(1-c2^m)B_S/2^m$, the minimum cannot be $B_S/2^m$, so $R_B\le\varrho/(1-c2^m)$. For the error label, $p\,\mathbb{E}[\min\{B,-\log\nu_\pi(y^*\mid x)\}\mid x\in E]\le R_B$, and by Jensen's inequality

$$\mathbb{E}[\nu_\pi(y^*\mid x)\mid E]\ge\mathbb{E}[\max\{\nu_\pi(y^*\mid x),2^{-B}\}\mid E]-2^{-B}\ge 2^{-R_B/p}-2^{-B}.$$

(iv). By (D1) with $t=1$ and $R(\pi)\ge H(p)$, $J_c(\pi)\ge c(d-4-\Lambda_1)+H(p)>c(d-4-\Lambda_1)+c(\lceil L_S\rceil+\kappa_b+4+\Lambda_1)=c(d+\lceil L_S\rceil+\kappa_b)\ge J_c(\pi_b)$. □

Numbers. All numbers of Section 5 and this appendix come from deterministic or seeded scripts in the accompanying repository, with saved outputs. The script for Table 2 computes the windows in closed form and by bisection on $J_g\le\Phi_2$ (the two agree to 10^{-15}), the identification bounds with $\delta_S\in\{0,10,20\}$ and g optimised over $1,\dots,40$, and the memoriser band; $L_S=11{,}710.273$ bits comes from the exact binomial coefficient, and the crossover $G_1=H(p)$ at $B=\infty$ is $p=0.05596$. A direct computation over 2830 random instances confirms the reduction to two endpoints in (o): $\min_{t\ge 2}$ of the exact $c(d-s(t)-\Lambda_1)^+ + G_{t-1}$ never falls below $\Phi_2(c)$. For the three hypotheses of Remark 5.1, the scripts of the earlier report give the same window $(1.119\cdot 10^{-6}, 8.989\cdot 10^{-3})$, with ratio 8037, and the same $\lambda=1$ switch at $n=894{,}054$.

In the memoriser regime of the running example ($B=\infty$, $\epsilon=0$, $\delta_S=0$), part (iii) gives the following lower bounds on e for every near-optimal hypothesis ("–": (iii) does not apply, since at $\kappa=30$ it needs $c<0.989\cdot 2^{-m}$); rows with $\kappa=0$ are illustrations.

$c\,2^m$	0.5	0.8	0.9	0.94	0.97	0.99
$\kappa = 0$	0.955	0.831	0.658	0.483	0.223	0.010
$\kappa = 30$	0.917	0.706	0.457	0.256	0.060	–

C.4 The full truth table

Proof of Corollary 5.10. With $m = d$ the error set is $E = S$ and $2^m = N$. On the whole truth table the empirical loss equals N times the population risk under uniform inputs, so $J_\lambda(\pi; D) = \lambda\ell(\pi) + NR(\pi) = N\,J_{\lambda/N}(\pi)$, and a λ-MAP minimises $J_{\lambda/N}$ exactly.

(i) Put $c = \lambda/N$. Then $c2^m = \lambda > 1$, and the hypothesis $J_g(c) < \Phi_2(c)$ holds for every $c < c_{\mathrm{hi}}$, by the form of Φ_2. A λ-MAP has $J_c(\pi) = \Phi(c)$, so Theorem 5.7(ii) applies with $\epsilon = 0$, $B = \infty$. Substituting $c2^m - 1 = \lambda - 1$ and $c/(c2^m - 1) = \lambda/(N(\lambda - 1))$ gives the stated forms of (a)–(d).

(ii) At $\lambda = 1$ the objective is $\ell + NR$. First, $\ell(\pi_g) + NR(\pi_g) = \ell(\pi_g) + NH(p) \le d + \kappa_g + L_E + \log(N + 1)$, because $NH(p) \le L_E + \log(N + 1)$. Next, $\ell(\pi_b) + NR(\pi_b) \le d + \lceil L_E \rceil + \kappa_b$. Every program of level 1 has $\ell + NR \ge \ell + NR_B \ge d + B_E - 4 - \Lambda_1 = d + L_E - \delta_E - 4 - \Lambda_1$, by (D2). Every program of level at least 2 has $\ell + NR \ge NR \ge NG_1$ by Lemma 5.4, and by the hypotheses $G_1 > H(p)$ and $N(G_1 - H(p)) > d + \kappa_g + \log(N + 1)$ this exceeds $\ell(\pi_g) + NR(\pi_g)$, since $L_E \le NH(p)$; so the optimum is attained at level 1. Here $d = \log N$, and κ_g, κ_b and Λ_1 are $O(\log N)$, so the optimum and the two values lie within $O(\log N) + \delta_E$ of each other. □

At $d = m = 20$, $p = 2^{-10}$, $B = \infty$, a direct computation (code in the accompanying repository, with g optimised) gives the window $\lambda \in (1, 11945)$ with the machine constant set to 0 (an illustration) and $(1, 6456)$ at $\kappa = 30$, and the following bounds for every λ-MAP. Near $\lambda = 1$ the deficiency enters the length and information bounds at full weight, through $\Lambda_2/(\lambda - 1)$.

		$\lambda = 2$				$\lambda = 105$		
κ	δ_E	$u \ge$	$\ell \le$	$I \le$	$e \le$	$\ell \le$	$I \le$	$e \le$
0	0	0.991	$d + 58.2$	70.8	0.220	$d + 15.7$	28.3	0.185
0	20	0.991	$d + 78.2$	90.8	0.251	$d + 15.9$	28.5	0.204
30	0	0.991	$d + 148.2$	190.8	0.320	$d + 46.3$	88.9	0.261
30	20	0.991	$d + 168.2$	210.8	0.341	$d + 46.5$	89.1	0.275

In an exact small instance with structured product families ($d = m = 8$, $|E| = 8$), the good model and the memoriser exchange places at $\lambda = 0.936$, an illustration of how lower-order terms decide at $\lambda \approx 1$.

C.5 Finite samples

Lemma C.2 (Finite sample)**.** *Let $m \le d$ and $B < \infty$. Draw $n \le 2^d$ inputs uniformly without replacement and label them by y^*; let $\hat{J}_c(\pi) := c\,\ell(\pi) + \frac{1}{n}\sum_{i \le n} -\log \nu_\pi(y^*(x_i) \mid x_i)$. Let π_{ref} be a fixed program whose per-input losses lie in $[0, A]$. With probability at least $1 - \delta$, every minimiser $\hat{\pi}$ of $\hat{J}_c$ satisfies*

$$J_c^B(\hat{\pi}) \le J_c(\pi_{\mathrm{ref}}) + \epsilon_1 + B\sqrt{\frac{\hat{L}\ln 2 + \ln(2/\delta)}{2n}}, \qquad \epsilon_1 := A\sqrt{\frac{\ln(2/\delta)}{2n}}, \quad \hat{L} := \frac{J_c(\pi_{\mathrm{ref}}) + \epsilon_1}{c}.$$

Proof. Each sampled input is marginally uniform. By Hoeffding's inequality for sampling without replacement (Hoeffding, 1963), with probability at least $1 - \delta/2$, $\hat{J}_c(\pi_{\mathrm{ref}}) \le J_c(\pi_{\mathrm{ref}}) + \epsilon_1$. For each

program π the truncated losses lie in $[0, B]$; apply the same inequality with confidence $(\delta/2)2^{-\ell(\pi)}$ and take a union bound, which costs at most $\delta/2$ by Kraft's inequality. With probability at least $1 - \delta/2$, simultaneously for all π,

$$\hat{J}_c(\pi) \geq c\,\ell(\pi) + \frac{1}{n}\sum_{i\leq n} \min\{B, -\log \nu_\pi(y^*(x_i) \mid x_i)\} \geq J_c^B(\pi) - B\sqrt{\frac{\ell(\pi)\ln 2 + \ln(2/\delta)}{2n}}.$$

On both events, a minimiser satisfies $c\,\ell(\hat{\pi}) \leq \hat{J}_c(\hat{\pi}) \leq \hat{J}_c(\pi_{\mathrm{ref}}) \leq J_c(\pi_{\mathrm{ref}}) + \epsilon_1$, so $\ell(\hat{\pi}) \leq \hat{L}$, and the two displays combine to the claim. □

The bound is one-sided and uses truncation, so no noise floor on the hypotheses is needed; this adapts the empirical-selection lemma of the earlier report. The value of c is fixed for the given n, so the lemma also applies when $\lambda = \lambda_n$ follows any schedule; Proposition 6.1(b) uses this.

Proof of Corollary 5.11. A λ-MAP with $\lambda = cn$ minimises $J_\lambda/n = \hat{J}_c$. Apply Lemma C.2 with $\pi_{\mathrm{ref}} = \pi_g$, whose per-input losses $-\log(1-p)$ and $-\log p$ lie in $[0, \log(1/p)]$, so $A = \log(1/p)$. Then $J_c^B(\hat{\pi}) \leq J_c(\pi_g) + \epsilon_n \leq J_g(c) + \epsilon_n$, which is the hypothesis of Theorem 5.7(ii) with $\epsilon = \epsilon_n$. □

At $d = 100$, $m = 20$, $p = 2^{-10}$, $B = 30$, $\delta = 0.01$, $\delta_S = 0$, $g = 7$ and the machine constant set to 0 (an illustration), a direct computation (code in the accompanying repository) gives:

c	n	ϵ_n/c (bits)	$u \geq$	$\ell \leq$	$e \leq$
10^{-4}	10^{11}	9.2	0.9825	$d + 29.8$	0.212
10^{-4}	10^{13}	0.9	0.9830	$d + 21.5$	0.204
10^{-3}	10^{11}	0.7	0.9117	$d + 20.7$	0.274

At $n = 10^9$ and $c = 10^{-4}$ the excess is still 92.5 bits ($\ell \leq d + 113.9$, $e \leq 0.284$).

C.6 The total-bits version of the example

Example C.3 (The total-bits version of the example)**.** By Theorem 5.7 with $d = 100$, $m = 10$ and $p = 2^{-6}$, so that $s_0 = 16$ and $L_S = 115.6$ bits, a direct computation (code in the accompanying repository) gives the following. At $n = 10^4$ the model saves $n(1 - H(p)) = 8839$ bits for its 100 bits (88 bits per bit) and the error rule saves $nH(p) = 1161$ bits for its 116 bits (10.0 bits per bit): the sizes and rates of the total-bits version in Remark 5.1. The window is $c \in (2^{-10}, c_{\mathrm{hi}}]$ with $c_{\mathrm{hi}} = 5.80 \cdot 10^{-3}$ with the machine constant set to 0 (an illustration) and $2.53 \cdot 10^{-3}$ at $\kappa = 30$; at $n = 10^4$ and $\kappa = 0$ this is $\lambda \in (9.77, 58.0]$, against $(10, 100)$ for the three hypotheses. Only coarse identification is proved at this size: near-optimal hypotheses have $u > \frac{3}{4}$, and at $\kappa = 0$, $\delta_S = 0$, $c = 5.5 \cdot 10^{-3}$ they have $\ell \leq d + 28.3$ and $I \leq 45.6$ bits out of 115.6; near the lower end the information bound exceeds L_S, and (e) is vacuous because $s_0 = 16$ is too small. At $n = 10^4$ samples Corollary 5.11 gives nothing.

In more detail, the computation gives $L_S = 115.58$ bits, $H(2^{-6}) = 0.1161$, and the window ends $c_{\mathrm{hi}} = 5.803 \cdot 10^{-3}$, $4.774 \cdot 10^{-3}$, $4.055 \cdot 10^{-3}$, $3.116 \cdot 10^{-3}$ and $2.530 \cdot 10^{-3}$ at $\kappa = 0, 5, 10, 20, 30$, so the window is nonempty for every $\kappa \leq 30$ checked. At $\kappa = 0$, $\delta_S = 0$, $\epsilon = 0$ and $B = \infty$, (c) and (d) give $\ell \leq d + 103.7$ and $I \leq 121.0$ at $c = 1.5 \cdot 10^{-3}$ (vacuous, since $L_S = 115.6$), $\ell \leq d + 47.8$ and $I \leq 65.0$ at $c = 2.5 \cdot 10^{-3}$, $\ell \leq d + 33.2$ and $I \leq 50.5$ at $c = 4 \cdot 10^{-3}$, and $\ell \leq d + 28.3$ and $I \leq 45.6$ at $c = 5.5 \cdot 10^{-3}$. Part (a) gives $u \geq 0.800$ at $c = 1.5 \cdot 10^{-3}$ and only $u > \frac{3}{4}$ for $c \geq 2.5 \cdot 10^{-3}$, where $2^{-(J_g + \epsilon)} - p$ falls below $\frac{3}{4}$; the bound of (e) exceeds 1 throughout.

The threshold of (iv) is $c = 6.917 \cdot 10^{-4}$. For the finite sample at $n = 10^4$: every admissible $B \geq \log(1/p) = 6$ makes the excess of Lemma C.2 at least $6\sqrt{(d - 4 - 2\log d)\ln 2/(2n)} = 0.321$ at $\kappa = 0$, because $\hat{L} \geq \ell(\pi_g) \geq d - 4 - 2\log d - \kappa$ by (D1), whose proof uses only the terms $2\log d + \kappa$ of Λ_1; the margin $\Phi_2 - J_g$ is at most 0.224 anywhere in the window (its largest value is at $B = \infty$).

For comparison, a full-table instance that matches the error rule's size but not the model's ($d = 13$, $m = 10$, $p = 2^{-6}$, $N = 2^{13}$) has the window $\lambda \in (8, 86.2]$ at $\kappa = 0$, $(8, 62.0]$ at $\kappa = 10$ and $(8, 39.7]$ at $\kappa = 30$.

C.7 Concentration of the power posterior

The power posterior with $\lambda = cn$, not only its maximiser, concentrates on near-optimal programs. Let $\hat{J}_c(\pi) := c\,\ell(\pi) + \frac{1}{n}\sum_{i \leq n} -\log \nu_\pi(y_i \mid x_i)$ (Remark 3.7) and $\hat{J}_g := \hat{J}_c(\pi_g)$.

Proposition C.4 (The power posterior concentrates on near-minimisers)**.** *Let $\lambda = cn$ with $cn \geq 1$, and let D be any data set of size n. For every $\epsilon > 0$,*

$$\frac{\Pi_\lambda(\{\pi : \hat{J}_c(\pi) \geq \hat{J}_g + \epsilon\} \mid D)}{\Pi_\lambda(\pi_g \mid D)} \;\leq\; 2^{-n\epsilon + (\hat{J}_g + \epsilon)/c}.$$

Consequently, for data sampled as in Corollary 5.11, with probability at least $1 - \delta$ all but this fraction of the posterior mass lies on programs that satisfy Theorem 5.7(ii) with $\epsilon' := \epsilon + \epsilon_1 + B\sqrt{(\hat{L}' \ln 2 + \ln(2/\delta))/(2n)}$ in place of ϵ, where $\hat{L}' := (J_g(c) + \epsilon_1 + \epsilon)/c$, provided $J_g(c) + \epsilon' < \Phi_2(c)$.

Proof. For every π, $2^{-\lambda\ell(\pi)}\nu_\pi(D) = 2^{-n\hat{J}_c(\pi)}$. Write $\hat{R}(\pi) := \frac{1}{n}\sum_i -\log \nu_\pi(y_i \mid x_i) \geq 0$ for the empirical risk. Then $n\hat{J}_c(\pi) - \ell(\pi) = (\lambda - 1)\ell(\pi) + n\hat{R}(\pi) \geq (1 - 1/\lambda)(\lambda\ell(\pi) + n\hat{R}(\pi)) = (1 - 1/\lambda)\, n\hat{J}_c(\pi)$. So, on the set $\{\pi : \hat{J}_c(\pi) \geq \hat{J}_g + \epsilon\}$,

$$2^{-n\hat{J}_c(\pi)} = 2^{-\ell(\pi)}\, 2^{-(n\hat{J}_c(\pi) - \ell(\pi))} \leq 2^{-\ell(\pi)}\, 2^{-(1-1/\lambda)\, n(\hat{J}_g + \epsilon)},$$

and by Kraft's inequality the sum over the set is at most $2^{-(1-1/\lambda)n(\hat{J}_g + \epsilon)}$. Divide by $2^{-n\hat{J}_g}$, the unnormalised weight of π_g, and use $n/\lambda = 1/c$: the ratio is at most $2^{-n\epsilon + (\hat{J}_g + \epsilon)/c}$.

For the consequence, work on the event of Lemma C.2 with $\pi_{\mathrm{ref}} = \pi_g$ and $A = \log(1/p)$, on which $\hat{J}_g \leq J_g(c) + \epsilon_1$ and the uniform deviation bound holds for every program. A program with $\hat{J}_c(\pi) < \hat{J}_g + \epsilon$ has $\ell(\pi) \leq \hat{J}_c(\pi)/c < \hat{L}'$, hence

$$J_c^B(\pi) \leq \hat{J}_c(\pi) + B\sqrt{\frac{\hat{L}' \ln 2 + \ln(2/\delta)}{2n}} \leq J_g(c) + \epsilon',$$

and Theorem 5.7(ii) applies with ϵ' when $J_g(c) + \epsilon' < \Phi_2(c)$. The bad set has posterior mass at most the ratio times $\Pi_\lambda(\pi_g \mid D) \leq 1$. □

Near-minimisers can be ϵ/c bits longer than the minimiser, which is why the length budget is $\hat{L}'$ and not the $\hat{L}$ of Lemma C.2. For instance, with $\hat{J}_g$ replaced by its population value $J_g = 0.0232$, $c = 10^{-4}$, $\epsilon = 10^{-3}$, $n = 10^6$ and the machine constant set to 0 (an illustration), $-n\epsilon + (J_g + \epsilon)/c = -1000 + 241.7 = -758.3$, so the ratio is at most 2^{-758}. At $c = 10^{-4}$, $\epsilon = 10^{-3}$, $n = 10^{13}$, $B = 30$, $\delta = 0.01$ and $\kappa = 0$, a direct computation (code in the accompanying repository) gives $\hat{L}' = 241.7$ bits and $\epsilon' = 1.0933 \cdot 10^{-3}$, and the condition $J_g + \epsilon' < \Phi_2$ holds. Two further consequences: the steeper predictive, averaged over E, gives the erroneous label probability at most $\sup e(\pi) + 2^{-n\epsilon + (\hat{J}_g + \epsilon)/c}$, the supremum over the programs just described; averaged over the unseen error inputs $E \setminus \mathrm{dom}\, D$ instead, the bound is multiplied by $|E|/(|E| - n)$, which is negligible at $d = 100$.

D Proofs for Section 6 (limits of steeper selection)

The setting is Definition 5.2 unless stated otherwise; (F1)–(F5) are the facts of Appendix A.1.

D.1 Plain Bayes, MDL and SI

Proof of Proposition 6.1. (a) The memoriser π_b is deterministic and consistent with every D_n, so $P(D_n) \geq 2^{-\ell(\pi_b)-\kappa}$. Since P is a shared hypothesis, its sequential log loss telescopes: $\sum_{i\leq n} -\log P(y^*(x_i) \mid x_i, D_{i-1}) = -\log P(D_n) + \log P(\emptyset) \leq \ell(\pi_b) + \kappa$. Each conditional is at most 1, so each term is nonnegative, and a label with probability below $\frac{1}{2}$ adds more than one bit; normalised conditionals are at least the raw ones. On an error input the teacher's label is the erroneous one. The memoriser can be given $s_0 = 2^{m-r}$ through r, so $\ell(\pi_b) \leq d + \lceil L_S \rceil + O(\log d + \log m)$. If all 2^d inputs are revealed there are $p2^d$ error inputs, and at most $d + \lceil L_S \rceil + \kappa_b + \kappa$ of them get the erroneous label with probability below $\frac{1}{2}$. At $d = 100$, $m = 20$, $p = 2^{-10}$ and $\kappa = 0$ there are $1.238 \cdot 10^{27}$ error inputs and at most 11,853 exceptions, a fraction $9.57 \cdot 10^{-24}$ (code in the accompanying repository). For M_F, $P \geq 2^{-K(\nu_\pi)-O(1)}\nu_\pi$ because product laws are machine laws, and $K(\nu_\pi) \leq \ell(\pi) + \kappa$ by (F1).

(b) Apply Lemma C.2 with $\pi_{\mathrm{ref}} = \pi_b$, whose per-input losses are 0, so $A = 0$ and $\epsilon_1 = 0$, and with $c = \lambda_n/n$; then its $\hat{L}$ is $\ell(\pi_b) \leq d + \lceil L_S \rceil + \kappa_b$. A λ-MAP minimises $\hat{J}_c$, so with probability at least $1 - \delta$, $R_B(\hat{\pi}) \leq J_c^B(\hat{\pi}) \leq c\,\ell(\pi_b) + B\sqrt{(\hat{L}\ln 2 + \ln(2/\delta))/(2n)}$. The lemma holds for each n separately, so λ may depend on n. The error-label bound is the last step of the proof of Theorem 5.7(iii): $p\,\mathbb{E}[\min\{B, -\log \nu\} \mid E] \leq R_B$ and Jensen's inequality. If $(\lambda_n/n)\hat{L} \to 0$ and $\hat{L}/n \to 0$, with δ fixed, both terms of the bound on R_B tend to 0, and $e(\hat{\pi}) \geq 1 - 2^{-B} - o(1)$. On a fixed instance the bound $2^{-R_B/p} - 2^{-B}$ is informative only if R_B is small compared with p, and the bound on R_B is at least $(\lambda_n/n)\hat{L} \geq (\lambda_n/n)(d + L_S)$. For $\lambda_n = n^{0.9}$ at the example's sizes, $\lambda_n/n = n^{-0.1} \geq 2^{-10} = p$ for every $n \leq 2^{100}$, so $(\lambda_n/n)(d + L_S) > 100p$ and the bound says nothing for any $n \leq 2^d$. Numbers (code in the accompanying repository; the machine constant set to 0, $B = 30$, $\delta = 0.01$): for $\lambda = 1$, $e \geq 0.255$, 0.649, 0.958, 0.996 at $n = 10^{12}, 10^{13}, 10^{15}, 10^{17}$; for $\lambda_n = \sqrt{n}$, $e \geq 0.045$, 0.734, 0.970 at $n = 10^{13}, 10^{15}, 10^{17}$. At $n = 10^{12}$, $\lambda_n/n = 10^{-6}$ still lies in the window of Theorem 5.7, and the bound says nothing.

(c) Encode the interleaved sequence with fixed-length blocks; with uniform i.i.d. inputs and $y_i = y^*(x_i)$ its law $\mathbb{P}$ is computable from a, S, d, m and r, so it has a program $\pi_{\mathbb{P}} \in \mathcal{H}_{\mathrm{SI}}$ with $\ell(\pi_{\mathbb{P}}) \leq d + \lceil L_S \rceil + O(\log d + \log m)$. Since $M \geq 2^{-\ell(\pi_{\mathbb{P}})}\mathbb{P}$, the chain-rule argument in the proof of Theorem 4.2(d) gives $\mathbb{E}_{\mathbb{P}} \sum_{i\leq n} -\log M(y_i \mid x_1 y_1 \cdots x_i) \leq \ell(\pi_{\mathbb{P}})$. For $\lambda > 1$ apply Theorem 4.2(d) itself, with $\kappa_0 = 0$ in $\mathcal{H}_{\mathrm{SI}}$. □

D.2 Rate blindness: the criterion in the machine constant

Proof of Proposition 6.4. (i) $J_c(\pi_b) < J_c(\pi_g)$ if and only if $c < H(p)/(\ell(\pi_b) - \ell(\pi_g))$, and $J_c(\pi_{\mathrm{coin}}) \leq J_c(\pi_g)$ if and only if $c \geq (1 - H(p))/(\ell(\pi_g) - \ell(\pi_{\mathrm{coin}}))$; under the hypothesis these two ranges cover all $c > 0$. (ii) Let $u(\pi) > \frac{3}{4}$, so $\mathrm{lev}(\pi) = 1$, and $R(\pi) \geq H(p)$. Below the left side of $(\star)$, Theorem 5.7(iv) gives $J_c(\pi_b) < J_c(\pi)$. At or above it, $c\,(d - 4 - \Lambda_1 - \kappa) \geq 1 - H(p)$ by $(\star)$, so (D1) gives $J_c(\pi) \geq c(d - 4 - \Lambda_1) + H(p) \geq 1 + c\kappa \geq J_c(\pi_{\mathrm{coin}})$. (iii) is Theorem 5.7(iii). □

Proof of Lemma 6.5. (i) With $B = \infty$, $\Lambda_1 = 2\log d + 2\log m + \kappa$ and $\kappa_b = 2\log d + 2\log m + 2\log s_0 + \kappa$. The first denominator of $(\star)$ is positive. When the second is positive, $(\star)$ is equivalent to $H\,(d - 4 - \Lambda_1 - \kappa) \geq (1 - H)(\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1)$. Dividing by H, the difference of the two

sides is

$$d-4-2\log d-2\log m-2\kappa-q\big(\lceil L_S\rceil+4\log d+4\log m+2\log s_0+2\kappa+4\big)=d-\alpha_0\log d-\beta_0=F(d).$$

Conversely, $F(d)\ge 0$ implies $d-4-\Lambda_1-\kappa=d-4-2\log d-2\log m-2\kappa\ge F(d)+4q\log d+q(\lceil L_S\rceil+4)>0$, so the side condition holds as well.

(ii) $F'(d)=1-\alpha_0/(d\ln 2)$, so F is strictly increasing on $[\alpha_0/\ln 2,\infty)$ and tends to ∞. On $[1,\alpha_0/\ln 2]$, $F(1)=1-\beta_0<0$ because $\beta_0\ge 4$, and for $d>1$, $F(d)<d-\beta_0\le\alpha_0/\ln 2-\beta_0\le 0$. So $F<0$ on $[1,\alpha_0/\ln 2]$, and $\{d\ge 1: F(d)\ge 0\}=[d_0,\infty)$ for the unique root $d_0>\alpha_0/\ln 2$.

(iii) β_0 is affine in κ with slope $2+2q$, α_0 does not depend on κ, and $d_0=\beta_0+\alpha_0\log d_0$.

Taking all three constants equal to a common upper bound is legitimate because $(\star)$ only becomes harder to satisfy when any of them grows. □

A direct computation (code in the accompanying repository) gives $d_0(\kappa)$, the least integer d with $F(d)\ge 0$, for five cheap error rules. In every row $\beta_0\ge\alpha_0/\ln 2$; $(\star)$ fails at d_0-1, holds at d_0, and holds at each of about 400 values of d in $[d_0,10^6]$.

error rule	$\kappa=0$	10	20	30	50	100
$m=3$, $p=1/8$ (one 3-bit prefix)	49	90	130	169	245	434
$m=4$, $p=1/16$ (one 4-bit prefix)	106	172	236	299	422	727
$m=6$, $p=1/64$ (one 6-bit prefix)	450	639	823	1005	1364	2248
$m=7$, $p=1/128$ (one 7-bit prefix)	901	1231	1554	1873	2505	4063
$m=6$, $p=1/8$ (eight 6-bit prefixes)	89	129	168	206	282	470

Illustrations at $\kappa=0$. These are not proved for any reference machine. In the three-hypothesis form Proposition 6.4(i), with the lengths set equal to their upper bounds $d+\kappa_g$ and $d+\lceil L_S\rceil+\kappa_b$, at $d=100$ a single 6- or 7-bit prefix is never selected, with margins of 9.6 and 1.4 bits (the conclusion reverses if $\ell(\pi_b)-\ell(\pi_g)$ is larger by that much), while an 8-bit prefix leaves a window. In the all-hypothesis form, the 3-bit rule satisfies $(\star)$ at $d=100$ for $\kappa\le 10$ and fails for $\kappa\ge 20$. In the memoriser regime of a single 6-bit prefix at $c=10^{-4}$, near-optimal programs give the error label probability at least 0.833.

D.3 Separable components, guard erosion, non-identifiability, input law

Proof of Proposition 6.6. The objective $c\,\ell+R$ is a sum over regions of (extra length times c) plus (risk on the region), and the choices in different regions are independent, so it is minimised region by region; in each region Lemma A.1 applies. With two options, component j lowers the objective if and only if $c\,k_j<r_j$. If a valid component with rate ρ_v is kept, then $c<\rho_v\le\rho_e$ (up to ties), so an error component with rate ρ_e is kept as well. Finally, J_c depends on the data only through the teacher's channel and the input law, which coincide in the two worlds. □

Proposition D.1 (Guard erosion)**.** *In a region of contexts a schema σ is one of n_{st} possible steps, and a guard G splits the region into G-contexts (input mass $\mu(G)$), where the teacher plays σ with probability $1-\epsilon_G+\epsilon_G/n_{\mathrm{st}}$ and every other step with probability $\epsilon_G/n_{\mathrm{st}}$, and $\neg G$-contexts (mass $\mu(\neg G)$), where it plays one fixed step other than σ. Over a common base, "none" is uniform (length 0), "unguarded σ" predicts the G-context law everywhere (length k_σ), and "guarded σ" does so in G-contexts and is uniform over the other $n_{\mathrm{st}}-1$ steps elsewhere (length $k_\sigma+k_G$). With H_G the entropy of the G-context law, $g_G:=\log n_{\mathrm{st}}-H_G$ and $\Delta_G:=\log(n_{\mathrm{st}}/\epsilon_G)-\log(n_{\mathrm{st}}-1)$, put $\rho_\sigma:=(\mu(G)g_G-\mu(\neg G)\log(1/\epsilon_G))/k_\sigma$ and $\rho_G:=\mu(\neg G)\Delta_G/k_G$. If $\rho_G<\rho_\sigma$, then for every*

$c \in (\rho_G, \rho_\sigma)$ *the unguarded rule is the unique selection, and it applies* σ *where the teacher never did. Otherwise the selection goes from guarded to none.*

Proof of Proposition D.1. Per context of the region, the expected log losses are: for "none", $\log n_{\mathrm{st}}$ everywhere; for "unguarded σ", H_G in G-contexts and $\log(n_{\mathrm{st}}/\epsilon_G)$ in $\neg G$-contexts, where the teacher's fixed step has probability $\epsilon_G/n_{\mathrm{st}}$ under the G-context law; for "guarded σ", H_G in G-contexts and $\log(n_{\mathrm{st}} - 1)$ in $\neg G$-contexts. Relative to "none", the unguarded option lowers the risk by $\mu(G)g_G - \mu(\neg G)\log(1/\epsilon_G) = k_\sigma\rho_\sigma$ for k_σ bits, and the guard lowers it by a further $\mu(\neg G)\Delta_G = k_G\rho_G$ for k_G more bits. So the three options are the points $(0, 0)$, $(k_\sigma, -k_\sigma\rho_\sigma)$ and $(k_\sigma + k_G, -k_\sigma\rho_\sigma - k_G\rho_G)$ in the (length, risk) plane, after subtracting the common base. Adding back the risk $\log n_{\mathrm{st}}$ of "none" per context shifts all three objectives by the same constant and makes the risks nonnegative, so Lemma A.1 applies. The middle point is a vertex of the hull if and only if the slope $-\rho_\sigma$ of the first segment is steeper than the slope $-\rho_G$ of the second, that is, $\rho_\sigma > \rho_G$. By Lemma A.1(iii) it is then the unique minimiser for $c \in (\rho_G, \rho_\sigma)$. Otherwise the hull has only the two outer points, and the selection switches at the slope between them, $(k_\sigma\rho_\sigma + k_G\rho_G)/(k_\sigma + k_G)$. □

Numbers (code in the accompanying repository). With $n_{\mathrm{st}} = 64$ and $\epsilon_G = 0.05$, the G-context law puts 0.95078 on σ, $H_G = 0.5773$, $g_G = 5.4227$ and $\Delta_G = 4.3446$. For $\mu(G) = 0.05$, $\mu(\neg G) = 5 \cdot 10^{-4}$, $k_\sigma = 22$, $k_G = 12$: $\rho_\sigma = 1.2226 \cdot 10^{-2}$, $\rho_G = 1.8103 \cdot 10^{-4}$, and as c increases the selection goes guarded, unguarded (on $(1.8103 \cdot 10^{-4}, 1.2226 \cdot 10^{-2})$), none. For $\mu(G) = \mu(\neg G) = 0.002$: $\rho_\sigma = 1.0007 \cdot 10^{-4} < \rho_G = 7.2411 \cdot 10^{-4}$, and the selection goes from guarded to none at $c = 3.2032 \cdot 10^{-4}$. The teacher has noise ϵ_G in G-contexts; what it never does is apply σ outside G.

Proposition D.2 (λ cannot be chosen from imitation data)**.** *(i) Every strictly proper scoring rule has expected score on the teacher's data uniquely maximised by the teacher's channel* $P^*(\cdot \mid x)$*, errors included (Gneiting and Raftery, 2007), so validation by such a score prefers the teacher's channel to the intended model. (ii) Along the population selection path the held-out log loss* $R(h_c)$ *is nonincreasing as* c *decreases (Lemma A.1(ii)), so log-loss validation over a grid of values of* c *is minimised at the smallest, error-memorising, value. (iii) In the twin worlds of Proposition 6.6, every rule that chooses* c *from imitation data alone is wrong in one of them.*

For instance, for the three hypotheses of Remark 5.1, with the machine constant set to 0 (an illustration), a direct computation (code in the accompanying repository) gives the held-out losses 1.0, 0.0112, 0.0112 and 0 at $c = 10^{-1}, 10^{-3}, 10^{-5}, 10^{-7}$.

Proof of Proposition D.2. (i) A scoring rule is strictly proper if, for each x, the expected score of a forecast Q under the label law $P^*(\cdot \mid x)$ is uniquely maximised at $Q = P^*(\cdot \mid x)$ (Gneiting and Raftery, 2007); average over x. (ii) The held-out log loss of the population selection h_c is $R(h_c)$, and by Lemma A.1(ii), $c < c'$ implies $R(h_c) \le R(h_{c'})$. (iii) An imitation-only rule sees the same data distribution in both worlds, since the teacher's channel and the input law are the same, and Proposition 6.6 gives the same selection at every c; the selection that is right in one world keeps an error or drops a valid rule in the other. □

Proposition D.3 (The selection depends on the input law)**.** *In the setting of Proposition 6.6, let the inputs have law* μ *and let* Δ_j *be component* j*'s expected log-loss reduction per input of* X_j*. Then component* j *is selected if and only if* $\mu(X_j)\Delta_j > c\,k_j$*. So the window that keeps a valid component* j *and drops an error component* j' *is* $c \in (\mu(X_{j'})\Delta_{j'}/k_{j'},\ \mu(X_j)\Delta_j/k_j)$*; it moves with*

μ and is empty once $\mu(X_{j'})/\mu(X_j) \geq \Delta_j k_{j'}/(\Delta_{j'} k_j)$, and after a shift to μ' a c chosen in μ's window keeps the error if and only if $\mu'(X_{j'})\Delta_{j'} > c\, k_{j'}$.

Proof of Proposition D.3. The population risk is $\sum_j \mu(X_j)\mathbb{E}[\text{loss} \mid x \in X_j]$, so choosing component j changes J_c by $c\,k_j - \mu(X_j)\Delta_j$; the other claims are rearrangements of this criterion. □

A direct computation (code in the accompanying repository) confirms the thresholds of Proposition 6.6 by brute force in 8000 pairs of an instance and a value of c.

D.4 Error envelopes

Proof of Proposition 6.3. The envelope. Off E_1 the inputs with prefix in S_2 are the error inputs, a fraction p_2 of the inputs off E_1, so $R(\pi_e) = (1-p_1)H(p_2)$. Since $p = p_1 \cdot 1 + (1-p_1)p_2$, concavity of H gives $H(p) \geq p_1 H(1) + (1-p_1)H(p_2) = R_e$. The program π_e needs a, d, m, a description of S_1 given m, and s_2 (from which p_2 follows), which gives the length bound.

(i) The error set still has density p, so Theorem 5.7(o) applies: programs of level at least 2 have $J_c \geq J_c^B \geq \Phi_2(c) > J_e(c) + \epsilon$, so $\mathrm{lev}(\pi) = 1$. Then (D1) gives $\ell(\pi) \geq d - 4 - \Lambda_1$, and $c(d - 4 - \Lambda_1) + R(\pi) \leq J_c(\pi) \leq c(d + k_1 + \kappa_e) + R_e + \epsilon$, that is, $R(\pi) \leq R_e + \zeta$.

(ii) Write $R(\pi) = p_1 R^{(1)} + (1-p_1)R^{(2)}$, where $R^{(1)}$ and $R^{(2)}$ are the risks conditional on E_1 and off E_1, and put $x := (1-p_1)R^{(2)}$.

Coding S_2 given S_1. For $z \notin S_1$ define $q_z(b)$ as in the proof of Lemma 5.5 with $B = \infty$. Then $Q(S') := \prod_{z \notin S_1} q_z(\mathbf{1}_{S'}(z))$ is a lower-semicomputable semimeasure on subsets of $\{0,1\}^m \setminus S_1$, given (π, a, m, S_1). For $z \notin S_1$ the teacher's label is $f_a \oplus \mathbf{1}_{S_2}(z)$, so $-\log Q(S_2) = (2^m - s_1)R^{(2)} = 2^m x$. By (F2), $K(S_2 \mid a, \pi, S_1) \leq 2^m x + 2\log m + \kappa$.

Chain. By (F3), (F1) and Lemma 5.3 with $t = 1$, $K(a, S_2 \mid S_1) \leq \ell(\pi) + 4 + 2\log d + 2^m x + 2\log m + \kappa$. By (F3) and (F4), $K(a, S_2 \mid S_1) \geq K(a, S_2) - K(S_1) - \kappa \geq d + B_2 - K(S_1) - \kappa$. With $\Lambda_1 = 2\log d + 2\log m + \kappa$ and (D1),

$$\ell(\pi) \;\geq\; d - 4 - \Lambda_1 + \max\{0,\; B_2' - 2^m x\}, \qquad B_2' := \big(B_2 - K(S_1)\big)^+.$$

Combination. $J_c(\pi) = c\,\ell(\pi) + p_1 R^{(1)} + x \leq J_e(c) + \epsilon$ gives

$$p_1 R^{(1)} \leq \zeta + (1-p_1)H(p_2) - \big[x + c\max\{0, B_2' - 2^m x\}\big].$$

Over $x \geq 0$ the bracket is piecewise linear: for $x \leq B_2'/2^m$ it equals $cB_2' + (1 - c2^m)x$, and for larger x it equals x. So its minimum is $B_2'/2^m$ (at $x = B_2'/2^m$) if $c2^m \geq 1$, and cB_2' (at $x = 0$) if $c2^m < 1$; that is, $\min(c, 2^{-m})B_2'$. Hence $R^{(1)} \leq \bar{R}_1$, and Jensen's inequality gives $e_1(\pi) \geq 2^{-R^{(1)}} \geq 2^{-\bar{R}_1}$.

For $c \geq 2^{-m}$, use $(2^m - s_1)H(p_2) \leq \log\binom{2^m - s_1}{s_2} + \log(2^m + 1)$, which gives $\bar{R}_1 \leq \big(\zeta + (\delta_2 + \log(2^m+1) + K(S_1))/2^m\big)/p_1$ with $\delta_2 := \log\binom{2^m - s_1}{s_2} - B_2$. □

Numbers (code in the accompanying repository). A direct computation also confirms the minimisation identity on 2000 random cases and, on 40 exact small instances ($d = 6$, $m = 4$), that $-\log Q(S_2) = (2^m - s_1)R^{(2)}$ and that Q has total mass at most 1. At $d = 100$, $m = 20$, $p = 2^{-10}$, with $S_1 = \{z : z_1 = \cdots = z_j = 0\}$, so that $K(S_1 \mid m) \leq 2\log(j+1) + \kappa$, and S_2 uniformly random (so δ_2 is small with high probability), at $\kappa = 0$ unless stated:

instance	$\log\binom{2^m - s_1}{s_2}$	$H(p) - R_e$	$10^5 c_{\mathrm{env}}$ ($\kappa = 0/10/30$)	share ($\kappa = 0/30$)
$j = 11$ $(512 + 512)$	6,364	0.00510	6.98/4.94/3.12	0.47 / 0.42
$j = 13$ $(128 + 896)$	10,418	0.00123	1.64/1.17/0.746	0.31 / 0.25

lower bound on e_1 ($\delta_2 = 0$)	$c = 1.01 \cdot 2^{-m}$	$2 \cdot 10^{-6}$	$5 \cdot 10^{-6}$	10^{-5}	$3 \cdot 10^{-5}$
$j = 11$	0.862	0.774	0.567	0.338	0.042
$j = 13$	0.545	0.350	0.097	0.012	0.000

The values of e_1 are computed from the relaxed bound for $c \geq 2^{-m}$ at the end of the proof, with $K(S_1) \leq k_1 + 2\log m + \kappa$. In both instances $J_e < \Phi_2$ on $(0, c_{\mathrm{env}}]$, as part (i) requires. With $\delta_2 = 20$ the first column becomes 0.839 and 0.489. Here $j = 11$ means $s_1 = 512$ cheap and $s_2 = 512$ random points, and the share is the share of the window's logarithmic width below c_{env}, $\log(c_{\mathrm{env}} 2^m)/\log(c_{\mathrm{hi}} 2^m)$, with the window $(2^{-m}, c_{\mathrm{hi}}]$ of Theorem 5.7 for an incompressible rule. Against $L_S = 11{,}710$ the deficiencies of the two rules are about 5,300 and 1,300 bits.

A three-hypothesis illustration ($\kappa = 0$, upper-bound lengths treated as exact; not proved). For $j = 11$, the envelope beats the good model for $c < 1.87 \cdot 10^{-4}$ with the generic code lengths above; a computation with explicit code lengths ($\ell(\pi_g) = d + 19.9$, $\ell(\pi_e) = d + 28.9$) gives $c < 5.7 \cdot 10^{-4}$, where the envelope gives the erroneous label average probability 0.50 on E, against 0.001 for the good model. A memoriser of the whole rule wins only below $9.5 \cdot 10^{-7}$.

Envelopes with density $\theta < 1$. If the cheap region contains errors with density θ, the envelope predicts the error there with probability θ, and the proof of part (i) goes through with $R_e = p_1 H(\theta) + (1 - p_1) H(p_2)$ and the corresponding length. For $\theta < \frac{1}{2}$ the intended label is still the more likely one inside the region; for $\theta > \frac{1}{2}$ the envelope is a learned systematic error. Part (ii) is proved only for $\theta = 1$.

D.5 Unseen properties

Proof of Proposition 6.7. (i) Let V' be the prefix machine that reads a self-delimiting code z of a pair (decider q for Q, program q' for b), then a program $\pi' \in \operatorname{dom} V$, and outputs an index of the conditional semimeasure that is the point mass at $b(x)$ (or 0 while $b(x)$ is undefined) if q accepts x, and $\nu_{\pi'}(\cdot \mid x)$ otherwise; this is lower semicomputable. By closure under prefix-free simulation (Definition 3.4) there is σ with $V(\sigma z \pi') = V'(z\pi')$; let $\kappa := |\sigma|$ and take $|z| = K(Q, b)$. Distinct π' give distinct programs $\sigma z \pi'$. Since $\operatorname{dom} D \cap Q = \emptyset$, $\nu_{\sigma z \pi'}(D) = \nu_{\pi'}(D)$, and since $Z \subseteq Q \cap \operatorname{dom} b$, $\nu_{\sigma z \pi'}(D \cup b|_Z) = \nu_{\pi'}(D)$. So

$$\sum_{\pi' \in \operatorname{dom} V} 2^{-\lambda|\sigma z \pi'|}\, \nu_{\sigma z \pi'}(D) = 2^{-\lambda(K(Q,b)+\kappa)} \sum_{\pi'} 2^{-\lambda|\pi'|} \nu_{\pi'}(D) = 2^{-\lambda(K(Q,b)+\kappa)}\, \xi_\lambda(D),$$

which is the posterior mass claim, and the same sum with $D \cup b|_Z$ in place of D is a lower bound for $\xi_\lambda(D \cup b|_Z)$.

(ii) Equal likelihoods cancel in the ratio of posteriors, which leaves the ratio of prior weights $2^{-\lambda\ell(\pi_1)}/2^{-\lambda\ell(\pi_2)}$. □

On a conjunction ladder (Hänni, 2026a, Theorem 9.10 and Remark G.11), deterministic selection of a least-size consistent formula, a λ-MAP rule, makes exponentially many mistakes on some ordering for every tie rule, while the $\lambda \to \infty$ power posterior, uniform over those formulas, makes $O(d^2)$ Gibbs mistakes. The danger in (ii) is therefore sharpest for single-hypothesis selection.

D.6 A limitation of the method

Remark D.4 (A limitation of the method). The window of Theorem 5.7 needs $G_1 > H(p)$, that is, $p < 0.056$. The sharper floor of Lemma C.1 gives the kink for every $p < 1/4$ (Proposition 8.3), but does not extend the window as the argument stands: with it and no deviation term the level-2

floor $1-\log(3/2-p)$ exceeds $H(p)$ only for $p<0.123$; and a program of level at least 2 has $I(\pi)$ bounded only through its length, which near $c=2^{-m}$ allows $I(\pi)\approx L_S$ and a deviation term of order $\sqrt{H(p)/p}$. Whether a window exists for $p<0.123$ and c well above 2^{-m} is part of Question 6.8.

E Proofs for Section 7 (the model–randomness split)

The proof of Theorem 7.3 is given in the main text.

E.1 Proof of the Ville lemma

We prove Lemma 7.2. Let $Z_n := \xi(D_n)/\nu(D_n)$. Given the past and the next input x_{n+1}, which the input rule chooses before y_{n+1} is drawn,

$$\mathbb{E}[Z_{n+1}] = \sum_y \nu(y \mid x_{n+1}) \frac{\xi(D_n + (x_{n+1}, y))}{\nu(D_n)\,\nu(y \mid x_{n+1})} \le \frac{\sum_y \xi(D_n + (x_{n+1}, y))}{\nu(D_n)} \le Z_n,$$

the sum running over labels of positive ν-probability, by the one-point inequality. So (Z_n) is a nonnegative supermartingale with $Z_0 = \xi(\emptyset) \le 1$, also when the input rule is adaptive or randomised. Ville's maximal inequality (Ville, 1939) gives $\mathbb{P}[\sup_n Z_n \ge 2^j] \le Z_0\, 2^{-j} \le 2^{-j}$. For the consequence, $\xi_{\mathcal{C}} \ge 2^{-\ell(\pi)}\nu_\pi$, so $\nu_\pi(D_n) \le 2^{\ell(\pi)}\xi_{\mathcal{C}}(D_n) \le 2^{\ell(\pi)+\delta^*}\nu(D_n)$. □

E.2 Fill-ins from stored stream prefixes

Lemma E.1 (Fill-ins from stored stream prefixes)**.** *Let the class be $\mathcal{H}_\times$, let ν be a sampler with a program π_ν of length k, and let D be a data set with n inputs, all of length at most d, with $\beta_D := \min_{(x,y)\in D} \nu(y \mid x) > 0$. Put $i_D := \lceil \log(6n/\beta_D) \rceil$ and $t_D := \lceil \log(2d + 2 + 2\log(i_D + 1)) \rceil$. Then*

$$W_{\det}(D) \;\ge\; 2^{-k-2\log(t_D+1)-\kappa}\,\nu(D).$$

Since $\beta_D \ge \nu(D)$, $i_D \le \log(6n) + \log(1/\nu(D)) + 1$; so the cost $2\log(t_D+1)$ is doubly logarithmic in d and triply logarithmic in n and in the sampled randomness $-\log\nu(D)$.

Proof. Order $Y = \{y^{(1)}, \dots, y^{(|Y|)}\}$.

Private streams. Let $j(x)$ be the length-lexicographic index of x, so $j(x) \le 2^{d+1} - 2$ for $|x| \le d$, and let $\langle a, b\rangle := (a+b)(a+b+1)/2 + b$ be the Cantor pairing, which is injective and increasing in b. For a finite string w (bits indexed from 0), the *stream prefix* of x in w is $w_{\langle j(x),0\rangle} w_{\langle j(x),1\rangle} \cdots$, continued while the positions are below $|w|$. Distinct inputs read disjoint positions.

Certification. For an input x put $F_h := \sum_{h' \le h} \nu(y^{(h')} \mid x)$, so $F_0 = 0$ and $F_{|Y|} = 1$. At depth $i \ge 1$ compute, by a fixed algorithm from the program for ν, rationals $q_{i,h}$ with $|q_{i,h} - F_h| \le 2^{-i-2}$ ($q_{i,0} = 0$, $q_{i,|Y|} = 1$). A stream prefix $b_1 \cdots b_i$ names the dyadic interval $I = [0.b_1 \cdots b_i,\ 0.b_1 \cdots b_i + 2^{-i})$. It *certifies* $y^{(h)}$ at depth i if $q_{i,h-1} + 2^{-i-2} \le \min I$ and $\sup I \le q_{i,h} - 2^{-i-2}$. Then $I \subseteq [F_{h-1}, F_h]$.

Consistency. Suppose the stream of x certifies $y^{(h)}$ at depth i and $y^{(h')}$ at depth i', with $h < h'$. The two intervals are nested, since one stream prefix extends the other; the inner one has positive length and lies in $[F_{h-1}, F_h] \cap [F_{h'-1}, F_{h'}]$. For $h < h'$ this intersection is empty or the single point F_h, a contradiction. So $\varphi_w(x)$, defined as the label certified at some depth within the stream prefix of x in w (undefined if none), is a well-defined partial computable function, and an index of it is computable from $(V(\pi_\nu), w)$.

Probability. Every depth-i dyadic interval inside $[F_{h-1} + 2^{-i-1}, F_h - 2^{-i-1}]$ certifies $y^{(h)}$, because $q_{i,h-1} + 2^{-i-2} \le F_{h-1} + 2^{-i-1}$ and $q_{i,h} - 2^{-i-2} \ge F_h - 2^{-i-1}$. These intervals cover that segment except for at most $2 \cdot 2^{-i}$, and the segment has length $\nu(y^{(h)} \mid x) - 2^{-i}$. So a uniform stream prefix of length at least i certifies $y^{(h)}$, and hence gives $\varphi_w(x) = y^{(h)}$, with probability at least $\nu(y^{(h)} \mid x) - 3 \cdot 2^{-i}$.

Independence. For w uniform on $\{0,1\}^L$, the stream prefixes of distinct inputs are independent and uniform. If $L \ge (2^{d+1} + i)^2$, every input of length at most d has a stream prefix of length at least i, because $\langle j(x), i-1\rangle < (j(x)+i)^2 \le (2^{d+1}+i)^2$. Take $i = i_D$; then $3 \cdot 2^{-i_D} \le \beta_D/(2n)$, and

$$\Pr_w[\varphi_w \supseteq D] \;\ge\; \prod_{(x,y)\in D} \big(\nu(y \mid x) - 3 \cdot 2^{-i_D}\big) \;\ge\; \nu(D)\Big(1 - \frac{1}{2n}\Big)^n \;\ge\; \frac{\nu(D)}{2},$$

using $\nu(y \mid x) \ge \beta_D$ and Bernoulli's inequality.

Programs. Let V' be the prefix machine that reads π_ν (by simulating V, which consumes exactly π_ν), then a self-delimiting code of an integer t of length at most $2\log(t+1)+1$, then $L_t := 2^{2^t}$ bits w, and outputs an index of the deterministic conditional semimeasure φ_w. By closure under prefix-free simulation (Definition 3.4) there is σ with $V(\sigma u) = V'(u)$. The programs $\pi_w := \sigma\, \pi_\nu\, \mathrm{code}(t)\, w$ are distinct for distinct w. With $t = t_D$ we have $2^{t_D} \ge 2d + 2 + 2\log(i_D+1)$, so $L_{t_D} \ge \big(2^{d+1}(i_D+1)\big)^2 \ge (2^{d+1}+i_D)^2$.

Sum.

$$W_{\det}(D) \ge \sum_w 2^{-|\pi_w|}\mathbf{1}[\varphi_w \supseteq D] = 2^{-|\sigma|-k-|\,\mathrm{code}(t_D)|} \Pr_w[\varphi_w \supseteq D] \ge 2^{-k-2\log(t_D+1)-|\sigma|-2}\,\nu(D). \;\square$$

A direct computation (code in the accompanying repository) confirms the certification step on 600 random samplers with adversarial approximations $q_{i,h} = F_h \pm 2^{-i-2}$, and the pairing and length facts for $d \le 12$, $i \le 64$. The cost $2\log(t_D+1)$ is 5.6 bits at $d = 20$, $n = 10^3$ ($\beta_D = 1/3$); 6.3 bits at $d = 100$, $n = 10^6$ (for $\beta_D = 1/3$ and for $\beta_D = 2^{-10}$); 7.2 bits at $d = 1000$, $n = 10^{12}$ ($\beta_D = 10^{-3}$); and 8.9 bits at $d = 10^6$, $n = 10^{15}$ ($\beta_D = 1/3$). The hashing cost of Lemma E.2 at the same sizes is 31.9, 56.3 (with $r = 2$) or 60.1 (with $r = 10$), 106.6 and 142.7 bits.

For Lemma 7.2, a simulation (code in the accompanying repository; 4000 runs) with an adaptive input rule that puts the next input where ξ deviates most from ν gives the empirical frequencies $\mathbb{P}[\delta^* \ge j] = 0.355, 0.163, 0.069, 0.035, 0.0065$ for $j = 1, 2, 3, 4, 6$, all below 2^{-j}.

E.3 An alternative for dyadic samplers: hashing

Lemma E.2 (Derandomising a dyadic sampler by hashing)**.** *Let the class be $\mathcal{H}_\times$, let ν be a sampler whose values are multiples of 2^{-r}, with a program π_ν of length k, and let D be a data set with n inputs, all of length at most d. Then*

$$W_{\det}(D) \;\ge\; 2^{-k-2\log(n+1)-2\log(d+1)-2\log(r+1)-\kappa}\,\nu(D).$$

Proof. Exact values. From the index $V(\pi_\nu)$ and r one computes ν exactly: enumerate lower approximations $\nu_s(y \mid x) \uparrow \nu(y \mid x)$ until $\sum_y \nu_s(y \mid x) > 1 - 2^{-r}$, which happens because ν is a measure and Y is finite. Then every deficit $\nu - \nu_s$ is below 2^{-r}, and $\nu(y \mid x)$ is the least multiple of 2^{-r} that is at least $\nu_s(y \mid x)$.

Canonical sampler. For $b \in \{0, \dots, 2^r - 1\}$ let $\mathrm{lab}(x, b)$ be the label y whose cumulative interval $[\sum_{y'<y} \nu(y' \mid x), \sum_{y' \le y} \nu(y' \mid x))$ contains $b2^{-r}$. Then $\Pr_b[\mathrm{lab}(x,b) = y] = \nu(y \mid x)$ for uniform b.

Hash. Let $q := \max(d+1, r)$, fix (computably from q) an irreducible polynomial of degree q over GF(2), and embed $\{0,1\}^{\le d}$ injectively into $\mathrm{GF}(2^q)$ by mapping x to the element with

coefficient vector $1x$. For a seed $s \in \mathrm{GF}(2^q)^n$ let $h_s(x)$ be the first r coefficient bits of $\sum_{j<n} s_j\, x^j$, computed in $\mathrm{GF}(2^q)$ with x embedded. For distinct $x_1, \dots, x_n$ the map $s \mapsto (\sum_j s_j x_i^j)_{i \le n}$ is a bijection (a Vandermonde matrix with distinct nodes), so for uniform s the values $h_s(x_i)$ are independent and uniform on $\{0,1\}^r$.

Programs, one per seed. Let V' be the prefix machine that reads π_ν (by simulating V), then self-delimiting codes of n, d and r, then qn bits s, and outputs an index of the deterministic conditional semimeasure $x \mapsto \mathrm{lab}(x, h_s(x))$ on $\{0,1\}^{\le d}$, undefined elsewhere. By closure under prefix-free simulation there is σ' with $V(\sigma' u) = V'(u)$. The programs $\pi_s := \sigma' \pi_\nu \operatorname{code}(n) \operatorname{code}(d) \operatorname{code}(r)\, s$ are distinct for distinct s, deterministic, and of length $L_0 + qn$ with $L_0 \le k + 2\log(n+1) + 2\log(d+1) + 2\log(r+1) + \kappa$.

Sum. $\sum_s 2^{-|\pi_s|} \mathbf{1}[\nu_{\pi_s} \supseteq D] = 2^{-L_0} \Pr_s[\forall i : \mathrm{lab}(x_i, h_s(x_i)) = y_i] = 2^{-L_0} \prod_i \nu(y_i \mid x_i) = 2^{-L_0} \nu(D)$. □

Distinct seeds often give the same function on X; the sum is over programs, one per seed, so this does not matter. A direct computation (code in the accompanying repository) confirms the independence and the full statement exactly over small fields. The hashing idea for fair coins and the program mixture is in Hänni (2026a, Remark B.3).

E.4 Solomonoff induction at $\lambda = 1$

Proposition E.3 (SI at $\lambda = 1$)**.** *Let a string $z_{1:n}$ be drawn from a computable measure ν with a program $\pi_\nu \in \mathcal{H}_{\mathrm{SI}}$ of length k, and let $\delta^* := \sup_n \log\big(M(z_{1:n})/\nu(z_{1:n})\big)$, where $M = \xi_1$ is the mixture of $\mathcal{H}_{\mathrm{SI}}$. (i) $\Pi_1(\pi_\nu \mid z_{1:n}) \ge 2^{-k-\delta^*}$ for all n, and $\mathbb{P}[\delta^* \ge j] \le 2^{-j}$. (ii) The deterministic hypotheses consistent with $z_{1:n}$ (point masses on computable infinite extensions) have total weight at least $2^{-k-K(n)-\kappa}\nu(z_{1:n})$. (iii) In the deterministic-program representation of M every hypothesis is a fill-in; the sampler appears only as the programs "decoder for ν, then a code of z", of total weight at least $2^{-k-\kappa}\nu(z)$.*

Proof. (i) Under ν, $Z_n := M(z_{1:n})/\nu(z_{1:n})$ is a nonnegative supermartingale with $Z_0 \le 1$, because $M(z0) + M(z1) \le M(z)$; Ville's inequality gives the tail bound, and $\Pi_1(\pi_\nu \mid z_{1:n}) = 2^{-k}\nu(z_{1:n})/M(z_{1:n}) \ge 2^{-k-\delta^*}$. (ii) Let $w := z_{1:n}0^\infty$ and let $\mathbf{1}_w$ be the point mass on its prefixes. Given n and π_ν, the restriction of ν to $\{0,1\}^n$ is a computable measure, so (F2) gives $K(z_{1:n} \mid n, \pi_\nu) \le -\log \nu(z_{1:n}) + \kappa$, and (F1), (F3) give $K(z_{1:n}) \le K(n) + k - \log \nu(z_{1:n}) + \kappa$. An index of $\mathbf{1}_w$ is computable from $z_{1:n}$, so the class contains a program for $\mathbf{1}_w$ of length at most $K(z_{1:n}) + \kappa$, and $\mathbf{1}_w$ is consistent with $z_{1:n}$. (iii) The programs "a fixed decoder, a shortest program for ν, then a Shannon–Fano code of z under ν" have total weight at least $2^{-k-\kappa}\nu(z)$; this is the usual proof of $M \ge 2^{-K(\nu)-O(1)}\nu$ (Li and Vitányi, 2008). □

E.5 Proof of the private-randomness theorem

We prove Theorem 7.4. Write $D = D_n$.

(i). $J_\lambda(\hat\pi; D) \le J_\lambda(\pi_\nu; D) = \lambda k - \log \nu(D)$. By Lemma 7.2, $-\log \nu_{\hat\pi}(D) \ge -\log \nu(D) - \ell(\hat\pi) - \delta^*$. Substituting gives $(\lambda-1)\ell(\hat\pi) \le \lambda k + \delta^*$. The upper bound on the loss is $J_\lambda(\hat\pi; D) \le J_\lambda(\pi_\nu; D)$ with $\lambda \ell(\hat\pi) \ge 0$.

(ii). $\sum_{\ell(\pi) > L} 2^{-\lambda \ell(\pi)} \nu_\pi(D) \le 2^{-(\lambda-1)L} \xi_{\mathcal{C}}(D) \le 2^{-(\lambda-1)L + \delta^*} \nu(D)$, and the normaliser $\xi_\lambda(D)$ is at least the sampler's term $2^{-\lambda k}\nu(D)$.

(iii). A consistent deterministic π has $1 = \nu_\pi(D) \le 2^{\ell(\pi)+\delta^*}\nu(D)$, which is the length bound. For the power posterior, $\sum_{\mathrm{det}} 2^{-\lambda\ell(\pi)} = \sum_{\mathrm{det}} 2^{-\ell(\pi)} 2^{-(\lambda-1)\ell(\pi)} \le 2^{-(\lambda-1)(-\log \nu(D) - \delta^*)} W_{\mathrm{det}}(D)$,

the sums running over consistent deterministic programs; with $W_{\det} \le \xi_{\mathcal{C}} \le 2^{\delta^*}\nu(D)$ and the normaliser at least $2^{-\lambda k}\nu(D)$ this gives the bound. For the tempered posterior, the fill-ins' total weight is $W_{\det}(D) \le 2^{\delta^*}\nu(D)$, and the normaliser is at least the sampler's term $2^{-k}\nu(D)^{\eta}$.

(iv), likelihood ratios. Fix $\pi \in \mathcal{C}$ and let $\Lambda_n := \nu_\pi(D_n)/\nu(D_n)$ and $g_n := d_{\mathrm{H}}^2(\nu(\cdot \mid x_n), \nu_\pi(\cdot \mid x_n))$; g_n is determined before y_n is drawn. Given the past and x_{n+1},

$$\mathbb{E}\sqrt{\Lambda_{n+1}/\Lambda_n} = \sum_y \nu(y \mid x_{n+1})\sqrt{\nu_\pi(y \mid x_{n+1})/\nu(y \mid x_{n+1})} = 1 - g_{n+1}.$$

If $g_{n+1} = 1$, then almost surely $\nu_\pi(y_{n+1} \mid x_{n+1}) = 0$ and Λ stays 0 from then on. Let $M_n := \sqrt{\Lambda_n}/\prod_{i\le n}(1-g_i)$ while the product is positive, and $M_n := 0$ afterwards. Then (M_n) is a nonnegative supermartingale, so it converges almost surely to a finite limit. On the event $\{\sum_i g_i = \infty\}$ either some $g_i = 1$, and $\Lambda_n = 0$ eventually, or $\prod_{i\le n}(1-g_i) \to 0$, and $\sqrt{\Lambda_n} = M_n \prod_{i\le n}(1-g_i) \to 0$. This is the argument behind Kakutani's dichotomy (Kakutani, 1948); it holds for adaptive inputs because g_n is predictable.

(iv), MAPs. There are countably many programs, so almost surely, simultaneously, $\delta^* < \infty$ and $\Lambda_n(\pi) \to 0$ for every π that is not ν-equivalent on the run. By (i), every λ-MAP lies in the finite set $\{\pi : \ell(\pi) \le L^*\}$ with $L^* := (\lambda k + \delta^*)/(\lambda - 1)$. A λ-MAP $\hat\pi$ has $\log \Lambda_n(\hat\pi) \ge \lambda\ell(\hat\pi) - \lambda k \ge -\lambda k$, from $J_\lambda(\hat\pi) \le J_\lambda(\pi_\nu)$. Each non-equivalent program in the finite set eventually has $\Lambda_n < 2^{-\lambda k}$, so eventually none of them is a λ-MAP.

(iv), posterior. For every L, using (ii) for the long programs and the sampler's term for the normaliser,

$$\Pi_\lambda(\text{not equivalent} \mid D_n) \;\le\; 2^{\lambda k} \sum_{\ell(\pi)\le L,\ \text{not equivalent}} 2^{-\lambda\ell(\pi)}\Lambda_n(\pi) \;+\; 2^{\lambda k + \delta^* - (\lambda-1)L}.$$

The first term is a finite sum of terms that tend to 0. Let $n \to \infty$, then $L \to \infty$. □

A direct computation (code in the accompanying repository) in an explicit product class of samplers, fill-ins and partial memorisers with Kraft weights confirms (i)–(iii) and the upper end of Theorem 7.3(iii) in 1440 cases.

E.6 Proof of the shared-randomness theorem

We prove Theorem 7.6. (i) Since $U(o) > 0$, $J_\lambda(\pi_U; o) < \infty$. By (Dom), $-\log U(o) \le k + \kappa_0 - \log P_{\pi_\nu}(o)$, so $J_\lambda(\pi_U; o) = \lambda\ell_U - \log U(o) \le \lambda\ell_U + \kappa_0 + k - \log P_{\pi_\nu}(o)$, which is below $\lambda k - \log P_{\pi_\nu}(o) = J_\lambda(\pi_\nu; o)$ exactly when $(\lambda-1)k > \lambda\ell_U + \kappa_0$; the same condition says $\ell^*_\lambda < k$, and every λ-MAP has length at most ℓ^*_λ by Theorem 4.2(a). (ii) $\sum_{\pi\in A} 2^{-\lambda\ell(\pi)}P_\pi(o) \le 2^{-(\lambda-1)\min_A \ell}\sum_\pi 2^{-\ell(\pi)}P_\pi(o) = 2^{-(\lambda-1)\min_A \ell}\xi_1(o) \le 2^{-(\lambda-1)\min_A \ell + \kappa_1}U(o)$, and the normaliser $\xi_\lambda(o)$ is at least $2^{-\lambda\ell_U}U(o)$. For SI with adaptive or non-computable inputs the sampler is not an SI hypothesis, since it does not model the inputs; in $\mathcal{H}_{\mathrm{chr}}$ it is a chronological semimeasure, and M_H dominates every lsc chronological semimeasure on every input sequence (Hänni, 2026a, Corollary 6.17(a)), so (Dom) holds there with $U = M_H$. □

A direct computation (code in the accompanying repository) in an explicit shared class that contains its own mixture confirms (ii) and Theorem 4.2(a) and (c1) in 300 cases.

Example E.4 (Noisy parity, full table: an illustration)**.** Let $d = 12$ and let the data be the full table ($N = 4096$ inputs) of a random parity with i.i.d. flips at rate 1/8 (509 flips). A small product class contains the coin, parity-plus-noise samplers ($\ell = 24.2$ bits) and the memoriser (parity plus flip set); a "shared class" adds the finite Bayes mixture U over all parities and a grid

of noise rates, with an *assigned* length $\ell_U = 12$. A direct computation (code in the accompanying repository) gives: the product-class MAP is the sampler at $\lambda = 1, 1.1, 1.5, 2, 3, 6$, and at $\lambda = 1$ the memoriser is only 8.2 bits worse, a tie up to $O(\log N)$; the shared-class MAP is the sampler for $\lambda < 1.315$ and U above, the threshold being $(-\log U(D) + \log \nu(D))/(k - \ell_U)$ (the rounded values 2234.0, 2218.0, 24.2, 12 give 1.311). Since $\ell_U = 12$ is assigned and this finite class does not satisfy (Dom), the example illustrates the mechanism of Theorem 7.6; it is not an instance of it.

Remark E.5 (No split in SI's program representation). Solomonoff's M is a mixture over deterministic programs. Tempering does not change deterministic hypotheses (Lemma 3.6(iii)), and a power prior $2^{-\lambda \ell(\pi)}$ charges sampled bits and model bits alike, so it is indifferent between "decoder for ν plus a code of the randomness" and a literal fill-in of the same length. The split needs a representation in which the randomness is not part of the program: a product hypothesis with private coins.

F Proofs for Section 8 (algorithmic statistics)

F.1 Steeper codes over sets and products, and sets from products

Proposition F.1 (Steeper two-part codes over sets)**.** *Let $\lambda > 1$ and let A_λ minimise $\lambda K(A) + \log |A|$ over $A \ni z$. Then, for every $b \geq 0$,*

$$K(A_\lambda) \leq \operatorname{soph}_b^{\mathrm{set}}(z) + \frac{b + \kappa}{\lambda - 1},$$

and A_λ is $\big(b + (\lambda - 1) \operatorname{soph}_b^{\mathrm{set}}(z)\big)$-sufficient.

Proof. Let A_b attain $\operatorname{soph}_b^{\mathrm{set}}(z)$. Then $\lambda K(A_\lambda) + \log |A_\lambda| \leq \lambda K(A_b) + \log |A_b| \leq (\lambda - 1) \operatorname{soph}_b^{\mathrm{set}}(z) + K(z) + b$. Subtract the sufficiency line for the first claim; use $K(A_\lambda) \leq \lambda K(A_\lambda)$ for the second. A minimiser exists because only finitely many A have $K(A) \leq L$. □

Remark F.2 (λ near 1). The bounds do not show that λ slightly above 1 selects an approximately minimal sufficient statistic: as $\lambda \to 1$ the length bound becomes vacuous, and as λ grows the sufficiency deficiency grows. The proposition is useful at $\lambda - 1 = \Theta(1)$ when the sophistication is small.

Proposition F.3 (Steeper two-part codes over products)**.** *Every λ-MAP over $\mathcal{H}_\times$ for D has length at most $\operatorname{soph}_b^{\times}(D) + (b + \kappa)/(\lambda - 1)$ and is $\big(b + (\lambda - 1) \operatorname{soph}_b^{\times}(D)\big)$-product-sufficient.*

Proof. As for Proposition F.1, with Lemma 3.8 as the sufficiency line. The same trade-off applies (Remark F.2). □

Proposition F.4 (Sets from products)**.** *Let π be product-b-sufficient for D, and let ν_π be a computable conditional measure with rational values. Put $r_D := \lceil -\log \nu_\pi(D) \rceil$ (of order n) and $A := \{y' \in Y^{\operatorname{dom} D} : \nu_\pi(\operatorname{dom} D \mapsto y') \geq 2^{-r_D}\}$. Then $y_D \in A$, $K(A \mid \operatorname{dom} D) \leq \ell(\pi) + 2 \log(r_D + 1) + \kappa$, and A is $(b + 2 \log(r_D + 1) + \kappa)$-sufficient conditionally on $\operatorname{dom} D$: $K(A \mid \operatorname{dom} D) + \log |A| \leq K(y_D \mid \operatorname{dom} D) + b + 2 \log(r_D + 1) + \kappa$. Consequently $\operatorname{soph}_{b'}^{\mathrm{set}}(y_D \mid \operatorname{dom} D) \leq \operatorname{soph}_b^{\times}(D) + 2 \log(r_D + 1) + \kappa$ with $b' = b + 2 \log(r_D + 1) + \kappa$, provided a computable rational product hypothesis attains $\operatorname{soph}_b^{\times}(D)$. Against the unconditional set sophistication of the string y_D, add $K(\operatorname{dom} D)$.*

Proof. Since Y is finite and ν_π is computable with rational values, the test $\nu_\pi(\mathrm{dom}\, D \mapsto y') \geq 2^{-r_D}$ is decidable, so A is computable from $(\pi, \mathrm{dom}\, D, r_D)$. By (F1), (F3) and (F5), $K(A \mid \mathrm{dom}\, D) \leq \ell(\pi) + 2\log(r_D+1) + \kappa$. It contains y_D because $\nu_\pi(D) \geq 2^{-r_D}$. Since $\sum_{y'} \nu_\pi(\mathrm{dom}\, D \mapsto y') \leq 1$, $|A| \leq 2^{r_D}$. Hence

$$K(A \mid \mathrm{dom}\, D) + \log|A| \leq \ell(\pi) + 2\log(r_D+1) + \kappa + r_D \leq \ell(\pi) - \log\nu_\pi(D) + 1 + 2\log(r_D+1) + \kappa,$$

and product-b-sufficiency of π gives the claim. The consequence follows by taking π to attain $\mathrm{soph}_b^\times(D)$. Against the unconditional set sophistication of the string y_D, describe $\mathrm{dom}\, D$ first, at cost $K(\mathrm{dom}\, D)$. □

Question F.5 (lower-semicomputable product models)**.** Does Proposition F.4 hold for lower-semicomputable product hypotheses, for which the set A is only computably enumerable?

F.2 Proof of the noisy-parity proposition

We prove Proposition 8.3, in the notation of Section 8.3.

(i), injectivity and size. If $f_{a'} \oplus \mathbf{1}_{E'} = f_{a''} \oplus \mathbf{1}_{E''}$ with $a' \neq a''$, then the nonzero parity $f_{a'} \oplus f_{a''}$, which has $N/2$ ones, equals $\mathbf{1}_{E'} \oplus \mathbf{1}_{E''}$, which has at most $2pN < N/2$ ones. So $a' = a''$ and then $E' = E''$. Hence $|A_0| = 2^d\binom{N}{pN} = 2^{d+L_E}$.

(i), complexity of A_0 and of z. A_0 is computable from d and r, so $K(A_0) \leq 2\log d + 2\log r + \kappa$ by (F5). The table z determines a as the unique parity within distance $pN < N/4$ of z, and then $E = \{x : z(x) \neq f_a(x)\}$; d is determined by $|z| = N$. So $K(z) \geq K(a, E) - \kappa$, and by symmetry of information (F4), $K(a,E) = K(a) + K(E \mid a, K(a)) + O(1) \geq d + L_E - \delta_E - \kappa$. Therefore $K(A_0) + \log|A_0| \leq K(z) + \delta_E + 2\log d + 2\log r + \kappa$.

(i), flatness. For an integer $0 \leq v \leq d$ let A_v be the subset of A_0 whose parities a' agree with a on the first $d - v$ coordinates. Then $\log|A_v| = v + L_E$ and $K(A_v) \leq d - v + 2\log d + 2\log(v+1) + 2\log r + \kappa$, since A_v is computable from d, r, v and those $d - v$ bits. So every A_v has two-part length $d + L_E + O(\log d)$, within $\delta_E + O(\log d)$ of the sufficiency line. With the sufficiency line itself as the lower bound, the structure function h_z lies within $\delta_E + O(\log d)$ of the sufficiency line for every α from $O(\log d)$ up to $d + O(\log d)$.

(i), λ-optimal set models. By Proposition F.1 with $b = \delta_E + 2\log d + 2\log r + \kappa$, $K(A_\lambda) \leq 2\log d + 2\log r + \kappa + (b+\kappa)/(\lambda - 1)$. By (F3), $K(a) - K(a \mid A_\lambda) \leq K(A_\lambda) + \kappa$, which is the claim.

(ii), upper bound. Since $|E| = pN$ exactly, $-\log\nu_{\pi_g}(D) = NH(p)$, and $NH(p) \leq L_E + \log(N+1)$ by the standard bounds on binomial coefficients (Cover and Thomas, 2006). Also $K(y_D \mid \mathrm{dom}\, D) \geq K(z \mid d) - \kappa \geq K(z) - 2\log d - \kappa \geq d + L_E - \delta_E - 2\log d - \kappa$ by (i). So

$$\ell(\pi_g) - \log\nu_{\pi_g}(D) \leq d + \kappa_g + L_E + \log(N+1) \leq K(y_D \mid \mathrm{dom}\, D) + \delta_E + 2\log d + \kappa_g + \log(N+1) + \kappa,$$

and π_g is product-b-sufficient for every b in the range. Hence $\mathrm{soph}_b^\times(D) \leq \ell(\pi_g) \leq d + \kappa_g$.

(ii), lower bound. The memoriser gives $K(z \mid d) \leq d + 2\log N + L_E + \kappa \leq d + NH(p) + 2\log N + \kappa$: describe a, then $|E|$, then the index of E among the sets of that size. Let π be product-b-sufficient. If $\mathrm{lev}(\pi) \geq 2$, then $R(\pi) \geq G_1$ by Lemma 5.4 (with $B = \infty$), so

$$\ell(\pi) + NR(\pi) \geq NG_1 > d + NH(p) + 2\log N + \kappa + b \geq K(y_D \mid \mathrm{dom}\, D) + b,$$

by the upper end of the range of b, which contradicts sufficiency. So $\mathrm{lev}(\pi) = 1$, and (D1) of Lemma 5.6 with $s(1) = 4$ gives $\ell(\pi) \geq d - 4 - \Lambda_1$.

(ii), the kink. If $\ell(\pi) < d - 4 - \Lambda_1$, then $\mathrm{lev}(\pi) \neq 1$ by (D1), so $R(\pi) \geq G_1$ and $-\log\nu_\pi(D) = NR(\pi) \geq NG_1$.

(iii) follows from (i) and (ii), and from (i) and (iv) for every $p < 1/4$.

(iv). Let $\ell(\pi) < d - s(t) - \Lambda_1$. If $\mathrm{lev}(\pi) = t' \le t$, then (D1) and the monotonicity of s give $\ell(\pi) \ge d - s(t') - \Lambda_1 \ge d - s(t) - \Lambda_1$, a contradiction; so $\mathrm{lev}(\pi) > t$ and $u(\pi) \le \frac{1}{2} + 2^{-t-1}$. Lemma C.1 with $m = d$, $B = \infty$ and the information budget $I(\pi) \le \ell(\pi) + \kappa < d + \kappa$ applies with a deviation term at most γ_d, because the deviation term is increasing in the budget. It gives

$$\mathbb{E}_x \nu_\pi(y^*(x) \mid x) \le p + (1-2p)\Big(\frac{1}{2} + 2^{-t-1}\Big) + p\big(2\gamma_d + 2^{1-g}\big) = \frac{1}{2}\Big(1 + (1-2p)2^{-t} + 2p\big(2\gamma_d + 2^{1-g}\big)\Big),$$

and Jensen's inequality gives $R(\pi) \ge F_t$. For the sophistication: a product-b-sufficient π has

$$\ell(\pi) + NR(\pi) \le K(y_D \mid \mathrm{dom}\, D) + b \le K(z \mid d) + \kappa + b \le \ell(\pi_b) + 2\kappa + b \le d + \lceil L_E \rceil + \kappa_b + 2\kappa + b,$$

so $NR(\pi) < NF_t$ in the stated range of b, and the kink forces $\ell(\pi) \ge d - s(t) - \Lambda_1$. The upper bound and its range of b are those of (ii), whose proof uses only minimum-distance decoding, valid for $p < 1/4$, and the good model, whose description of p costs $K(p)$. For the limit: $j_d = d + \delta_E + O(\log d) + O(g)$ while $pN = p2^d$, so $\gamma_d \to 0$ when $\delta_E = O(d)$; taking $g \to \infty$ slowly, $F_t \to 1 - \log(1 + (1-2p)2^{-t})$, which tends to $1 > H(p)$ as $t \to \infty$, so some fixed t_p has $F_{t_p} > H(p)$ for all large d. □

F.3 Numbers for the noisy parity at every flip rate below 1/4

Table 3 evaluates Proposition 8.3(iv) for four dyadic flip rates of small complexity ($K(p) \le 2\log(h'+1) + 2\log(e'+1) + \kappa$ for $p = h'/2^{e'}$), with $\delta_E = 0$, $t = 2$ (so $s(t) = 7.2$) and the least g for which the range of b is nonempty. A direct computation (code in the accompanying repository) produced the values, with every occurrence of κ set to 0 or to 30 (so the constant in j_d is 2κ). The rows with $\kappa = 0$ are illustrations, not results about a particular reference machine; the rows with $\kappa = 30$ show how the bounds degrade. For this choice of g, at $d \ge 30$, $F_2 - H(p)$ lies between 0.016 (for $p = 15/64$) and 0.064 (for $p = 1/8$). For comparison, the floor of Lemma 5.4 at large level, $1 - \log(1 + 2p)$, exceeds $H(p)$ only for $p < 0.1525$, and with it no admissible t exists for $p \ge 3/16$ at these sizes.

Table 3: Bounds on product sophistication for the full-table noisy parity (Proposition 8.3(iv)), $\delta_E = 0$, $t = 2$. The interval for $\mathrm{soph}_b^{\times}(D)$ holds for every b in the stated range. Rows with $\kappa = 0$ set the machine constant to 0 (an illustration). The upper ends of the ranges depend on the choice of g.

κ	d	lower bound	upper bound $d + \cdots$ for $p =$				range of b
			1/8	3/16	7/32	15/64	($p = 1/8$; $p = 15/64$)
0	20	$d - 24.5$	14.6	17.3	19.8	22.3	$[43, 5.8 \cdot 10^4)$; $[51, 2.8 \cdot 10^3)$
0	30	$d - 26.8$	15.8	18.5	21.0	23.4	$[56, 6.8 \cdot 10^7)$; $[63, 1.6 \cdot 10^7)$
0	60	$d - 30.8$	17.8	20.5	23.0	25.4	$[90, 7.3 \cdot 10^{16})$; $[97, 1.8 \cdot 10^{16})$
0	100	$d - 33.7$	19.3	21.9	24.5	26.9	$[133, 8.1 \cdot 10^{28})$; $[140, 2.0 \cdot 10^{28})$
30	20	$d - 54.5$	44.6	47.3	49.8	52.3	$[103, 5.6 \cdot 10^4)$; $[111, 8.1 \cdot 10^3)$
30	30	$d - 56.8$	45.8	48.5	51.0	53.4	$[116, 6.8 \cdot 10^7)$; $[123, 1.6 \cdot 10^7)$
30	60	$d - 60.8$	47.8	50.5	53.0	55.4	$[150, 7.3 \cdot 10^{16})$; $[157, 1.8 \cdot 10^{16})$
30	100	$d - 63.7$	49.3	51.9	54.5	56.9	$[193, 8.1 \cdot 10^{28})$; $[200, 2.0 \cdot 10^{28})$

At $d = 20$ the lower bound is negative, so the interval says nothing there; the $O(\log d)$ terms dominate. Upper ends of the b-ranges are rounded down, lower ends to the nearest integer.

A smaller computation (code in the accompanying repository) illustrates (i)–(ii) at $d = 10$, $|E| = 16$ ($p = 1/64$): product list mixtures that leave v bits of a unknown have risk rising from $H(p) = 0.116$ at $v = 0$ to about 1 by $v = 7$, while the per-sample code lengths of the corresponding shared mixtures stay between 0.116 and 0.126.

G Proofs for Section 9 and the correspondence with the earlier report

Proof of Proposition 9.1. We go through the proofs and list, for each, the property of the class it uses and the constant that changes. Since $\mathcal{H}'$ is c.e. and prefix-free, (F1) holds for it with a constant κ_2 that depends on $\mathcal{H}'$: $K(\pi) \le \ell(\pi) + \kappa_2$.

- Lemma 5.3: the agreement $u(\pi; a')$ is lower semicomputable uniformly in (π, a', d), as in every class with uniformly lower-semicomputable likelihoods; the constant of the enumeration of the list depends on the class.
- Lemma 5.5: q_z is lower semicomputable uniformly in (π, a, m, B), so (F2) applies with a constant that depends on the class.
- Lemma 5.4: Jensen's inequality only; no constant changes.
- Lemma 5.6: the two lemmas above and (F1) with κ_2.
- Theorem 5.7: the lemmas above, and the reference programs only through J_g and J_b, whose constants are κ_g and κ_b of π'_g and π'_b.
- Theorem 7.4: Ville's inequality for the class's mixture $\xi_{\mathcal{C}}$ (Lemma 7.2), Kraft's inequality and countability; no constant enters.
- Proposition 4.7(ii): the coding theorem for the semimeasure $\tilde{R}$ built from the class, which holds with a constant that depends on the class. □

Proof of Proposition 9.3. (a) Each term $2^{-\lambda \ell(\pi)} P_\pi(D)$ is lower semicomputable uniformly in (π, D), because λ is computable, and the sum of uniformly lower-semicomputable nonnegative terms is lower semicomputable.

(b) Identify $\mathbb{N}$ with infinitely many inputs. Suppose the raw conditional were computable. Define $f(t)$, for $t \in \mathbb{N}$, as the first $y \in \{0, 1\}$ that a dovetailed search certifies to satisfy $\xi_\lambda(y \mid D_t, t) < 0.6$, where $D_t := \{(s, f(s)) : s < t\}$. The two conditionals sum to at most 1, so one of them is at most $\frac{1}{2}$ and the search halts. Then f is total computable, and by telescoping $\xi_\lambda(D_n) \le \xi_\lambda(\emptyset) \cdot 0.6^n \le 0.6^n$. But the class contains a deterministic program for f of length at most $K(f)+\kappa$, so $\xi_\lambda(D_n) \ge 2^{-\lambda(K(f)+\kappa)}$ for every n, a contradiction for large n. For the normalised conditional, run the same construction with it in place of the raw one; then its cumulative loss on D_n is at least $n \log(1/0.6)$. Each normaliser $\sum_{y'} \xi_\lambda(D_t + (t, y'))$ is at most $\xi_\lambda(D_t)$, so the normalised conditional is at least the raw one, and the normalised cumulative loss is at most the raw one, $-\log \xi_\lambda(D_n) + \log \xi_\lambda(\emptyset) \le \lambda(K(f) + \kappa)$. This is again a contradiction for large n.

(c) Lower-semicomputable reals are computable in the limit, and the denominators are positive, because every finite data set has a consistent deterministic program. □

G.1 Correspondence with the earlier report

The earlier report (Claude (Anthropic), 2026) works with a general imitation channel; its results are named here as they are named there.

earlier result	here	change
per-input versus once-only pricing	Proposition 4.4	exact minimax value kept
the sequence pathology	Theorem 4.2(a)	same threshold; adds (b)–(d)
universal conditional semimeasure	Remark B.1	–
per-block pricing	Proposition B.2	countable families with computable weights
kink: list-size and floor lemmas	Lemmas 5.3 and 5.4	semimeasures, via the lower-semicomputable agreement; truncation
kink: error segment	Lemma 5.6(D2)	–
kink: shared randomness	Proposition 8.3(i)	every λ-optimal set model
randomness of E removes the $2p$ slack	Lemma C.1	semimeasures; used for the kink at every $p < 1/4$
identification in the parity example	Corollary 5.10	adds information and error-label bounds; c_{hi} for $t^* < 2$; rigorous $d = 20$ window
empirical selection follows population selection	Lemma C.2	one-sided, truncated, without replacement
selection traces the lower hull	Lemma A.1	–
rate threshold, separable classes; rate blindness	Proposition 6.6	–
steepness not identifiable	Proposition D.2	–
guard erosion	Proposition D.1	teacher model stated; exact crossing
error envelopes	Proposition 6.3	all hypotheses; cheap errors reproduced
input-law dependence	Proposition D.3	separable case only

Not reused: the idealised rate threshold for arbitrary hypotheses and the realisability of every frontier shape, because Theorem 5.7 covers one model with one error rule over all hypotheses and Propositions 6.6 and D.3 cover separable components; the treatment of inference rules, which is not about function induction; and the pathologies of parasitic errors and language relativity, which hold for steeper SFI by the same arguments but add nothing specific to it. The scripts of the earlier report reproduce its numbers (Appendix C.3).